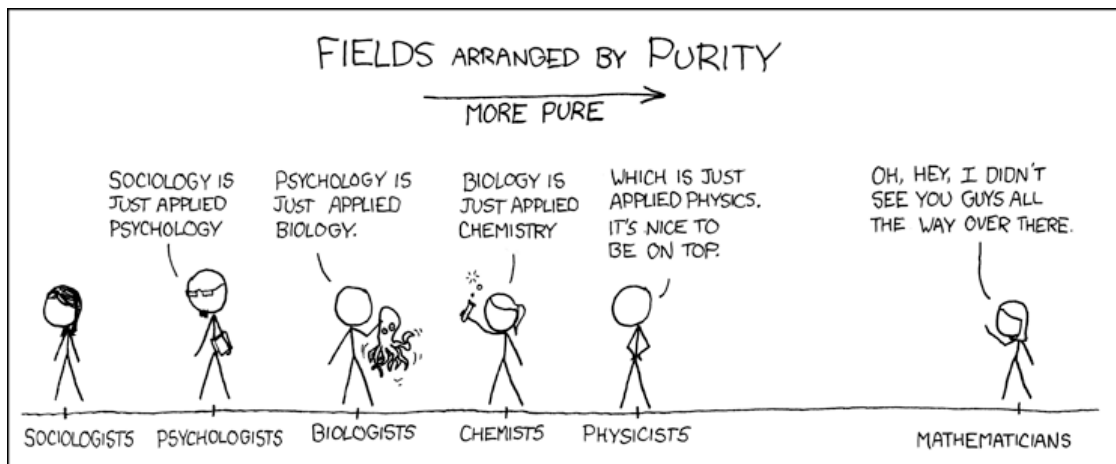


Science Mathematics

For the Unconvinced

CH1204 Semester 1 Workbook

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Introduction

If you're reading this, you're a student doing a degree in a science subject studying mathematics as a part of the degree. Perhaps you haven't really done maths since your GCSE days, and are fairly unconvinced about this whole thing. Science promised to be all about medicines, and drugs, and solving the climate crisis, and all kinds of things that seem more exciting than algebra. Maybe you've even questioned if this choice of subject was right for you, having flicked through the problem sheets and got a bit spooked.

If this is you, please don't worry, because I've written this course with you in mind.

On the other hand, maybe you already like maths. If this is case, I'm glad – and I hope you'll enjoy this course too.

Mathematics plays a central role in all science, and especially in chemistry and the chemistry-adjacent sciences like pharmacy and medicine, so you do need to learn it. However, it's also true that maths in school tends to be taught in a way that makes it seem much more boring than it actually is. Schools normally focus on the *machinery* behind the maths (like, how to solve quadratic equations), but don't tend say much about *why* you would actually want to do that (because lots of equations that describe the world around us are quadratic).

Here's an example. Imagine you are a pharmaceutical scientist interested in knowing how long a drug will hang around in the bloodstream. You need to know that in order to know what kind of dosage is sensible, so it's important to get it right. Under certain conditions, which we'll learn about later, the concentration, C , of the drug will decay over time, t according to the equation below:

$$C = C_0 e^{-kt}$$

Where C_0 is the initial amount of drug, and k is related to the how long it takes for the drug to decompose. By the end of this course, you'll understand:

- How to make sense of this equation, and understand what it physically means.
- How to manipulate this equation, and turn it into a form that is useful to you.
- How to derive this equation, so you can apply the same ideas to different problems.

Overall, our approach to mathematics is going to be emphasising the *substance* and *physical meaning* behind the equations. There are times when we will focus on the machinery, especially at first, but we will always try to make it clear *why* this stuff is useful. Understanding that is what will make you an excellent scientist.

This semester, we will start from a point which might seem very basic. The point is not that we race to the complex stuff, but that we start somewhere where everyone is comfortable. Then we will develop things up to a point where we are all familiar with the basic mathematics that any scientist needs. The level of the content this semester is mostly below A level standard, except for some things in chapters 7 and 8 which are not in the A level. However, things are presented differently from how they were in school, and there are new concepts in every chapter, so you will benefit from working through all of it. My aim is to teach you how to think like a scientist more than it is to teach you lots of methods that you know the steps for, but don't really understand.

Learning Resources

This handout has been written to be self-contained, so it will contain everything you need. However, you might benefit from reading other people's presentations of the same material, as everyone's brain works differently. Some good options for other mathematics books are:

- *Maths for Chemistry* by Monk and Munro.
- *Foundations of Science Mathematics* by Sivia and Rawlings.
- *Engineering Mathematics* by Stroud.

Many physical chemistry textbooks, such as those by Atkins, will also contain good, concise, explanations of fundamental mathematics.

There are also lots of good resources online, especially on YouTube. The channel *3blue1brown* is one of the best for explaining concepts in linear algebra and calculus (next semester).

Assessment

The module assessment will largely be by exam (75%), with a piece of written coursework in SEM2 (25%):

- Exam 1, in January, will be on the material from SEM1.
 - Two hours. Answer four questions from a choice of five.
- Exam 2, in May, will be on all material, but focussing on SEM2.
 - Two hours. Answer five questions, no choice.
- The written coursework will be introduced in SEM2 and will be a task in applied science mathematics.

The real aim of this module is to ensure that everyone has the baseline level of mathematical competency that will allow them to complete the rest of the degree. If you attend all the sessions and complete the workbooks and problem sheets, you should have no problem in passing the exams.

How To Use This Handout

This handout is structured into 8 **chapters**. Each chapter corresponds roughly to a week of the semester of term, and is structured as series of around 30 **steps**, separated by horizontal lines, and each step ends with a question for you to answer before moving on to the next step. The answer to the previous step is given at the top of the next step. The idea is that by working through it step by step, you will gradually learn what you need to know.

I suggest that you work through this one step at a time, using a bit of paper to cover up the step after the one you are on. Solve the problem in the current step, and then move the paper to check if you got it wrong or right. There should be enough space around the margins for you to write answers and working in, but at times you might benefit from having some extra paper for working too.

The only way to really learn science mathematics is to do it, and solve problems.

The most important thing you can do in this course is tackle the problems, and struggle with them until you get them right. That's why this handout is designed to be worked through, not just read. Do not just read the handout, assuring yourself that the answer stated is *what you would have got if you'd done it*, and looking at my explanation of the solution and feeling confident you'd be able to do that on your own without writing it down. **You need to solve the problems on your own.** You cannot learn to ride a bike by reading a lot of books about riding bikes. Mathematics is the same, but with a smaller risk of knee injuries.

After each chapter, there is a problem sheet **which you should complete before moving on to the next chapter**. The steps in each chapter introduce all the concepts you'll need for the problem sheets, but there will also be concepts introduced in the problem sheets that will be useful. **Doing the workbook without doing the problem sheets is not enough.**

In the taught course, there'll be weekly workshops and lectures where I will go through certain things, and be able to help you out with things you're stuck on. I recommend that you approach this in the following way:

- Work through the step-by-step part of each chapter before the workshop each week.
- Attempt the problem sheets before each workshop, but don't worry if you find some questions tricky.
- Come to the workshops and lectures prepared with any questions you have.

It won't be enough to just rely on working through everything in the timetabled sessions. You will need to do significant independent study as well. But that's why you're at university.

We're a broad church in university chemistry, and some of you have probably done maths or further maths at A level, and will find some of this course trivial. For you, there are **challenge problems** at the end of each sheet. These are not necessarily harder, but are less guided and may require some lateral thinking. If you want more maths, I have plenty – come and ask me.

Email (jp598@le.ac.uk) or come and find me (GP 0.22) if you have questions. Otherwise, let's get into it.

Chapter 1

Numbers and Arithmetic

To start with, we will recap some things that you've covered already in GCSE maths. Most of this should already be familiar, but we'll try and make it as interesting as possible.

1.1 Obviously, in science, we will be dealing with **numbers** fairly often. Therefore it's useful to know a few words that describe different types of numbers that you'll hear about:

- An **integer** is a whole number, that can be either positive, negative, or zero.
 - Examples of integers are: 1, 2, -43, 0, 102.
- A **rational number** is a number that can be created by dividing two integers. You can express rational numbers in **decimal form**.
 - Examples of rational numbers are: $\frac{1}{2} = 0.5$, $\frac{-3}{12} = -0.25$, $\frac{4}{1} = 4$.
 - All integers are also rational numbers.
- An **irrational number** is a number that you *cannot* get by dividing two integers, but that you can write in decimal form.
 - Examples of irrational numbers are: $\pi = 3.14159\dots$, $e = 2.718\dots$, $-\sqrt{2} = -1.41\dots$
- A **real number** is any number that is either an integer, rational, or irrational, but is not a *complex number*. If you've never heard of a complex number, then any number you can imagine is a real number.
 - All the numbers previously mentioned are real numbers.

Classify the numbers below as either integer, rational, or irrational:

$$1, -2\pi, 0.1, 100, \frac{1}{3}$$

1.2 Integer, irrational, rational, integer, rational.

We're not number theorists, so aren't going to have to classify lots of numbers, before you decide this is all very dull. It's more useful to just know the definitions. For example, someone might tell you that an atomic nucleus has 'integer spin', or that you need the 'real solution' to an equation. Now you know that this means the spin is a whole number, and that the equation might have complex solutions.

Here are some other things that it's useful to know about numbers:

- We can express any real number as a **decimal**, where there is a **decimal point** that separates the integer part from the fractional part (something like 10.0, or 1.23). Sometimes, especially in the context of computing, you will hear these (somewhat erroneously) called **floating point numbers**, or **floats**.

- We can take the **absolute value** of a number, which for any real number returns the positive version of the same number (if the number was negative). The notation for this is two vertical lines $|-3| = 3$. The absolute value of a positive number is just itself.
- In the previous step I mentioned **complex numbers**. These are numbers that contain something called the *imaginary unit*, i . We are not going to work with complex numbers in this course.

What is the absolute value of the numbers:

$$-4, 5.6, -0.0023$$

1.3 4, 5.6, 0.0023.

The **absolute value** of a number is sometimes called the **modulus** or **magnitude** of the number. These all mean the same thing for our purposes, but we will talk more about this in chapter 8.

Let's think about rational numbers. How else could I write the number $\frac{3}{6}$?

1.4 As 0.5, or $\frac{2}{4}$, or $\frac{1}{2}$.

It'll be useful if we can manipulate fractions. In the fraction $\frac{3}{6}$, 3 is the **numerator** and 6 is the **denominator**. We can freely manipulate fractions as long as we **do the same thing to both the numerator and denominator**, so I could divide both of those by 3 and get:

$$\frac{\frac{3}{3}}{\frac{6}{3}} = \frac{1}{2}$$

But you knew that already if you got the answer to the last step right.

What is the result if I add the two fractions below?

$$\frac{1}{2} + \frac{1}{2} = ?$$

1.5

$$\frac{2}{2} = 1$$

So, if fractions **have the same denominator**, we can add them together by **adding the numerators** and leaving the denominator.

If two fractions have different denominators, we can't add them directly. We need to manipulate one of them so it has the same denominator as the other. Can you add the two fractions below?

$$\frac{1}{4} + \frac{1}{2} = ?$$

1.6

$$\frac{3}{4}$$

We get there by multiplying the top and bottom of $\frac{1}{2}$ by 2 to get $\frac{2}{4}$. Then we add the numerators to get $\frac{3}{4}$. Multiplying the top and bottom of a fraction by the same number doesn't change the fraction, as it's equivalent to multiplying the fraction by 1 (you can think about why).

What about the example below?

$$\frac{1}{3} + \frac{1}{4} = ?$$

1.7

$$\frac{7}{12}$$

Here we can't just multiply top and bottom of one fraction by a number to get the same denominators. Instead, the thing we can do is multiply the top and bottom of the first fraction by the denominator of the second, and vice versa:

$$\left(\frac{1}{3} \times \frac{4}{4}\right) + \left(\frac{1}{4} \times \frac{3}{3}\right) = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$$

This method always works to add fractions that have different denominators.

To multiply fractions, we just multiply both numerators **and** both denominators. Try it now:

$$\frac{3}{5} \times \frac{2}{3} = ?$$

1.8

$$\frac{6}{15}$$

Just by multiplying both numerators ($3 \times 2 = 6$) and both denominators ($3 \times 5 = 15$).

What about dividing fractions? A useful trick is that dividing one fraction by another is the same as flipping the second fraction upside down and multiplying them, try it on the below:

$$\frac{3}{5} \div \frac{2}{3} = ?$$

1.9

$$\frac{9}{10}$$

Because:

$$\frac{3}{5} \div \frac{2}{3} = \frac{3}{5} \times \frac{3}{2} = \frac{9}{10}$$

Where we flipped the $\frac{2}{3}$ to $\frac{3}{2}$. Easy, innit?

Finally on fractions, if you multiply a fraction by a constant, you just multiply the numerator by the constant. What is:

$$4 \times \frac{3}{4} = ?$$

Remember that you can think of a number, like 4 as just $\frac{4}{1}$.

1.10 3, because:

$$4 \times \frac{3}{4} = \frac{4}{1} \times \frac{3}{4} = \frac{4 \times 3}{1 \times 4} = \frac{12}{4} = 3$$

That's enough fractions for now.

Here's a leading question about numbers: is the number 2000000 an integer?

1.11 Yes, it is.

Though writing out 2000000 is a bit cumbersome, as you have to count the zeros to know what it is, and that starts to give me a headache when there are more than about three zeros. Infact, we often have to deal with very large or very small numbers in science, so it will be useful to have a way to write out these numbers more neatly.

Standard form is the way we do this. In standard form, you write the number 2000000 as:

$$2 \times 10^6$$

Then we can easily see that:

- The number starts with a 2.
- There are 6 zeros after the 2 (and $10^6 = 1000000$).

I find it much easier to compare 2×10^6 to 2×10^8 , than to compare 2000000 to 200000000.

Write the number 50000 in standard form.

1.12 5×10^4

Easy, right? Just count the zeros. What about the number 42500?

1.13 4.25×10^5 .

Maybe you wrote 425×10^2 . That isn't *wrong*, but usually we choose to write the first number in standard form as a decimal between 0 and 10. This makes it easier to compare numbers with each other. To illustrate this, which of the numbers below is bigger?

$$3.65 \times 10^6 \text{ or } 365.0 \times 10^5$$

1.14 365.0×10^5 .

But you probably had to think about it for a few seconds.

In contrast, the same two numbers written in standard form:

$$3.65 \times 10^6 \text{ or } 3.65 \times 10^7$$

makes that easier to see, because you just need to look at the power on the 10. Hence, we usually keep the first number between 0 and 10.

Standard form also works for numbers lower than 1, by making the power on the 10 negative.

How would you write the number 0.1 in standard form?

1.15 1×10^{-1}

So negative powers of 10 give us numbers lower than 1, and $10^{-1} = 0.1$.

How would you write 0.01 in standard form?

1.16 1×10^{-2}

0.01 is a factor of 10 smaller than 0.1, so the number in the power of our standard form gets smaller by 1.

What about the number 0.0002, can you write that in standard form?

1.17 2×10^{-4}

You are hopefully getting the idea by now.

Anyway, we said standard form is useful in science, because often we deal with huge or tiny numbers, so here's an example:

The mass of a proton is 1.67×10^{-27} kg. Write this number out as a decimal.

1.18 0.000 000 000 000 000 000 000 016 7 kg.

If you actually did that, rather than turning over the page in frustration, I'm impressed with your resolve. The point is that standard form makes this easier to see. We will see it again many times.

Time for another trivial leading question:

$$\sqrt{100} = ?$$

1.19 $\sqrt{100} = 10$

$\sqrt{x}$ is a **function**. Functions are a fundamental part of science mathematics. Simply, a function takes an input (the **argument**), does some kind of calculation, and gives you an output (the **result**). In the example above:

$$\sqrt{100} = 10$$

We have the function $\sqrt{x}$, the argument x is 100, and the result is 10. Often we denote functions with the letter f , and use the letter x to denote an argument we haven't yet specified. For example, we could write:

$$f(x) = \sqrt{x}$$

For the example above.

I have a new function:

$$f(x) = x + 3$$

What is the result if input the argument $x = 0$ to this function?

1.20 The result is 3.

The function above, $f(x) = x + 3$, was just a function that says '*take the input, add three to it, and return the result*'. You might think that that seems like a bit of a long-winded way to describe 'adding three to a number', and you'd be right, but other functions can be more complicated. What about:

$$f(x) = x^2 + 3x + 2$$

What is the result if you put in $x = 1$ to that function?

1.21 The result is $1 + 3 + 2 = 6$.

All functions have associated **inverse functions**, which means you can 'undo' the effect of a function. Take our 'adding three to the input function' from before:

$$f(x) = x + 3$$

What would be the inverse of this function? We want to 'undo' the addition of three to the result.

1.22 $f^{-1}(x) = x - 3$

Subtracting three from the input. Note the notation $f^{-1}(x)$ that means 'inverse function of $f(x)$ '.

So you can see that **subtraction** is the inverse of **addition** and vice-versa. Here's another function, $g(t)$. Note we're use new letters to denote the function (g) and argument (t), because there is nothing sacred about f and x :

$$g(t) = 2t$$

What is the result if you input the argument $t = 6$ to $g(t)$?

1.23 The result is $g(6) = 2(6) = 12$.

So the function $g(t)$ was just ‘multiply the argument by two’.

What is the inverse function of $g(t) = 2t$?

1.24 The inverse is ‘divide by two’:

$$g^{-1}(t) = \frac{t}{2}$$

If you wrote $t \div 2$, that’s ok too. It’s just more common for us to use the fraction notation than the old-school divide sign $\div$.

So we can see that **division** is the inverse of **multiplication** and vice versa. All functions have inverses, but some have interesting inverses. What about this function, the **reciprocal function**:

$$f(x) = \frac{1}{x}$$

Think about what you get if you put in, say, $x = 2$. Can you find the inverse function?

1.25 $f^{-1}(x) = \frac{1}{x} = f(x)$

So the reciprocal function **is its own inverse**. That’s fun, right? (*he looks around hopefully*)

Sometimes there are restrictions on the allowed values that a function can take. What if I try and put the value $x = -3$ into the function:

$$f(x) = \sqrt{x}$$

What is the result?

1.26 If your answer is ‘you can’t take the square root of a negative number’ – that is fine and all we need to know for this course.

If your answer is ‘it’s a complex number’ or ‘ $\sqrt{3}i$ ’ – that is also fine, and I am glad you’ve done A-level Further Maths! For this course though, we’ll say we can’t take the square roots of negative numbers (or at most, we’ll just say ‘the answer is a complex number’).

Other functions, notably the **logarithm** functions we will see later, also don’t take negative number inputs.

Anyway, there are lots of different kinds of functions that we will see, and that you have may have already seen before – functions like $\sin(x)$, e^x , and $\log(x)$. The important things to remember are:

- Functions take an **input** (argument), do some kind of manipulation, and give an **output** (result)
- All functions have an **inverse function** which undoes the effect of the original function.
- Sometimes there are restrictions on the allowed inputs to a function.

Time for a new question, what is the answer to the following:

$$(3 + 2) \times 6 - 1 = ?$$

1.27 The answer is 29. If you didn’t get that, read on. (If you did get it, still read on please).

The calculation $(3 + 2) \times 6 - 1$ involves **arithmetic operations**, things like $+$, $-$, $\times$, $\div$. The things that were on the cover of Ed Sheeran’s albums, until he ran out of them (I should write to him and suggest using $\int$, and $\otimes$).

There is an **order of priority** for these operations. You maybe learnt this in school as **BIDMAS**:

- **B**rackets

- Indices
- Division
- Multiplication
- Addition
- Subtraction

All over the world there are a load of different acronyms: PEMDAS, BODMAS, BEDMAS, BOMDAS, PEDMAS...

Basically it means you evaluate anything inside brackets first, then do anything raised to a power (like 2^3 – these are called **indices** in old mathematics books, hence the name), then do division/multiplication, and then addition/subtraction.

What is the result of the below:

$$4 \div (2 \times (1 + 1)) = ?$$

1.28 The answer is 1.

We do the part in the innermost brackets first to get $1 + 1 = 2$, then the multiplication to get $2 \times 2 = 4$, and finally the division to get $4 \div 4 = 1$. You might have seen these kinds of questions on clickbait YouTube shorts titled things like *'99% of people fail to get this right!!'*. You're now one of the 1%. Enjoy it.

In practice, if there was an equation written out without any brackets, like:

$$4 + 5 \times 3 = ?$$

It would normally be considered a bit 'untidy', because without BIDMAS there is some ambiguity about whether you do $4 + 5$ first, or 5×3 first. What is the answer, using BIDMAS?

1.29 19, because we do the multiplication first, then the addition.

Usually though, we can avoid this by just adding brackets wherever we might need them. For example, it is much clearer to avoid any ambiguity by writing the equation from the previous frame as:

$$4 + (5 \times 3) = ?$$

Then, all we have to remember is that the brackets get done first. This becomes very important when you start doing calculations on a calculator – I always recommend that you put in an excessive number of brackets to ensure that the calculation gets done as you expect.

Here's a classic example, courtesy of approximately 800 chemistry undergraduates from the past few years. What is -3^2 ?

1.30 With BIDMAS, the answer is -9, because the power (index) gets done before the multiplication by -1 .

However, you actually might *mean* to calculate $(-3)^2$, in which case the answer is 9, not -9 . In words, the difference is:

- -3^2 is 'minus, three squared' (i.e. do 3^2 and then make the result negative).
- $(-3)^2$ is 'minus three, *all* squared (i.e. $(-3) \times (-3)$).

You might think this seems stupidly simple, but people get it wrong all the time when typing numbers into their calculators.

Here's another question: rearrange the equation below to find x . We are looking for a value of x that makes the equation below true.

$$x + 2 = 4$$

1.31

$$x = 2$$

Maybe that was easy, but maybe it wasn't. Let's unpack it.

We had the equation $x + 2 = 4$, so whatever x is, we know that if we add 2 to x , we get 4. To get back to just having x , we apply the inverse function (subtracting 2) **to both sides of the equation**:

$$x + 2 - 2 = 4 - 2$$

$$x + 0 = 2$$

$$x = 2$$

It is important that we always **do the same thing to both sides of the equation**. If we change one side and not the other, then we don't have our original equation anymore. As long as we do the same thing to both sides, nothing can go wrong.

Rearrange the equation below to find y :

$$2y + 4 = 10$$

1.32 $y = 3$.

That was a bit harder, as there were two steps (first subtract the 4, then divide by 2). You can think of this as applying two inverse functions, but that always seems a bit confusing to me. I always think of the process as:

- The equation $2y + 4 = 10$ tells us that there is a specific number y that satisfies that relation. I need to manipulate it so it looks like $y = \dots$
- If I subtract 4 from both sides, I get rid of the 4 to find $2y = 6$.
- If I then divide both sides by 2, I get $y = 3$.

That works for me and my brain. You might find another way that works better for you.

Rearrange the equation below to find t :

$$t^2 = 4$$

1.33 $t = 2$ or $t = -2$.

There are two options here, because $2^2 = 4$ **and** $(-2)^2 = 4$. So the equation $t^2 = 4$ has **two solutions**. This is totally allowed, and in general equations can have any number of solutions. Many have an infinite number of solutions.

To get to this, we either could think of taking the square root of both sides (what I'd do), or another way to think of it could be:

- I want a number t that, when I square it, becomes 4. What numbers satisfy that relationship?
- ...think for a bit...
- Realise that $t = 2$ and $t = -2$ satisfy this relationship.

Equations are called *equations* because they tell you how things *equate* or *relate* to each other. So, with the equation $t^2 = 4$, we are told that whatever t is, t^2 and 4 have to be equal. We then search for the value of t that makes that true.

All of this might seem a bit abstract, but it is more useful in the long run than just learning the mechanics of 'add the same thing to both sides' and so on. In science, we do maths because it **tells us real, physical, tangible things about the world around us**. Understanding what equations are telling us, and what they tell us about the world, is more important than knowing how to play with them (though that is important too).

If I had an equation that told me that the concentration (c) of a solution was related to the number of moles of solute (n) and volume of solvent (V):,

$$c = \frac{n}{V}$$

What happens to c if I (i) increase n ?, (ii) decrease V ?

1.34 In both cases, c increases. If I increase n then c has to increase (numerator gets bigger), and if I decrease V then c also has to increase, as the denominator gets bigger so I am dividing by a smaller number.

Does this make physical sense? Explain why, or why not.

1.35 It does, because we'd expect that more moles of solute leads to a more concentrated solution (increased n), and a smaller volume of solvent (decreased V) has the same effect.

You can hopefully start to see how being able to appreciate the *substance* behind the equation is really what is useful to us.

Anyway, back to learning the mathematical machinery. Here's a slightly harder question. I want to know what the values of x and y are that satisfy the equations:

$$\begin{aligned}x + y &= 1 \\x + 2y &= 2\end{aligned}$$

Now I have two unknown things (x and y), but also two equations to satisfy ($x + y = 1$ and $x + 2y = 2$). The two equations have to be satisfied **simultaneously** by the two values of x and y , so these are called **simultaneous equations**.

You did do these in school, so have a go. Don't worry if you've forgotten - you'll learn more by trying and failing than by just looking at the answer.

1.36 $x = 0$ and $y = 1$ satisfy both equations simultaneously. Plug in the values and confirm it.

To solve these equations, there are a few methods we can use. We will look at two now, and mention a third one later. The two we will look at all revolve around finding a way to manipulate the two equations so that we can eliminate one of the variables (either x or y), and then be left with *one* variable and *one* equation: much simpler.

Firstly, we can rearrange one of the equations and plug it into the other one. Let's label the equations A and B:

$$\begin{aligned}x + y &= 1 & \text{A} \\x + 2y &= 2 & \text{B}\end{aligned}$$

Now rearrange equation A so that x is the subject (making it $x = 1 - y$), and plug it into equation B:

$$(1 - y) + 2y = 2$$

What is the value of y that satisfies this equation?

1.37 $y = 1$.

Now we know this value of y satisfies *both* equations, because we used *both* equations to obtain it.

Now take this value, and plug it back into either equation A or B to find the corresponding value of x . What is x ?

1.38 $x = 0$

For example, putting $y = 1$ into equation A gives: $x + 1 = 1$, and so $x = 0$.

Our values of x and y that *simultaneously* solve our two equations are therefore $x = 0$ and $y = 1$, but it is worth checking that these values do actually solve our original equations (in case we've screwed something up). To do this, just plug the values you've found back into the equations and check that they still work, like this:

$$\begin{aligned} x + y = 1 &\longrightarrow 0 + 1 = 1 \longrightarrow 1 = 1 \longrightarrow \text{Nice...} \\ x + 2y = 2 &\longrightarrow 0 + 2(1) = 2 \longrightarrow 2 = 2 \longrightarrow \text{...nice.} \end{aligned}$$

If you do this, and you end up with something like $2 = 3$, or $-5 = 7$, it's an indication that you've messed something up somewhere.

Solve the two equations below by the same method (substitution). I've labelled the equations C and D just to help you:

$$\begin{aligned} y - x &= 2 & \text{C} \\ y + 2x &= 8 & \text{D} \end{aligned}$$

1.39 You should find that $x = 2$ and $y = 4$.

Another way to solve these equations is to add or subtract the equations in some way that will cancel out one of the variables. Sometimes this will be easier than doing the substitution, other times it won't be – but these are tools to have in your toolbox.

For the example we just did, we could subtract equation C from equation D. In practice, this means we subtract the right-hand side (RHS) of C from the RHS of D, and the same for the LHS.

Do this subtraction now (equation D – equation C), and write down the equation you end up with.

1.40

$$3x = 6$$

This equation obviously rearranges to $x = 2$. Now, we put this value back into either of the original equations to find y .

Do that now, and write down the value of y .

1.41 $y = 4$

So the final values are $x = 2$ and $y = 4$, the same as we got from the substitution method.

As a last thing, solve the two simultaneous equations below to find the variables t and s .

$$\begin{aligned} t + s &= 3 \\ s - t &= 2 \end{aligned}$$

1.42 You should find that $s = 2.5$ and $t = 0.5$.

You could also leave your answer as fractions, $s = \frac{5}{2}$, and $t = \frac{1}{2}$. Unless a question specifically says 'leave your answer as a fraction', I am *entirely not arsed* about the form you leave the answer in (as long as it's right...).

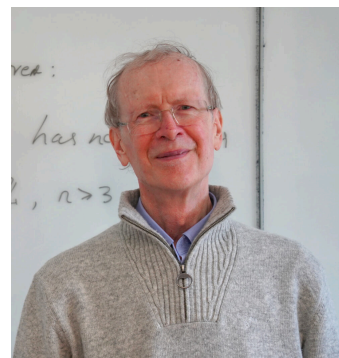
Now you can work through **problem sheet 1**, which you should do and confirm your answers for before continuing onto the next chapter.

Problem Sheet 1 - Numbers and Arithmetic

Numbers and arithmetic are the fundamental building blocks of mathematics. You'd think that by now we've figured out everything there is to figure out with them, but that's not the case. **Number theory** is a very active branch of mathematics research, and a famous unsolved problem in number theory was only relatively recently solved – the proof that there are no three integers a, b, c that can satisfy the expression below if $n > 2$.

$$a^n + b^n = c^n$$

This problem is known as **Fermat's Last Theorem** after it was posed around 400 years ago by French mathematician Pierre de Fermat. It was finally proved in 1994, when I was one year old, by the man pictured on the right, Andrew Wiles. My claim to fame is that I've once sat next to him at a lunch buffet.



You should be able to complete these problems having worked through chapter 1 of the workbook.

1. Without using a calculator, find answers to the following:

i $(7 + 6) \div 2$

iv $\frac{2}{3} + \frac{1}{4}$

ii $7 + (6 \div 2)$

v $\frac{3}{2} \div \frac{3}{7}$

iii $\frac{2}{5} \times \frac{3}{4}$

vi $(\frac{1}{4} + \frac{2}{8}) \times \frac{1}{3}$

2. Simplify the following expressions:

i $\frac{1}{2ab} - \frac{1}{3ab}$

iii $(a + b)(a - b)$

ii $(\frac{1}{a} + \frac{1}{b})(a - b)$

iv $(a + b)^2 - (a - b)^2$

3. Solve the following equations to find the value of x :

i $\frac{1}{x} + \frac{2}{5} = \frac{1}{2}$

iv $x + 5 = \frac{24}{x-5}$

ii $\frac{15}{24} - \frac{17}{60} = \frac{x}{360}$

v $x + 8 = \frac{1296}{x+8}$

iii $\frac{5}{x+2} = \frac{3}{x-4}$

vi $\frac{7y-2xy}{y} = 3$

4. The number n is an integer. Find the largest value of n that satisfies the relationship:

$$20 - n > 4n + 3$$

Note that you can treat the $>$ sign like an equals sign, in terms of manipulating the inequality: always do the same thing to both sides!

5. The ideal gas law relates the pressure (p), volume (V), and temperature (T) of an ideal gas:

$$pV = nRT$$

Where n is the number of moles of gas and $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ is a constant.

- i Find the number of moles of gas that occupy a room of volume 1 m^3 , at atmospheric pressure (101 325 Pa) and room temperature (20°C).

- ii How many molecules of gas are there in a volume of 1 cm^3 under the same conditions?

6. Real gases don't quite follow the ideal gas law, under some conditions. One improvement on the ideal gas law is the *van der Waals equation of state*:

$$\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

Rearrange the van der Waals equation to make p the subject.

7. Solve the following pairs of equations simultaneously:

i $2x + y = 4$ and $x + 2y = 5$

iii $a^2 = 4b$ and $a = b$

ii $7x + 2y = 11$ and $4x + y = 7$

iv $y - 2x = 7$ and $3y - 6x = 12$

8. Neon exists as two naturally occurring isotopes, ^{20}Ne and ^{22}Ne . The atomic masses of the isotopes are $19.992 \text{ g mol}^{-1}$ for ^{20}Ne and $21.991 \text{ g mol}^{-1}$ for ^{22}Ne .

Find the percentage abundances of the two isotopes, given that the average atomic mass of a sample of neon gas is $20.182 \text{ g mol}^{-1}$.

9. We can solve more than two equations simultaneously if we want to (I can feel your excitement). Solve the following three equations simultaneously to find the values of x , y , and z .

$$x + y + z = 4$$

$$2x + 3y - z = -2$$

$$x - y + 3z = 12$$

Challenge Problems

1. *Egyptian Fractions* are fractions used by the Ancient Egyptians, which all had 1 as the numerator. Express the sum of Egyptian fractions below as a single fraction, where the integers n , p , and q each appear only once.

$$\frac{1}{p(p+q)/n} + \frac{1}{q(p+q)/n}$$

2. Prove Pythagoras' Theorem.

3. Solve the simultaneous equations below:

$$y = x^2 + x - 2$$

$$y = x^3 + x - 2$$

4. A *centered square number* is a number that gives the number of dots in a square with a dot in the center and all other dots surrounding the center dot in successive square layers. The first three centered square numbers are 1, 5, and 13.

Derive a formula for the first n centered square numbers. You may find it helpful to draw a sketch.

Chapter 2

Powers and Polynomials

Now we will build on the fundamental bits of algebra we've just learned with two new things. We will learn how to manipulate **powers** (things like the 4 in x^4), and then how to solve more complex equations involving them (like $x^2 - x = 6$). These equations are called **polynomials**.

2.1 Powers are things like this:

$$x^2$$

It's useful to be able to manipulate these. How can you simplify the below?

$$x^2 + x^2 = ?$$

2.2

$$x^2 + x^2 = 2x^2$$

That was simple enough, hopefully. We had two lots of ' x^2 ' added together, so we just stick a 2 in front of the thing. This '2' is called the **coefficient** of the x^2 .

What about the below? Note that central dot ("·") means the same as the multiply symbol ("×"), but looks less like the algebraic x so is a bit clearer. We will use both, and sometimes omit any multiplication symbol where that doesn't make things unclear.

$$x^2 \cdot x^2 = ?$$

2.3

$$x^2 \cdot x^2 = x^4$$

The general rule is that if we multiply two things raised to powers, then the powers add:

$$x^n \cdot x^m = x^{m+n}$$

Simplify the expression below:

$$x^2 \cdot 3x^3 \cdot x^5$$

2.4

$$x^2 \cdot 3x^3 \cdot x^5 = 3x^{10}$$

Hopefully that makes sense. One important thing to note is that for this to work, the ' x ' has to be the same on all the parts we are multiplying. If I tried to simplify:

$$x^2 \cdot y^3$$

All we can do is write it more compactly as x^2y^3 . This is **not** the same as $(xy)^5$.

Simplify the below:

$$t^3(1 - t^2)$$

2.5

$$t^3(1 - t^2) = t^3 - t^5$$

Which we get by expanding the bracket into $1 \times t^3$ and $-t^2 \times t^3$.

What if I divided two powers, can you simplify the below?

$$\frac{x^2}{x} = ?$$

2.6

$$\frac{x^2}{x} = x$$

Remember that $x = x^1$. You can see that the powers subtract in division (the opposite of multiplication).

How about this one?

$$\frac{y^5}{y^{2.5}} = ?$$

2.7

$$\frac{y^5}{y^{2.5}} = y^{2.5}$$

Again, there is nothing sacred about using x , and also no reason that our powers need to be integers. We just subtract them ($5 - 2.5 = 2.5$).

How about the below?

$$\frac{y^6}{y^7}$$

2.8

$$\frac{y^6}{y^7} = y^{-1}$$

Negative powers are fine too, and actually $y^{-1} = \frac{1}{y}$. This is an example of another useful result we can use when manipulating powers:

$$x^{-n} = \frac{1}{x^n}$$

Using that rule, how can I rewrite the fraction below?

$$\frac{1}{x^8}$$

2.9

$$x^{-8}$$

This leads us to another useful rule: **dividing by a positive power is the same as multiplying by the corresponding negative power, and vice versa.**

Use this rule to simplify the below:

$$2^{-2} \times \frac{2^5}{2^3}$$

2.10

$$2^{-2} \times \frac{2^5}{2^3} = 2^{-2} \times 2^{-3} \times 2^5 = 2^{-5} \times 2^5 = 2^0 = 1$$

That's written out in a very long way, but hopefully you see how we got there. Remember that *anything* raised to the power of zero is equal to one.

Simplify the below:

$$x(x^{-\frac{1}{2}})$$

2.11

$$x(x^{-\frac{1}{2}}) = x^{\frac{1}{2}}$$

These fractional powers have an interesting meaning. It turns out that:

$$x^{\frac{m}{n}} = \sqrt[n]{x^m}$$

i.e. the n on the bottom of the fraction means we take the n -th root of everything else.

Calculate the below:

$$27^{\frac{1}{3}} = ?$$

2.12

$$27^{\frac{1}{3}} = 3$$

Because $27^{\frac{1}{3}} = \sqrt[3]{27} = 3$.

How about:

$$49^{0.5} = ?$$

2.13

$$49^{0.5} = 7$$

Because $49^{0.5} = \sqrt{49}$. We can write the 'half' as either a fraction or a decimal.

Another thing we might want to do is below. Do you know how to simplify this? Don't worry if not.

$$(x^2)^3 = ?$$

2.14

$$(x^2)^3 = x^6$$

So when we have a power **raised to another power**, we multiply the two powers:

$$(x^n)^m = x^{n \cdot m} = x^{nm}$$

How could you simplify:

$$(y^3)^{12}$$

2.15

$$(y^3)^{12} = y^{36}$$

Let's summarise the rules of powers:

- $x^n \cdot x^m = x^{n+m}$
- $x^{-n} = \frac{1}{x^n}$

- $(x^n)^m = x^{nm}$
- $x^{\frac{m}{n}} = \sqrt[n]{x^m}$
- $x^0 = 1$ for any value of x .

Other rules (like the one about division) can be derived from these.

You can chain these together to simplify more complex expressions. Simplify the below:

$$\left(\frac{x^2}{x^{-2}}\right)^{\frac{1}{2}}$$

2.16

$$\left(\frac{x^2}{x^{-2}}\right)^{\frac{1}{2}} = x^2$$

Take it step by step. The bit in brackets simplifies to $\frac{x^2}{x^{-2}} = x^4$. Then we take multiply the 4 by the $\frac{1}{2}$ to end up with 2.

We could even have powers of different variables - we just have to remember to simplify them each separately. Simplify the below:

$$\frac{6x^2y}{3xy^2}$$

2.17

$$\frac{6x^2y}{3xy^2} = 2xy^{-1}$$

Or you could write it as $\frac{2x}{y}$, either is fine. Just remember to do each variable (here x and y) separately.

A final example before we move on. Simplify the below:

$$(p^6q^4)^{0.5} \times (pq)^{-3}$$

2.18

$$(p^6q^4)^{0.5} \times (pq)^{-3} = q^{-1}$$

If you didn't get that, think carefully about each step, and use the rules in the list we bulleted above.

Now to use these powers for something interesting. The energy of interaction, $V(r)$, between two molecules separated by a distance r can be modelled as:

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^6 \right]$$

Where ϵ and σ are constants. The minimum energy $V_{\min}$ is found when $r = 2^{1/6}\sigma$, i.e $V_{\min} = V(2^{1/6}\sigma)$.

Derive an expression for the energy $V_{\min}$. Be methodical and write out each step in full.

2.19

$$V_{\min} = -\epsilon$$

If you found that hard, don't worry. Let's go through it step by step. First, we plug in the value of r given in the question ($r = 2^{1/6}\sigma$).

$$V_{\min} = 4\epsilon \left[\left(\frac{\sigma}{2^{1/6}\sigma}\right)^{12} - \left(\frac{\sigma}{2^{1/6}\sigma}\right)^6 \right]$$

From here, you can see that it's possible to cancel out the σ from each term:

$$V_{\min} = 4\epsilon \left[\left(\frac{1}{2^{1/6}} \right)^{12} - \left(\frac{1}{2^{1/6}} \right)^6 \right]$$

Then we can use one of our rules of powers (which one?) to rewrite the fractions:

$$V_{\min} = 4\epsilon \left[\left(2^{-1/6} \right)^{12} - \left(2^{-1/6} \right)^6 \right]$$

Then we use another rule of powers to simplify:

$$V_{\min} = 4\epsilon \left[\left(2^{-12/6} \right) - \left(2^{-6/6} \right) \right]$$

Which we can simplify even further to:

$$V_{\min} = 4\epsilon \left[\left(2^{-2} \right) - \left(2^{-1} \right) \right]$$

From here, I think you can get to the final answer. Come and find me if you get stuck.

The equation we just worked with is called the **Lennard-Jones Potential**. It's very useful for modelling the behaviour of intermolecular interactions, and is a nice model system for us learning maths for chemistry. We will see it a lot.

Now for a new topic. Solve the equation below:

$$x^2 = 4$$

2.20

$$x = \pm 2$$

We did this before, actually. Remember there are two solutions (+2 and -2).

That equation was an example of a **quadratic** equation. Quadratic equations are part of a broader class of equations known as **polynomial equations**. A polynomial equation is one that only contains terms that are powers of x (or t , or y , or whatever the variable is).

Which of the two equations below are polynomial equations?

$$y = x^3 - x^2 + 2x + 1$$

$$f = \sin(2t) + t^2$$

2.21 The first one, because the other one contains a function that is **not** simply a power of x (or t , or whatever. I'm labouring the point but *the specific symbols do not matter!*).

In general a polynomial equation, or polynomial function, in one variable has the form:

$$f(x) = c_0 + c_1x + c_2x^2 + c_3x^3 + \dots + c_nx^n = \sum_{i=0}^n c_i x^i$$

The $\sum_{i=0}^n$ symbol is just a shorthand for 'add up all the terms inside it, starting at $i = 0$ and ending at $i = n$ '. The c_i are **coefficients**, constant factors that don't depend on x .

This polynomial has x as the variable, so we say it is a *polynomial in x* . We could equally well have a polynomial in t , or s , or θ , or a mixture of variables. We will use x for now.

When we talk about polynomial equations we often talk about the **order** of the polynomial. The order of a polynomial is equal to the *highest power of the variable (here x)*.

What is the order of the polynomial below?

$$f(x) = x + 2$$

2.22 The order is 1, a **first-order** polynomial, because the highest power on the x is 1. We also call this a **linear** equation, because the value of y depends *linearly* on x : if I change x by some amount, y changes by the same amount. This means that a graph of $y = x + 2$ would look like a straight line, and *linear* is another word for straight.

What is the order of the polynomial below?

$$g(t) = t^2 + 3t$$

2.23 The order is 2, a **second-order** polynomial. We also call this a **quadratic** equation, which is *nonlinear*. The graph of a quadratic equation looks like a parabola – we will see this again later.

What is the order of the polynomial below?

$$y(s) = s^3 + 2s - 1$$

2.24 The order is 3, a **third-order** polynomial. We also call this a **cubic** equation.

We can carry on, with **quartic**, and **quintic** equations, and more. But there's a point to all of this. Look at the linear equation below – how many solutions are there to that equation, and what are they?

$$x + 2 = 3$$

2.25 There is one solution, $x = 1$.

What about the quadratic equation below?

$$x^2 = 1$$

2.26 There are two solutions, $x = \pm 1$.

So in general, a **linear** equation has one solution, and a **quadratic** equation can have two.

How many solutions do you expect a cubic equation could have?

2.27 Three. In general if the highest power in a polynomial is n , then there could be n possible solutions to an equation involving that polynomial.

How many solutions are there to the equation below, and what are they?

$$x^2 = 0$$

2.28 There is one solution, $x = 0$, even though the equation is quadratic. The point is that there could be two solutions, but there doesn't have to be. Strictly here we say that $x = 0$ is a solution twice – we call this a **repeated root** for reasons we will see later.

Anyway, this is all fun, but the main thing we need to do here is learn some techniques for solving polynomial equations. You did these in school, so solve the equation below:

$$x^2 - x - 2 = 0$$

2.29

$$x = -1 \text{ or } x = 2$$

You can get to this result in a number of ways. The first way we will see is called **factorisation**. A *quadratic* equation can be written as a product of two *linear* factors. In this case:

$$x^2 - x - 2 = (x + 1)(x - 2) = 0$$

For a quadratic equation where the x^2 has no other coefficient in front of it, these factors are always going to look like $(x + p)(x + q)$, we just need to find p and q . The trick to doing this is that you need pq to give you the constant term in the equation (here -2), and you need $p + q$ to give you the coefficient of the linear term (here -1).

You really just need to practice and get used to finding these factors so you develop a sense for it, rather than learning a method. So factorise the equation below:

$$x^2 - x - 6 = 0$$

2.30

$$x^2 - x - 6 = (x - 3)(x + 2) = 0$$

One thing we neglected to mention is how to go from these factors to the actual solutions to the equation? Can you write down the solutions to the equation just from $(x - 3)(x + 2) = 0$?

2.31 The solutions are $x = 3$ and $x = -2$.

We get this by noting that the *product* of our factors is equal to zero, which means either one of our factors must be equal to zero. So we have either:

$$(x - 3) = 0 \longrightarrow x = 3$$

Or:

$$(x + 2) = 0 \longrightarrow x = -2$$

These are our two solutions.

Solve the equation below by factorisation:

$$x^2 + 5x - 6 = 0$$

2.32 The equation factorises to $(x + 6)(x - 1) = 0$, so our solutions are $x = -6$ and $x = 1$.

Every quadratic can be factorised, but sometimes the factors are not easy to find. Can you solve the equation below?

$$x^2 + \frac{\sqrt{3}}{4}x - \frac{3}{8} = 0$$

2.33 This does factorise, to $(x + \sqrt{3}/2)(x - \sqrt{3}/4)$. But that was harder to find than the previous cases.

Helpfully, there is actually a general formula for solving quadratic equations. We can write a generic quadratic equation as:

$$ax^2 + bx + c = 0$$

Where a, b, c are the coefficients in front of the terms.

For the equation below, what are a, b , and c ?

$$x^2 + \frac{\sqrt{3}}{4}x - \frac{3}{8} = 0$$

2.34 $a = 1$, $b = \sqrt{3}/4$, $c = -3/8$.

If we take the general formula for a quadratic equation from earlier, we can rearrange it (see problems) to:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

This is then a formula for the values of x that solve an equation with a given a, b , and c . We get two solutions because of the $\pm$ sign that means '*plus or minus*'.

Use the quadratic formula to solve the equation from the previous frame.

2.35

$$x = -\frac{\sqrt{3}}{2} \text{ or } x = \frac{\sqrt{3}}{4}$$

Let's write that out in detail. We know that $a = 1$, $b = -\sqrt{3}/4$, and $c = -3/8$, so put those numbers into the formula:

$$x = \frac{\frac{\sqrt{3}}{4} \pm \sqrt{\frac{3}{16} + \frac{12}{8}}}{2}$$

I skipped a bit of the rearranging, which is hopefully fine. This leads us to:

$$x = \frac{\frac{\sqrt{3}}{4} \pm \sqrt{\frac{27}{16}}}{2}$$

Becoming:

$$x = \frac{\frac{\sqrt{3}}{4} \pm \frac{3\sqrt{3}}{4}}{2}$$

Which ends up as:

$$x = \frac{\sqrt{3}}{8} \pm \frac{3\sqrt{3}}{8}$$

Which simplifies to the result given (you can show that bit yourself).

Sometimes using this formula will be quicker than factorising. Solve the equation below:

$$2x^2 + 8x - 42 = 0$$

2.36 $x = -3$ or $x = 7$.

You could do this either by factorising or using the formula. Unlike at school, we don't care *how* you solve a problem as long as you solve it correctly. **There is always more than one way to do things.**

Sometimes you'll encounter equations that are written in a way that makes them look more complicated than they are. In this case, we can often manipulate them into a form that is easier to work with. Look at the equation below. Can you find a way to turn it into a 'normal' looking quadratic?

$$2x + 3 = \frac{2}{x}$$

2.37 Multiply both sides by x , and rearrange so it equals zero, to get:

$$2x^2 + 3x - 2 = 0$$

So it is just a quadratic equation, in disguise. Can you solve it?

2.38

$$x = -2 \text{ or } x = \frac{1}{2}$$

Here's another example, which is a bit harder. Can you make a *substitution* that will turn the equation below into a 'normal' looking quadratic?

$$4^x - 2^x - 2 = 0$$

2.39 The substitution $y = 2^x$ will do the job for us. We could write the original equation as:

$$(2^x)^2 - (2^x) - 2 = 0$$

And so the substitution gives:

$$y^2 - y - 2 = 0$$

Can you solve the equation now, for y , and then for x ?

2.40 Solving for y gives $y = 2$ or $y = -1$.

The first of these gives us a plausible (real) solution for x :

$$y = 2 \longrightarrow 2^x = 2 \longrightarrow x = 1$$

The other value of y would give us a solution that is not a real number, which leads us neatly on to our final note about quadratic equations.

Sometimes you might find an equation where it seems impossible to factorise, or the quadratic formula seems to not work, because the $\sqrt{b^2 - 4ac}$ term gives you a negative square root. In this case, the equation has a **complex solution** (in the sense of *complex numbers*, not that the solution is necessarily very *complicated*). We won't use that in this course, but it is worth knowing. The solution to the equation $2^x = -1$ from above would also be a complex number.

Can you solve the equation below? Try factorising out an x first, if you are stuck.

$$x^3 + 2x^2 - 3x = 0$$

2.41 $x = 0$, or $x = 1$, or $x = -3$.

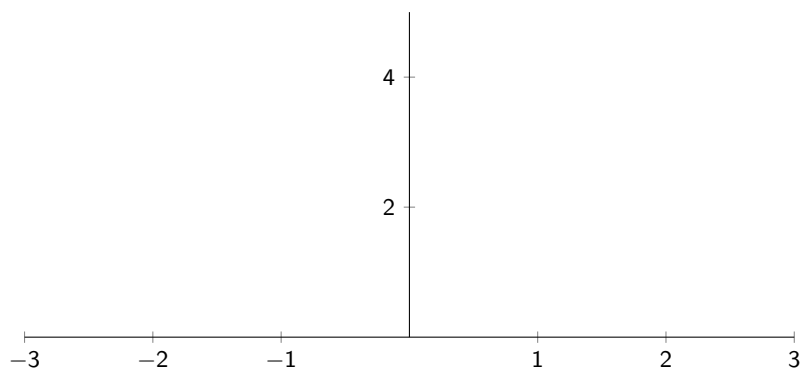
Now we have *three* solutions, because our equation was a *cubic* equation (the highest power of x was three. We get there by factorisation:

$$x^3 + 2x^2 - 3x = x(x^2 + 2x - 3) = x(x + 3)(x - 1) = 0$$

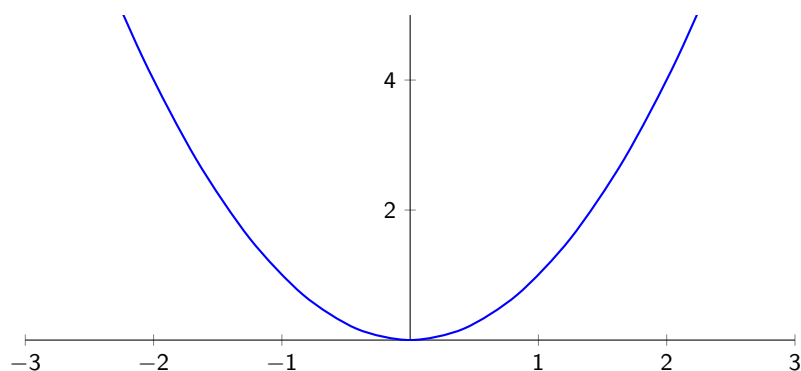
For higher order equations, factorisation is often the easiest way to solve them. There actually is a *cubic formula*, like the quadratic one – but it's very cumbersome to use (look it up and see...).

That's enough algebra for now. One last thing - we talked earlier about linear functions giving you a straight line graph, and quadratics giving you a parabola. This graphical interpretation is actually useful.

On the axes below, can you sketch the function $y = x^2$?



2.42



This is the **parabola** we talked about.

What are the solutions (or the *solution*) to the equation $x^2 = 0$?

2.43 The solution is $x = 0$.

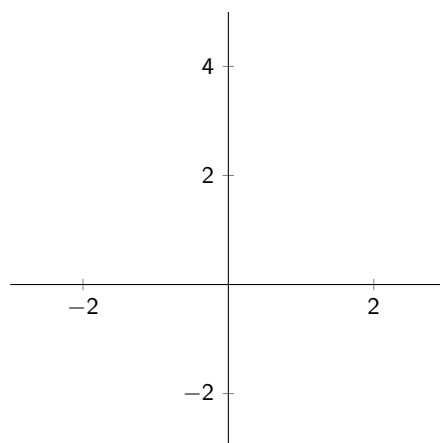
Can you identify where on the plot you just drew (of $y = x^2$) that that solution is?

2.44 It is where the curve crosses (or, in this case, touches) the x -axis.

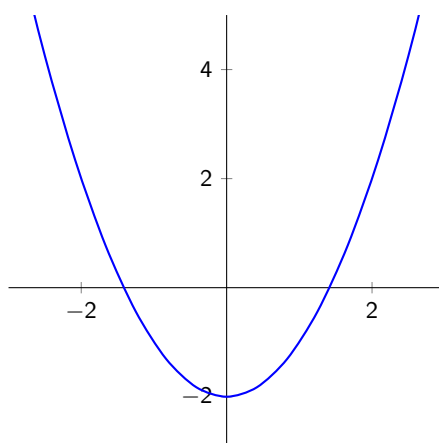
In general, if we have an equation $y = f(x)$, then solving the equation $f(x) = 0$ is equivalent to finding where the points on a graph of $f(x)$ cross, or touch, the x -axis (as this is where $y = f(x) = 0$).

The points where the graph of a polynomial cross the x axis are called the **roots** of that polynomial. These roots are what we have been finding by solving the polynomial equations algebraically.

Can you sketch, on the axes below, the graph of the function $y = x^2 - 2$?

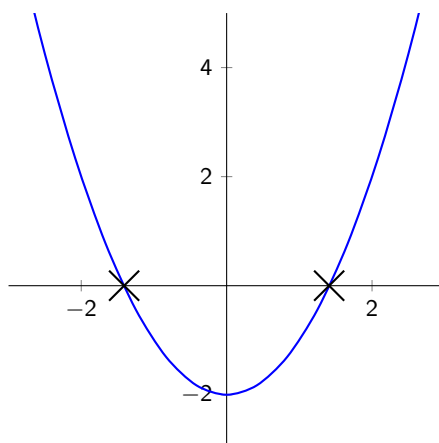


2.45



Where are the roots of the function $f(x) = x^2 - 2$? Annotate them on your plot.

2.46



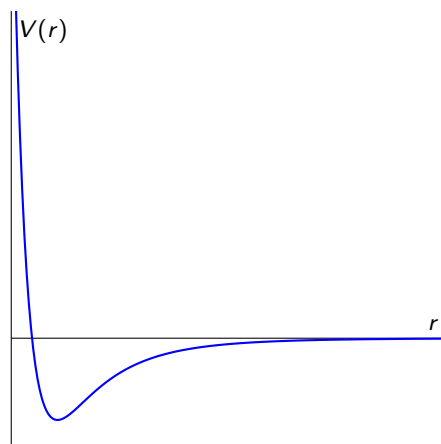
And what are the roots? What are the solutions to the equation:

$$x^2 - 2 = 0$$

2.47 $x = \pm\sqrt{2}$.

To emphasise, **roots** of a polynomial equation are what you get by setting the equation to zero and solving it. On a graph of the polynomial, they correspond to the points that cross the x-axis.

To finish off this chapter, below is a sketched plot of the **Lennard-Jones** potential we saw earlier.



The equation for the Lennard-Jones potential is:

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

Find the root of this equation (i.e. the value of r when $V(r) = 0$).

2.48 The root occurs at $r = \sigma$. You can get this by setting the whole equation equal to zero, and solving it for r . I will leave you to think about the details.

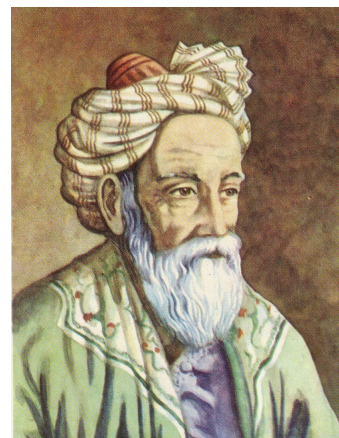
That's it for now. We will come back to polynomials and graphs again. Next time we will change the pace slightly and talk about doing calculations with **units**. Numbers in science usually mean something physical, and units are how we express that. It'll be fun.

Problem Sheet 2 - Powers and Polynomials

Humans have been studying polynomials for millenia. The first recorded example of this dates back to 2000 BC, with the ancient Babylonians, and it has been going on ever since.

An ancient mathematician who worked on them was **Omar Khayyam**, born in around 1050 AD in Persia (modern Iran). He developed the first ways of solving cubic equations, among many other things. He was also a renowned poet, especially for his four-line 'quatrains', of which one of my favourites is below. I think it's probably more profound in classical Persian.

Dreaming when Dawn's Left Hand was in the Sky
I heard a Voice within the Tavern cry
'Awake, my Little ones, and fill the Cup
before Life's Liquor in it's Cup be dry'.



1. Simplify the following:

i $\frac{1}{9^{-2}}$

ii $\frac{8z^2x^3}{2xz^{-2}}$

iii $(100)^{1/3} - \sqrt[3]{10^2}$

iv $\left(\frac{169}{49}\right)^{1/2}$

v $(\theta^5\phi^2)^2 \times (\theta\phi^2)^{-2}$

vi $\left(\frac{1}{2^{-3}}\right)^2 + 2^6$

2. Solve the following equations to find the value of x :

i $4 = \frac{x}{\sqrt{x}}$

ii $\frac{x^2+5}{x^2} = 6$

iii $\left(\frac{8}{x-2}\right)^2 = 256$

iv $\frac{x^{-1}}{x^5} = \frac{1}{64}$

v $10^x \times 10^4 = 10^7$

vi $2^x = 1024$

3. Kohlrausch's Law (below) relates the conductivity Λ of a solution to the concentration, c , for certain systems:

$$\Lambda = \Lambda_0 - K\sqrt{c}$$

Where Λ_0 and K are constants.

i Rearrange this equation to make c the subject.

ii What is the value of c , in the case that $\Lambda_0 - \Lambda = K$?

4. Factorise and solve the following quadratic equations:

i $x^2 - 5x = 0$

ii $x^2 = 36$

iii $x^2 - 11x = -30$

iv $2x^2 - 13x + 21 = 0$

Note for question (iv), the factorisation will be of the form $(2x + q)(x + p)$. Think about how the extra 2 changes things.

5. Solve the following quadratic equations by any method:

i $x^2 + 9x + 14 = 0$

ii $x^2 - 8x + 15 = 0$

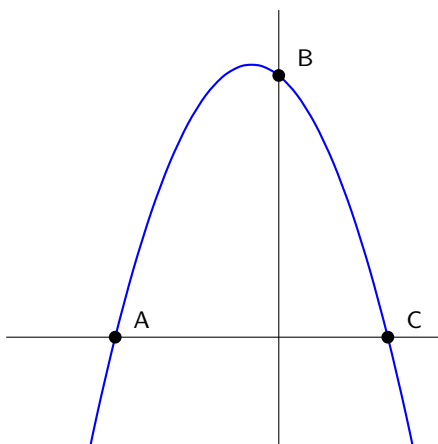
iii $2x^2 - 4.4x - 2.7 = 0$

iv $x^2 - 10x + 25 = 0$

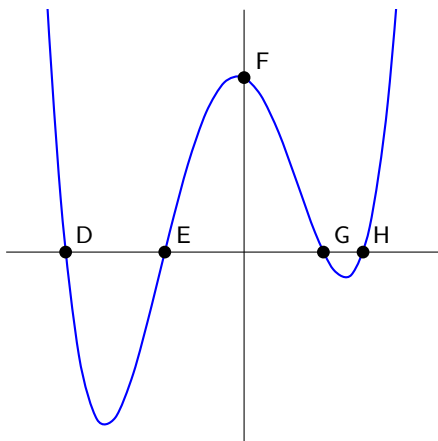
v $x - 6 + \frac{8}{x} = 0$

vi $\frac{x^2+1}{x^2-3} = 5$

6. The graph below shows the function $y = -x^2 - 2x + 24$. Find the coordinates of the points labelled A, B, and C.



7. The graph below shows the function $y = (x^2 - 4)(2x + 9)(x - 3)$. Find the coordinates of the points labelled D, E, F, G, and H.



8. Factorise and solve the following cubic equation:

$$x^3 - 11x^2 + 35x - 25 = 0$$

9. The **reduced mass**, μ , of a diatomic molecule is a quantity that is useful in spectroscopy and quantum chemistry. The reduced mass is given by:

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

Where m_1 and m_2 are the masses of the two atoms in the molecule. Derive an expression for m_1 , in terms of μ and m_2 .

10. In pharmaceutical chemistry, the **Hill Equation** gives the relationship between the fraction θ of drug bound to a protein and the concentration of drug $[D]$:

$$\theta = \frac{[D]^n}{K_d + [D]^n}$$

K_d is a dissociation constant and n is the *Hill coefficient*.

- i What form does the Hill equation take, in the case that $n = 0$?
- ii Derive an expression for $[D]$ in the case that $n = 1$.
- iii What are the possible values of $[D]$ in the case that $n = 2$? Are both of these values physically reasonable?

Challenge Problems

1. Derive the quadratic formula, starting from the general expression for a quadratic equation:

$$ax^2 + bx + c = 0$$

2. The total resistance, R , of n electrical resistors connected in parallel is given by:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

Derive expressions, expressed as a single fraction, for the resistance of (i) two resistors R_1 , and R_2 , and (ii) three resistors, R_1 , R_2 , and R_3 , connected in parallel.

3. The *Chebyshev Polynomials*, $T_n(x)$ are defined by a recurrence relation:

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$$

Where $T_0(x) = 1$ and $T_1(x) = x$.

Write down the first 5 Chebyshev polynomials. Show that:

$$T_n(-x) = (-1)^n T_n(x)$$

Chapter 3

Units and Dimensions

Now we have covered some basics, let's start to think about the application to science. Natural sciences are usually **quantitative**, which means that we try to use numbers to describe, measure, and explain things. Chemistry and pharmacy are especially quantitative – you need to understand how to manipulate units if you want to safely work out drug dosages, for example.

To give the numbers we calculate physical meaning, we give them **units**. Units describe the physical **dimensions** of what the number means, and give it some real meaning.

For example, the number 10 on its own is not that interesting to us. But knowing that something is 10 m long, or weighs 10 kg is more interesting, and gives the number a real meaning in the physical world around us. It is generally the case that numbers in science need to be quoted with some kind of unit, to give the number physical meaning. Sometimes we have *dimensionless* numbers, but we will talk about them later.

3.1 The international system of units (SI) recognises seven **SI base units** for the quantities described below. You should have met most of them before – can you fill in the blanks in the table below?

I've given you the last one, because as far as I know (and I am someone who spends a lot of time thinking about the light intensity of various things in research) no one ever uses it.

Length	Mass	Time	Temperature	Amount of Stuff	Electric Current	Light Intensity
						Candela

3.2 The table should be:

Length	Mass	Time	Temperature	Amount of Stuff	Electric Current	Light Intensity
Metre	Kilogram	Second	Kelvin	Mole	Amp	Candela

Hopefully you knew all this. The only slightly weird thing is that the *kilogram*, not the gram, is the SI unit of mass. Since 2019, the base units are all defined in terms of fundamental physical constants – which seems oddly recent for something so fundamental, but the real physical values of units didn't change much.

The point of these *base units* is that all the other units (Joules, Newtons, Pascals, etc...) can be derived from them, and we'll see that later. For now let's learn how to work with and manipulate units.

As a simple example, can you write down the relationship between mass, moles, and molar mass? You may have called molar mass something like *formula mass* at A level.

3.3

$$\text{mass} = \text{moles} \times \text{molar mass}$$

Can you now write down the units of those three quantities?

3.4 Mass is in kg, moles is in mol, and molar mass is in kg mol^{-1} .

If you said that mass was in g and molar mass was in g mol^{-1} , that's OK too. We usually use grams in chemistry because we usually don't work with massive amounts of substance, but to be *pedantic absolutely consistent with the SI base units* we would use kg.

Write down the equation again, but using the units of the quantities. Is there anything you notice about it?

3.5

$$\text{kg} = \text{mol} \times \text{kg mol}^{-1}$$

Notice that here the mol unit cancels out (we can treat it like any other algebraic quantity), and you're left with an equation:

$$\text{kg} = \text{kg}$$

Which is obviously correct (in words, it says: '1 kg is the same thing as 1 kg'). That this works like this means that our original equation (mass = mole $\times$ molar mass) is **dimensionally consistent**. Something being **dimensionally consistent** means that the units, or dimensions, of the quantities involved make physical sense.

To illustrate this, let me give you an equation which is **not** dimensionally consistent. Imagine we failed A-level chemistry and thought the relationship between mass, moles, and molar mass was:

$$\text{mass} = \frac{\text{moles}}{\text{molar mass}}$$

Can you show that this equation is **not** dimensionally consistent? Put in the units for each quantity and see what you end up with.

3.6

$$\text{mass} = \frac{\text{moles}}{\text{molar mass}} \rightarrow \text{kg} = \frac{\text{mol}}{\text{kg mol}^{-1}} \rightarrow \text{kg} = \text{mol}^2 \text{kg}^{-1}$$

Clearly, the end result here (that 1 kg is the same thing as $1 \text{ mol}^2 \text{kg}^{-1}$) is nonsensical, and so our original equation must have been wrong.

What we've done here is called **dimensional analysis**, and is one of the first things I recommend you do when you meet a new equation, or are trying to remember one for a calculation. If your equation ends up dimensionally inconsistent, then it's probably wrong. Dimensional analysis can also help you to understand what equations are physically telling you, and we'll see that as we go on.

From A level chemistry, do you recall the equation below? Can you write down the meanings of each quantity in it?

$$Q = mc\Delta T$$

3.7 Q is the energy (heat) added to a system, m is the mass of the system, c is the *specific heat capacity* of the system, and ΔT is the *change in temperature* of the system. The symbol Δ is commonly used to denote a *change* in something.

So, what does the equation $Q = mc\Delta T$ tell us? Write down the units of each quantity.

3.8 Q has units of energy, which are **Joules** (J), m is a mass (so in kg), ΔT is a temperature (so in K). What about c ?

You might just remember that the units of c are $\text{J kg}^{-1} \text{K}^{-1}$. But relying on memory is a dangerous game. I once misremembered the date of my wedding anniversary and it got me into a world of trouble. Instead, rely on calculation: could we work out the units of c , from the other units and the equation?

Try rearranging the equation to make c the subject, and work out the units of c .

3.9

$$c = \frac{Q}{m\Delta T} \longrightarrow \text{units of } c = \frac{\text{J}}{\text{kg} \times \text{K}} = \text{J kg}^{-1} \text{K}^{-1}$$

So, we see that for the original equation to be dimensionally consistent, the units of c **have** to be $\text{J kg}^{-1} \text{K}^{-1}$.

Put those units into our original equation now and check that it all makes sense.

3.10

$$Q = mc\Delta T \longrightarrow \text{J} = \text{kg} \times \text{J kg}^{-1} \text{K}^{-1} \times \text{K} \longrightarrow \text{J} = \text{J}$$

So it works - our equation is dimensionally consistent and therefore we've probably remembered it correctly. We will do this kind of analysis a lot. Especially in physical chemistry, I will often be asking you to tell me the units of things - it is a powerful tool for understanding the link between mathematics and the world around us.

Now is a good time to talk about the units of energy, the **Joule**. This seems like a really fundamental unit, and it is, but it wasn't one of our base units. The Joule is actually an **SI derived unit**. This means we can express it as, or derive it from, a combination of the base units. Let's see how this works with some dimensional analysis.

If you did physics in school (don't worry if you didn't), you might remember Newton's second law:

$$F = ma$$

Do you know what the quantities F , m , a , and their units, are? If you've never heard of it, just read on and don't worry.

3.11 F is force, in units of *Newton*s (N). m is mass (kg), and a is *acceleration*, which has units of m s^{-2} ('metres per second squared' or 'metres per second, per second').

Notice that m and a are both defined in terms of SI base units - kg, m, and s. Can you use the equation of Newton's second law to write down the definition of a Newton in terms of SI base units, i.e:

$$1 \text{ N} = 1(\text{what combination of SI base units?})$$

3.12

$$1 \text{ N} = 1 \text{ kg m s}^{-2}$$

So the unit of force, the Newton, is the same as the units kg m s^{-2} . This might not seem obviously physically meaningful, but you can think of it as the force needed to accelerate a 1 kg object to a speed of 1 m s^{-1} in 1 s.

Another equation you might remember from school is the relationship between *work*, *force*, and *distance*:

$$\text{Work} = \text{Force} \times \text{Distance}$$

Work is just a type of energy, with units of Joules, so can you put units into the equation above to work out how write Joules in SI base units?

3.13

$$1 \text{ J} = 1 \text{ N} \times 1 \text{ m} \longrightarrow 1 \text{ J} = 1 \text{ kg m s}^{-2} \times 1 \text{ m} \longrightarrow 1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$$

So $1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$. That's a bit of a mouthful, hence the shortening to 'Joule'. You can think about how those units physically represent energy. Meditate on it.

Knowing the units of a Joule in SI base units, or knowing how to derive it, is going to be useful. But I always say '*you'll be glad you know this later!*' and I feel like at this point there might be a limit to the faith you're willing to invest in me. So here's a direct example of a place where it will be useful in chemistry. In chemical

kinetics, it's often useful to know how fast molecules are moving. The average speed, c , of a molecule with mass m in a gas at temperature T is given by:

$$c = \sqrt{\frac{3k_B T}{m}}$$

Where k_B is the Boltzmann constant, which has a value of $1.38 \times 10^{-23} \text{ J K}^{-1}$. Plug in units for all the quantities in that equation, and show that the units of c are units of speed.

3.14

$$c = \sqrt{\frac{3k_B T}{m}} \rightarrow \sqrt{\frac{\text{J K}^{-1} \text{ K}}{\text{kg}}} \rightarrow \sqrt{\frac{\text{J}}{\text{kg}}} \rightarrow \sqrt{\frac{\text{kg m}^2 \text{ s}^{-2}}{\text{kg}}} \rightarrow \sqrt{\text{m}^2 \text{ s}^{-2}} \rightarrow \text{m s}^{-1}$$

So c has units of m s^{-1} , which is a unit of speed (distance over time). So our equation makes dimensional sense.

Note that we didn't have to include the units of the number 3 in our equation above. That is because the number 3 is **dimensionless**. There are no physical units associated with the concept of 'three', and so in our analysis we can just consider that it has units of '1' (a pure number). The same thing is true of other numbers, and of fundamental constants like π and e .

Other quantities are dimensionless because their units cancel out. You probably remember the **Arrhenius equation** from A level:

$$k = A \exp\left(-\frac{E_a}{RT}\right)$$

Where k is a rate constant, A is a constant, E_a is the activation energy, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, and T is temperature. If you don't know what the $\exp()$ bit means, don't worry – that's coming soon. The thing we need to know now is that *whatever is inside the $\exp()$ bit has to be dimensionless*. We will talk about why later on, but it means that the quantity:

$$-\frac{E_a}{RT}$$

Has to be dimensionless. What are the units of E_a that make this quantity dimensionless? Think carefully.

3.15 E_a must have units of J mol^{-1} .

We know that R has units of $\text{J K}^{-1} \text{ mol}^{-1}$, and that T has units of K . So RT has units of J mol^{-1} . For the ratio E_a/RT to be dimensionless, the units of E_a must be exactly the same as the units of RT , so that they cancel out:

$$-\frac{E_a}{RT} \rightarrow \frac{\text{J mol}^{-1}}{\text{J K}^{-1} \text{ mol}^{-1} \text{ K}} = 1 \rightarrow \text{dimensionless!}$$

These energy units, J mol^{-1} , are **molar energy units**. They're often more useful to chemists than simple Joules. They tell you how much energy a mole of molecules has, rather than the energy of just one molecule.

We can relate molar energy to normal energy using a fundamental constant, the **Avogadro number**, N_A . Write down the Avogadro number, and its units.

3.16

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

You probably knew the number, but maybe didn't know the units. The unit is mol^{-1} , or '*inverse moles*'. The way to think about this is that it means that there are 6.022×10^{23} particles *per mole* of substance.

Given the definition of N_A , write down the relationship between the molar energy E_m and normal energy E . Show that the relationship you write down is dimensionally consistent.

3.17

$$E_m = E \times N_A$$

Is it dimensionally consistent? Yes, because:

$$E_m = E \times N_A \longrightarrow \text{J mol}^{-1} = \text{mol}^{-1} \times \text{J} \longrightarrow \text{J mol}^{-1} = \text{J mol}^{-1}$$

Easy.

This is all good fun, and we'll use these dimensional analysis skills a lot going forward in the degree. But, usually when we do a calculation, we calculate some kind of number. That number will normally have a unit with it, to tell us what physical thing the number represents. The best thing for us to do is to keep the units with the number throughout the calculation, and check that *both* the number and the units make sense.

Here's an example: the molar energy of combustion of octane (a major constituent of petrol) is $-5470 \times 10^3 \text{ J mol}^{-1}$. What is the energy of combustion of one molecule of octane? Think about the units.

3.18 Using the equation above, we find:

$$E = \frac{E_m}{N_A} = \frac{-5470 \times 10^3 \text{ J mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}} = -9.08 \times 10^{-18} \text{ J}$$

So you can see, one molecule of octane doesn't produce much energy. But one mole is a massive number of molecules. To put it in context, if you had one molecule for every grain of sand *on earth*, you would still only have about 0.01% of a mole. Put another way, you'd need 10,000 earths to have enough grains of sand to make a mole. It's a big number (or, molecules are *tiny*).

The key thing we did in the calculation above though was **keep the units as part of the calculation**. If we do this, and manipulate the units alongside the numbers, we can check that in the end we get a sensible unit. In this case, we expected to get an answer in Joules, and we did – so this is a good indication that we did the calculation correctly.

Use the ideal gas law to calculate the temperature of one mole of gas in a 1 m^3 container held at a pressure of $1 \times 10^4 \text{ Pa}$. Show that the units of the answer make sense. Note that $1 \text{ Pa} = 1 \text{ J m}^{-3}$.

3.19

$$T = \frac{pV}{nR} = \frac{(1 \times 10^4 \text{ J m}^{-3}) \times (1 \text{ m}^3)}{(1 \text{ mol}) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1})} = 1202 \text{ K}$$

Hopefully you can see how by keeping the units there, we get an automatic check that our method was right. We expected to end up with a temperature, and we did. Nice.

Those of you who are interested in these things might be wondering why the unit of pressure, the Pascal (Pa), can be written as J m^{-3} . Before we answer this, can you write the definition of a Pascal purely in terms of the SI base units?

3.20

$$1 \text{ Pa} = 1 \text{ J m}^{-3} = 1 \text{ N m}^{-2} = 1 \text{ kg m}^{-1} \text{ s}^{-2}$$

We added an extra step there, for illustration, but these definitions are all equivalent. These units can help us understand what a 'Pascal' really means, and therefore what 'pressure' really means physically.

Think about the first definition (J m^{-3}). A cubic metre is a measure of volume, so the units are 'energy per volume'. If a gas is at high pressure, would you expect it to have a large or small energy in a given volume?

3.21 You'd expect it to have a large energy. High pressure implies that there's lots of energy bound up in the gas.

What about if the gas was in a huge volume? All other things being equal, if I have the same amount of gas in a big and a small volume, which one has a higher pressure?

3.22 The smaller volume has the higher pressure. This should be intuitive - if you have an inflated balloon and squeeze it (reducing the volume) it gets harder (high pressure) and might burst (pressure is so high it rips the container).

Can you see that these units *tell us about what the quantity means physically*? Pressure is a bit abstract, but thinking about it as *energy per unit volume* makes it more tangible. What about the definition $1 \text{ Pa} = 1 \text{ N m}^{-2}$ - can you make sense of this physically?

3.23 A Newton is a unit of force, and a square metre is an area, so 1 N m^{-2} is a *force per unit area*.

If I have a gas at low pressure, does it exert a low or a high force on the walls of the container it's in? Think about the balloon.

3.24 A low force, as it's not pushing very hard against the container walls. So another way to think about pressure is as the *force* that a gas presses with against the *area* of the walls of the container it's in. That's what the unit N m^{-2} tells us.

Hopefully you are getting the idea. Let's do another calculation.

A piece of zinc measures 5 cm by 2 cm by 0.2 cm, and weighs 14.26 g. What is the density (ρ) of zinc? Do the units you calculate make sense?

3.25

$$\rho = \frac{\text{mass}}{\text{volume}} = \frac{14.26 \text{ g}}{5 \text{ cm} \times 2 \text{ cm} \times 0.2 \text{ cm}} = \frac{14.26 \text{ g}}{2.00 \text{ cm}^3} = 7.13 \text{ g cm}^{-3}$$

The units make sense – they are *mass per volume*, which is the definition of density. You may also think that a dense object is one that is heavy for a given size, which is the thing as saying it has a high mass for a given volume, as the units imply.

Is the number we've calculated (7.13 g cm^{-3}) sensible? Does it seem reasonable?

3.26 It does. If you imagine a piece of metal that was a cubic centimetre, it weighing a few grams seems reasonable. If we'd calculated that the density was actually 7 kg cm^{-3} , then it would seem unreasonable (a cubic centimetre of metal weighing 7 kilograms seems like a lot!). **It is always a good idea to check any numbers you calculate seem reasonable within the units they have.**

We used units of centimetres rather than metres for length there. What is the definition of a centimetre?

3.27

$$1 \text{ cm} = 0.01 \text{ m} = 1 \times 10^{-2} \text{ m}$$

Or equivalently, there are 100 centimeters in a metre. The 'centi' part of 'centimetre' is called the **prefix**. We use these because often we will deal with numbers varying over many orders of magnitude, and having to include lots of powers of 10 in our definitions will get annoying. It's simpler to say 1 cm than it is to say $1 \times 10^{-2} \text{ m}$.

When manipulating units, we can just think of the prefix as giving us an additional factor in any calculations (highlighted below in blue):

$$1 \text{ centi metre} = 1 \times 10^{-2} \text{ metre}$$

Can you work out the volume of that piece of zinc from earlier, but in units of m^3 rather than cm^3 ?

3.28

$$\begin{aligned} 5 \text{ cm} \times 2 \text{ cm} \times 0.2 \text{ cm} &= 5 \times 10^{-2} \text{ m} \times 2 \times 10^{-2} \text{ m} \times 0.2 \times 10^{-2} \text{ m} \\ 5 \times 10^{-2} \text{ m} \times 2 \times 10^{-2} \text{ m} \times 0.2 \times 10^{-2} \text{ m} &= 5 \times 2 \times 0.2 \times (10^{-2})^3 \text{ m}^3 \\ &= 2 \times 10^{-6} \text{ m}^3 \end{aligned}$$

Can you see that we grouped together all the factors of 10^{-2} into $(10^{-2})^3 = 10^{-6}$, just like we grouped the three m's into one m^3 ?

Can you convert this volume back to cubic centimetres?

3.29 We already know the answer is 2 cm^3 , but let's see how to calculate it if we didn't know that.

We know the definition of a centimetre:

$$1\text{ cm} = 1 \times 10^{-2}\text{ m}$$

If we just cube both sides of that equation and rearrange:

$$\begin{aligned}(1\text{ cm})^3 &= (1 \times 10^{-2}\text{ m})^3 \\ 1^3\text{ cm}^3 &= (1 \times 10^{-2})^3\text{ m}^3 \\ 1\text{ cm}^3 &= 1 \times 10^{-6}\text{ m}^3\end{aligned}$$

Or equivalently, $1\text{ m}^3 = 1 \times 10^6\text{ cm}^3$ (multiply both sides by 10^6).

Our original volume in cubic metres was $2 \times 10^{-6}\text{ m}^3$, and we know the conversion above, so we can swap the m^3 for $1 \times 10^6\text{ cm}^3$ and get:

$$2 \times 10^{-6}\text{ m}^3 = 2 \times 10^{-6} \times 1 \times 10^6\text{ cm}^3 = 2\text{ cm}^3$$

In this way, we can always convert any unit we want, provided we know the prefixes. A prefix we use a lot in chemistry is 'deci' (d), which means 10^{-1} .

Convert the concentration 3 mol dm^{-3} to the units mol m^{-3} . Remember that $1\text{ dm} = 1 \times 10^{-1}\text{ m}$.

3.30

$$3\text{ mol dm}^{-3} = 3 \times 10^3\text{ mol m}^{-3}$$

If you didn't get that, try it again. If you're really stuck, take the equation $1\text{ dm} = 1 \times 10^{-1}\text{ m}$ and raise both sides to the power of -3 to create a conversion factor. If you're still stuck, meditate on it. This is how you learn.

There are more prefixes than just centi and deci. It's useful to just know them all, so there's a list of all the useful ones on the next page. Notice how apart from centi and deci, they all jump in factors of 1000 (10^3). Convention is that whenever you'd need to write more than 3 digits to express a quantity, you use a prefix instead. So it's better to write 100 km than it is to write $100\,000\text{ metres}$.

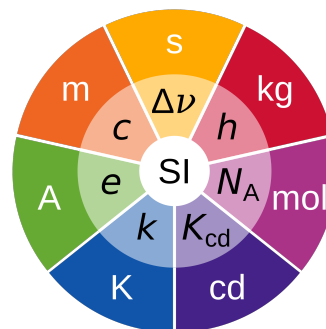
Learn the prefixes (or at least familiarise yourself with them) before you carry on to the problem sheet and next chapter.

Prefix	Symbol	Power of 10	Chemical Example
atto	a	10^{-18}	10 as $\approx$ time taken for an electron to orbit a nucleus
femto	f	10^{-15}	100 fs $\approx$ time taken for a chemical bond to vibrate
pico	p	10^{-12}	100 pm $\approx$ radius of an atom
nano	n	10^{-9}	400 nm = wavelength of blue visible light
micro	μ	10^{-6}	10 μ m $\approx$ diameter of a typical cell
milli	m	10^{-3}	1 mg = typical unit for talking about how much stuff you've synthesised
centi	c	10^{-2}	1 cm ³ = unit of volume, equivalent to 1 mL or one c.c.
deci	d	10^{-1}	1 dm ³ = unit of volume, equivalent to 1 L. For some reason always used to express molarity.
kilo	k	10^3	1 kg = now you're just showing off how much yield you got from that reaction
mega	M	10^6	1 MW = you're probably really into batteries or something
giga	G	10^9	100 GB = nice hard drive you've got there, in 2001.
tera	T	10^{12}	100 TB = data file size from the smallest cryo electron microscopy experiment
peta	P	10^{15}	10 PB = a tiny fraction of the data you use getting ChatGPT to do the problems for you
exa	E	10^{18}	1 Em $\approx$ the distance light would travel in a century

Problem Sheet 3 - Units and Dimensions

Not a person today, but an idea. The **Système Internationale d'Unités (SI System)** is a foundation stone of modern science. Before we had unified systems of units, every country had its own definitions of distance, mass, area, and so on. This was fine for general use, but a complete pain when you wanted to do science with other people and compare results. The image to the right shows the seven SI base units, and the fundamental constants they are defined by.

Today everyone doing science uses the SI system. Pretty much everyone in the world does for everything else too. The Americans are a notable exception, which is something to be aware of if you ever work in an American laboratory, but I think even they quietly accept the old systems of units are rubbish.



1. Rewrite the following very small or very large quantities using an appropriate SI prefix (see table at end of chapter). For example: width of a human hair = $0.000\,05\text{ m} = 50\text{ }\mu\text{m}$.

- i Energy of a chemical reaction: $100\,000\text{ J mol}^{-1}$.
- ii Wavelength of red light from the neon lamp in the periodic table in the foyer: $0.000\,000\,625\text{ m}$.
- iii The concentration of H_3O^+ in rain water: $0.000\,01\text{ mol dm}^{-3}$.
- iv The distance from Leicester to London: $15\,000\,000\text{ cm}$.
- v The volume of a gold nanoparticle: $0.000\,000\,000\,000\,000\,000\,000\,520\text{ m}^3$.

2. Convert the following quantities into the specified units:

- | | | |
|--|--|--|
| i 0.25 m into mm. | iv 4 days into seconds. | vii 25 kJ mol^{-1} into J. |
| ii 62 g into kg. | v 12.8 g cm^{-3} into kg cm^{-3} . | viii 15 min^{-1} into s^{-1} . |
| iii 375 cm^3 into dm^3 . | vi 60 km h^{-1} into m s^{-1} . | ix 10 mg cm^{-2} into g m^{-2} . |

3. The frequency of light (ν , in units of $\text{Hz} = \text{s}^{-1}$) is related to it's wavelength (λ , in units of m) by:

$$\nu = \frac{c}{\lambda}$$

Where c is the speed of light in vacuum ($2.998 \times 10^8\text{ m s}^{-1}$).

- i Show that this equation is dimensionally consistent.
 - ii Calculate the frequency of visible green light with a wavelength of 525 nm.
 - iii Calculate the wavelength of the light with frequency 3.45 GHz (the frequency that carries all the 5G data you're using to look up an online unit converter).
4. Write down units of all the quantities in the equations below, and show that the equations themselves are dimensionally consistent. You may need to look up quantities (or work them out from the others!) if you are unsure of what any symbols mean.
 - i The ideal gas law: $pV = nRT$
 - ii The definition of concentration: $c = \frac{n}{V}$
 - iii The Beer-Lambert law: $A = \epsilon cL$ (Hint: A is dimensionless, what must the units of ϵ be?)
 - iv Einstein's energy-mass equivalence relation: $E = mc^2$
 - v Relationship between photon energy and wavelength: $E = \frac{hc}{\lambda}$
 - vi The first law of thermodynamics: $\Delta U = q + w$

5. A useful rule of thumb, for people working in optical spectroscopy, is that light travels 'a foot per nanosecond'.

One foot is equal to 12 inches, and 1 inch is equal to 2.54 cm. Convert this rule of thumb into SI units, and compare it to the accepted speed of light given in Q3.

6. The concentration of bromide in seawater is $8.0 \times 10^{-4} \text{ mol dm}^{-3}$. Calculate the mass of potassium bromide that needs to be dissolved in 500 cm^3 of water to make a solution of this bromide concentration. [Data: the molar masses of potassium and bromine are 39.10 g mol^{-1} and 79.90 g mol^{-1} respectively.]
7. A solution of copper sulphate absorbs light with a wavelength of 500 nm. The absorbance of the solution is given by:

$$A = \epsilon c L$$

The absorbance of a solution with concentration $5 \times 10^{-2} \text{ mol dm}^{-3}$ measured in a cuvette with path length 10 mm is 0.89. Calculate the value of the absorption coefficient, ϵ , in units of $\text{m}^2 \text{ mol}^{-1}$.

8. The atmosphere of Earth is composed of 78% nitrogen, 21% oxygen, and 1% argon. The mass of the atmosphere is $5.27 \times 10^{18} \text{ kg}$. How many moles of gas are contained in the Earth's atmosphere? (*Hint: first calculate the average mass of one mole of 'air'.*)

[Data: Relative atomic masses of nitrogen, oxygen, and argon are 14.01, 16.00, and 39.95 g mol^{-1} .]

9. The radius of Earth is 6370 km. Assuming the planet is a perfect sphere, calculate the surface area of the Earth in units of square metres.
10. Pressure is defined as force exerted per unit area:

$$P = \frac{F}{A}$$

The force exerted on a mass M by gravity is given by $F = Mg$. On Earth, $g = 9.81 \text{ m s}^{-2}$. Use the mass calculated in Q8 and the area in Q9 to calculate the atmospheric pressure at the surface of the Earth.

11. An unknown gas has a density of 0.739 kg m^{-3} at a pressure of $98.7 \times 10^3 \text{ N m}^{-2}$ and a temperature of 273 K. Find the molar mass of the unknown gas, and hence propose a viable chemical formula for the gas. You may assume the gas behaves ideally.

Challenge Problems

1. Two beakers containing pure methanol and pure ethanol are placed on a bench. A 5 mL pipette is used to take 5 mL of the methanol and place it into the ethanol beaker to make a methanol-ethanol mixture. The same pipette is then used to take 5 mL of the methanol-ethanol mixture, putting it back into the methanol beaker.

Which beaker contains the most pure sample of either methanol or ethanol?

2. The Planck constant is $6.626 \times 10^{-34} \text{ J s}$. Why is it so small?
3. The average solar irradiance on Earth is about 1000 W m^{-2} . By making appropriate assumptions, determine the area of solar panels needed to boil enough water to make a reasonable cup of Yorkshire tea.

Chapter 4

Errors and Statistics

Most numbers that we deal with in science were, at some point, the result of an experiment. Experiments do not normally give us results that are perfectly precise, so it's important that we have some idea what the **error** or **uncertainty** is on any number that we quote.

4.1 We begin with some definitions. The number of **significant figures** in a number is the number of digits of that number **that contain useful information**. For instance, if I have a solution with a concentration of $0.023 \text{ mol dm}^{-3}$, given to two significant figures, it implies that those two significant figures are the ones that I am confident about, and that I know to some degree of certainty that the concentration is 0.023, not 0.024 or 0.022.

Usually in science we will talk about stating numbers to a certain number of significant figures, rather than decimal places – but you will encounter both. Decimal places just mean the number of digits *after* the decimal point.

The number of significant figures in a number is the number of digits, ignoring any leading zeros. For example, all of the numbers below are given to three significant figures:

- 123
- 0.0000123
- 12.0

How many significant figures are the following numbers given to?

1.2, 0.002, 0.00234, 12000

4.2

2, 1, 3, 5

The number of significant figures roughly reflects the precision to which you know the number. Taking the previous example of solution of concentration $0.023 \text{ mol dm}^{-3}$, in the absence of other information about the uncertainty on this number, the concentration $0.023 \text{ mol dm}^{-3}$ as written *implies* that the true solution concentration could be anywhere between 0.0225 and $0.0235 \text{ mol dm}^{-3}$. We call these numbers the **upper and lower bounds** on the number.

Sometimes, we might have additional information, like a calculated *standard deviation*, that tells us about what these bounds are precisely, but in the absence of that, the number of significant figures can be used to get a reasonable guess.

The mass of a substance was reported as 1.75 g. How many significant figures are given here, and what would be a reasonable guess at the upper and lower bound on the mass of this substance?

4.3 There are three significant figures, and a reasonable guess to the bounds would be that the mass is between 1.745 and 1.755 g.

It is important to take significant figures into account when doing calculations. For example, if I make up a solution using 22.0 mol of solute in 7.0 dm³ of solution, what is the concentration?

4.4

$$\text{Concentration} = \frac{22.0 \text{ mol}}{7.0 \text{ dm}^3} = 3.14285714286 \text{ mol dm}^{-3}$$

Is the number of significant figures in the answer given above sensible?

4.5 No, it is far too many!

Saying that the concentration is 3.14285714286 mol dm⁻³ implies that we know that concentration down to a precision of 0.00000000001 mol dm⁻³. The precision of our original quantities in the calculation was much less than this, and they were given to three and two significant figures respectively (22.0 mol and 7.0 dm³). The general rule is that:

You can **never** create precision through calculation.

So you should think about the significant figures of each component of the calculation, as the answer cannot have more significant figures than any of the components. A rule of thumb is that the number of significant figures in the result should be the same as the number in the least precise component.

What is the concentration of our solution, given to the correct number of significant figures?

4.6

$$\text{Concentration} = \frac{22.0 \text{ mol}}{7.0 \text{ dm}^3} = 3.1 \text{ mol dm}^{-3}$$

We give the result to two significant figures, because the least precise component (the volume) was given to two significant figures. We get there by **rounding** the original number. There are lots of ways of rounding, and if you're struggling to sleep one night you can read the Wikipedia article on the subject.

For our purposes, rounding to n significant figures means keeping the first n significant figures of the result. To work out the final significant figure, we look at the figure $n + 1$ and round up if it is equal to or more than 5, or round down if it is less than five. For example, if we round the following numbers to three significant figures:

$$1.234 \xrightarrow{\text{round}} 1.23 \text{ (rounded down)}$$

$$1.235 \xrightarrow{\text{round}} 1.24 \text{ (rounded up)}$$

Can you round the mass of a proton ($1.67262192595 \times 10^{-27}$ kg) to four significant figures?

4.7 1.673×10^{-27} kg.

In science, and specifically in chemistry, the precision to which we know different numbers varies a lot. If we are just measuring volume in a beaker, we might only know the result to the nearest millilitre, but in high-resolution spectroscopy it is not uncommon to know transition energies to 8 significant figures or more. For example, one of the transitions in the spectrum of molecular oxygen has a wavenumber of 68.652221 cm⁻¹, and we *know* that it's 68.652221 cm⁻¹, not 68.65222 cm⁻¹.

Thus, it is important that you think about the precision of any numbers you calculate. **Simply quoting all the digits on your calculator screen is wrong, and will be marked wrong!** You are more intelligent than your calculator.

Calculate the pressure of 1.0 mol of gas in a container of volume 2.5 m^3 at a temperature of 298 K. The gas constant, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$. Give your answer to the appropriate number of significant figures.

4.8

$$p = \frac{nRT}{V} = \frac{1.0 \text{ mol} \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 298 \text{ K}}{2.5 \text{ m}^3} = 991.0288 \text{ J m}^{-3} \xrightarrow{\text{round}} 990 \text{ Pa}$$

Where we round to two significant figures to reflect the precision of the least precise part(s) of the calculation (the pressure or the volume). Check that all the units make sense too, based on what we learnt last time.

I'm aware that at school, they make a huge song and dance about the correct number of decimal places to quote answers to. I am not really worried about it, as I have been working in physical science for my whole career without really worrying about it, and it hasn't caused me any problems. So, I don't want to spend ages on this, and I don't want you to be worrying about this too much (there are much more important things to understand). There are only two things you need to do to never have a problem:

1. **Do not** just write down all the digits from your calculator screen when answering a question. This is lazy, and wrong, and will be marked wrong.
2. **Do** just think about it for 2 seconds before you write down a number. Check the significant figures on the things that went into the calculation, and make sure the result doesn't have more.

Following these rules will mean you never have an issue, in my experience. Just think about what you are writing, before you write it down.

Let's now define what we mean by **precision**. If a measurement is precise, it means that repeated measurements of the same thing give me very similar results. For example, say I synthesised something in the lab 5 times and weighed the amount of product each time, and obtained the following masses:

1.540 g, 1.535 g, 1.546 g, 1.532 g, 1.545 g

What is the mean (average) of these numbers?

4.9 The mean is 1.5396 g (add all the numbers up and divide by the number of numbers).

However, do we actually know this number to five significant figures? Writing it as 1.5396 g implies that we *know* the last digit is 6, and that the number is 1.5396 g rather than 1.5395 g or 1.5397 g.

Looking at the data we collected, does it seem like we actually have that level of precision, down to 0.0001 g?

4.10 We don't. Because the measured numbers were a) measured to four significant figures, and b) clearly cover a range that is larger than 0.0001 g.

How many significant figures should the mean be given to, and what should it be?

4.11 The mean should be 1.540 g, given to four significant figures (rounding 1.5396 up to 1.540).

We can use these repeat measurements to calculate both the average (mean) value above, and also a more quantitative measure of the 'spread' of the data. We call this the **standard deviation**. For a set of data, the standard deviation σ is calculated as:

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n + 1}}$$

Where x_i is the i th measurement, $\bar{x}$ is the mean value of x we've calculated, and n is the total number of measurements. For our data above, $n = 5$, and $\bar{x} = 1.01 \text{ g}$. The symbol $\sum_{i=1}^n$ means 'take the sum of whatever is in front of that symbol, from $i = 1$ to $i = n$ '. This formula might look intimidating but if you try and think about the physical meaning of each part, it gets a bit simpler:

$$\text{Standard Deviation} = \sqrt{\frac{\sum_i (\text{Distance of } x_i \text{ from mean})^2}{\text{Number of measurements} + 1}}$$

So, if the measurements are further from the mean, then the standard deviation is higher. If we have more measurements, it gets lower. Does that make sense? I think it does, but convince yourself of it before moving on.

4.12 Hopefully you are now convinced. We can talk about why the denominator is $n + 1$ and not n another time.

Using this equation might look confusing, but it's easier if you just write out a few terms:

$$\sigma = \frac{(x_1 - \bar{x})^2}{6} + \frac{(x_2 - \bar{x})^2}{6} + \dots + \frac{(x_5 - \bar{x})^2}{6}$$

With this, can you calculate the standard deviation of the mass data given earlier?

4.13

$$\sigma = 0.006\,107\text{ g}$$

Given to four significant figures. Using the standard deviation, we can generate a good estimate of the **uncertainty in the mean**, or the **standard error**, α :

$$\alpha = \frac{\sigma}{\sqrt{n}}$$

Where σ is the standard deviation and n is the number of measurements we took originally.

What happens to the value of α as n increases? Does your answer make sense?

4.14 It goes down. This makes sense, because the more measurements we take, the more certain we are about the number we are calculating, and the smaller our error is.

Calculate the standard error for the mass data given earlier.

4.15

$$\alpha = \frac{0.006\,107\text{ g}}{5} = 0.001\,221\text{ g}$$

Where I have left the units in to show that this has the same units as the mean we calculated. The mean of our mass data was 1.540 g. The mean and the standard error together define the *best estimate* of our measured quantity:

$$\text{Best Estimate} = \text{Mean} \pm \text{Standard Error} = 1.540\text{ g} \pm 0.001\,221\text{ g}$$

There is one last thing to tidy up though. Look at the number of decimal places on the mean and the standard error - does it seem sensible that we quote the standard error down to the sixth decimal place, but the mean to the fourth?

4.16 No, it doesn't. In this case, we limit the decimal places on the standard deviation to be the same as those on the mean, so:

$$1.540\text{ g} \pm 0.001\text{ g}$$

Would be our final result. In general, whenever we quote a number that is the result of an experiment, we need to also give some estimate of the error, or uncertainty, on that number.

Here is another example for you to do yourself. You're measuring the concentration of a solution via absorption spectroscopy, and find the following concentrations (measured in mol dm^{-3}).

$$2.342, 2.362, 2.299, 2.327$$

Calculate the mean and standard error of the concentration of the solution (*Note: your calculator probably has an inbuilt function to calculate the standard deviation of a set of numbers.*)

4.17 Our mean and standard error are:

$$2.333 \pm 0.007 \text{ mol dm}^{-3}$$

We'll see some more examples in the problems, and you'll do this a lot in the labs and other modules over the next few years. The key point is that **measured quantities need to come with an estimate of the uncertainty in the measurement**. You need to not only state what you think the result is, but also how sure you are about your result – that is what the uncertainty estimate does.

To make use of these uncertainty estimates, we need to be able to use them in calculations, to see how the uncertainty in one quantity affects the calculation of a different quantity. This is called **error propagation**, the process amounts to carrying, or *propagating*, the error through the calculation alongside the original quantity. Let's see an example, using the concentration we just calculated.

Given that concentration ($2.333 \pm 0.007 \text{ mol dm}^{-3}$), I might want to know how many moles of substance were in my solution. The volume of the solution is $5.233 \pm 0.056 \text{ dm}^3$. Putting aside the uncertainty for now, how many moles of substance do I have?

4.18

$$\text{Moles} = 2.333 \text{ mol dm}^{-3} \times 5.233 \text{ dm}^3 = 12.207 \text{ mol}$$

That was easy. But what about the uncertainty?

Naïvely, we might expect that we just multiply the uncertainties in the concentration and volume to get the uncertainty in the number of moles. **This is wrong**, but let's try it to understand why. Multiply the uncertainties and see what you get.

4.19

$$\text{Uncertainty} = 0.007 \text{ mol dm}^{-3} \times 0.056 \text{ dm}^3 = 0.000392 \text{ mol}$$

So, just multiplying the uncertainties, we get a *much smaller* uncertainty in the number of moles than we had in either the concentration or the volume. A useful way to illustrate why this is wrong (and a useful thing to know how to do in general) is to calculate the **percentage error** in each quantity:

$$\text{Percentage Error} = \frac{\text{Error in Quantity}}{\text{Value of Quantity}} \times 100$$

For example, the percentage error in our original concentration is:

$$\frac{0.007}{2.333} \times 100 = 0.286\%$$

Calculate the percentage error in the volume we were given, and in the calculated number of moles using our naïve calculation of the error.

4.20 The percentage errors of all three quantities are:

$$\text{Concentration} = 0.286\%$$

$$\text{Volume} = 1.070\%$$

$$\text{Moles} = 0.003\%$$

Does the percentage error in the number of moles look correct?

4.21 No. It doesn't make sense that we can have errors on the order of 1% in our concentration and volume, but then have an error about 100 times smaller in the calculated number moles. Clearly, *our calculation of the uncertainty in the number of moles must have been wrong!*

To get this calculation right, we need to **propagate the errors** properly. Next year, we will learn a general way to do this – but it relies on knowing how to differentiate, which we won't learn until next semester. For now, here is a table of formulae you can use to propagate errors for some common cases.

In all cases we assume we have two quantities, A and B , with uncertainties ΔA and ΔB , that are combined to give a result R with uncertainty ΔR .

Calculation	Result (R)	Error in Result (ΔR)
Addition (or Subtraction)	$R = A \pm B$	$\Delta R = \sqrt{(\Delta A)^2 + (\Delta B)^2}$
Multiplication (or Division)	$R = A \times B$ or $R = \frac{A}{B}$	$\frac{\Delta R}{R} = \sqrt{\left(\frac{\Delta A}{A}\right)^2 + \left(\frac{\Delta B}{B}\right)^2}$
Raising to Power	$R = A^n$	$\frac{\Delta R}{R} = n \frac{\Delta A}{A}$

Can you use the formulae given above to work out the correct error in our number of calculated moles? As a reminder, the original concentration was $2.333 \pm 0.007 \text{ mol dm}^{-3}$, and the original volume was $5.233 \pm 0.056 \text{ dm}^3$.

4.22

$$\begin{aligned} \text{Error} &= 12.207 \text{ mol} \times \sqrt{\left(\frac{0.007 \text{ mol dm}^{-3}}{2.333 \text{ mol dm}^{-3}}\right)^2 + \left(\frac{0.056 \text{ dm}^3}{5.233 \text{ dm}^3}\right)^2} \\ &= 0.135 \text{ mol} \end{aligned}$$

So our calculated number of moles overall is:

$$12.207 \pm 0.135 \text{ mol}$$

Can you recalculate the percentage error in the number of moles using our new calculated error, and check that the result is more reasonable than what we had last time?

4.23

$$\% \text{ Error} = \frac{0.135}{12.207} \times 100 = 1.10\%$$

This seems more reasonable, remembering that the percentage errors in our original concentration and volume were 0.286% and 1.070% respectively.

One final thing to note before you just need to practice these calculations. Notice that the quantity:

$$\frac{\Delta R}{R}$$

Comes up in a lot of the equations in the table above, and in our equation for the percentage error. $\Delta R/R$ is called the **relative error** in R . Sometimes you'll hear that name used, so now you know what it means. It's just the error in a quantity divided by the quantity itself.

Let's do a final example before you move onto the problems. We have measured the length of both sides of a piece of paper to be:

$$29.7 \pm 0.05 \text{ cm}$$

$$21.0 \pm 0.05 \text{ cm}$$

Calculate the area, and the error in the area, of the piece of paper.

4.24 First the area: $29.7 \text{ cm} \times 21.0 \text{ cm} = 623.7 \text{ cm}^2$.

Then error propagation gives:

$$\text{Error in Area} = 623.7 \text{ cm}^2 \times \sqrt{\left(\frac{0.05 \text{ cm}}{29.7 \text{ cm}}\right)^2 + \left(\frac{0.05 \text{ cm}}{21.0 \text{ cm}}\right)^2} = 1.82 \text{ cm}^2$$

So we can write the overall area as $623.7 \pm 1.82 \text{ cm}^2$.

What is the relative error, and percentage error, in the calculated area?

4.25

$$\text{Relative Error} = \frac{1.82}{623.7} = 2.92 \times 10^{-3}, \quad \text{Percentage Error} = \text{Relative Error} \times 100 = 0.292\%$$

Note, if you haven't already, that the percentage error is just the relative error multiplied by 100.

That's it for errors and statistics in this module. Remember the golden rules to avoid losing marks:

1. **Do not just write down all the digits on your calculator screen.** This is just as wrong as giving too few significant figures, and will be marked wrong.
2. **Think about the numbers before you write them down.** Is the number of significant figures appropriate? If there is an error associated with the number, is that error reasonable?

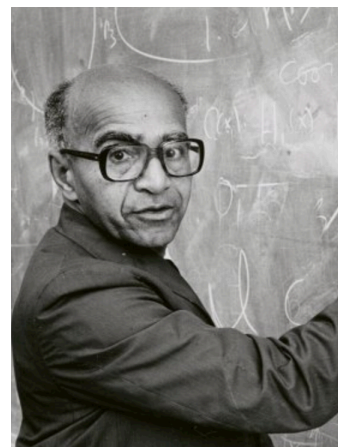
Just taking a few seconds to think before writing would sort out most of the problems I see when marking exams and coursework. It might seem fiddly now, but you will do this all the time in the labs and in other modules, so it will become second nature very quickly.

Next time, we will continue talking about functions, and learn about **logarithms**.

Problem Sheet 4 - Errors and Statistics

Statistics is a central part of experimental science. Our experiments always give us results that we have to infer physical meaning from in some way, using statistics and probability. There are two ways of viewing this process, either via **Frequentist** or **Bayesian** statistics.

The difference is beyond the scope of this course, but I'm a Bayesian statistics fanboy. David Blackwell (pictured) wrote one of the first textbooks on Bayesian statistics, in addition to developing a lot of other things in what would now be called **data science**. Although it's often made to sound like an exciting new frontier of science, data science is really just applied statistics. It has grown as a discipline, as experiments tend to produce more and bigger data as time goes on.



1. Write down the values of the following to two significant figures.

i $\frac{1}{3}$

iii 2π

v 0.12^3

ii $\sqrt{150}$

iv $34.695 + 66.305$

vi N_A

2. Given that $x = 3.265$ and $y = 12.5$, calculate the following quantities, giving your results to an appropriate number of significant figures using the 'rule of thumb' from the handout about significant figures in the absence of other information.

i x^3

iii $2y$

v y^x

ii $x + y$

iv $\frac{y}{x}$

vi $\sqrt[3]{y}$

3. Let $x = 3.265 \pm 0.0005$ and $y = 12.5 \pm 0.05$. Apply error propagation to calculate the numerical error on the results of the calculations (i), (ii), (iii), and (iv) from Q2.

Compare the propagated errors you find with the significant figures you gave in the answer to Q2. Was your choice of significant figures reasonable compared to the size of the propagated error?

4. Calculate the area, and error in the area, of a sheet of A3 paper. The dimensions of A3 paper are 42.0 ± 0.05 cm and 29.7 ± 0.05 cm.
5. A medicinal chemist is measuring the efficiency of a synthesis of a new drug molecule. They have measured 6 repeats of the mass of the produced drug (in milligrams)

12.32, 12.56, 11.89, 12.98, 12.83, 12.05

- Calculate the mean mass of the produced drug.
 - Calculate the standard deviation and standard error in the mean of the mass of the produced drug.
 - Calculate the relative error and percentage error in the mean of the mass of the produced drug.
 - The mass of the reactants put into the synthesis was 51.23 ± 0.01 mg. What is the percentage yield (in terms of mass) of the reaction, and the error in this percentage yield?
6. 5.00 ± 0.01 g of sodium chloride was dissolved in water in a volumetric flask with volume 100.0 ± 0.1 cm³. Calculate the molarity of the resulting solution, including an error derived through error propagation.

[Note: You may assume there is no error associated with the atomic masses of sodium and chlorine.]

7. A gas cylinder contains pure nitrogen at a pressure of $200,000 \pm 600 \text{ N m}^{-2}$. The volume inside the cylinder is estimated to be $0.050 \pm 0.002 \text{ m}^3$. The temperature is $290 \pm 1 \text{ K}$.

Calculate the number of moles of nitrogen inside the gas cylinder, and the uncertainty on this calculated number. You may assume that nitrogen behaves as an ideal gas under these conditions.

8. The volume of a cone, V is given by:

$$V = \frac{\pi h r^2}{3}$$

Where r is the radius of the base of the cone and h is the height of the cone.

A conical flask has a base radius of $7.0 \pm 0.1 \text{ cm}$ and a height of $19.5 \pm 0.2 \text{ cm}$.

Approximating the flask to be a cone, calculate the volume of the flask, and the uncertainty in this volume.

9. A industrial chemist is trying to decide if a prototype electrochemical cell for a battery meets the requirements of the application, which is that it has a voltage of 3.40 V .

Repeat measurements of the voltage are given below (in volts):

3.35, 3.45, 3.41, 3.38, 3.42

- Calculate the mean voltage.
- Calculate the error in this mean voltage.
- The industrial chemist's manager (who, by definition, has no scientific training at all) thinks that the battery does not meet the requirements, because he calculated the mean to be 3.401 V not 3.40 V . The chemist thinks that it does. Explain why the chemist is right, and her manager is wrong.

Challenge Problems

1. A fair coin is flipped three times. What is the probability of observing:

- One head, on any individual flip.
- Exactly two tails, across all three flips.

If the coin is flipped a very large number of times, what is the probability of observing more heads than tails?

2. Ultracold ($T = 0.38 \text{ K}$) molecules can be made by trapping them in droplets of superfluid helium. The probability, $P(k)$ for k molecules to be picked up by an individual droplet follows the *Poisson distribution*:

$$P(k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Where λ is the average number of events occurring.

- Calculate $P(k)$ for $k = 1$ to 6 , in the case that $\lambda = 1$. Sketch a graph of the results.
 - Calculate $P(k)$ for $k = 1$ to 6 , in the case that $\lambda = 3$. Sketch a graph of the results, and comment on the difference to the graph from the previous part question.
 - A research supervisor wants you to find a situation where there is *only* one molecule ever picked up by a droplet. Explain whether or not this is possible, and if it is not possible, describe the conditions that would best approximate this behaviour.
3. The formulae given for propagation of uncertainty make certain assumptions. Research and describe these assumptions, and explain where they may break down.

Chapter 5

Exponentials and Logarithms

Now we return to thinking about different kinds of functions. Previously we saw **polynomials**, which contained only powers of the function argument. For example, things like:

$$f(x) = x^3 + x^2 + x + 1$$

Remember that f is the function and x is the argument.

Over the next two chapters, we will expand the range of functions we know about. Today we are going to think about functions like:

$$f(x) = 2^x, \quad g(t) = e^t, \quad h(x) = \log x, \quad y(t) = \ln t$$

These kinds of functions are called **exponentials** (first two, f and g) and **logarithms** (last two, h and y).

5.1 Exponentials are familiar, so let's start there. What is the value of the function:

$$f(x) = 10^x$$

For $x = 0, 1, 2, 3, 4$?

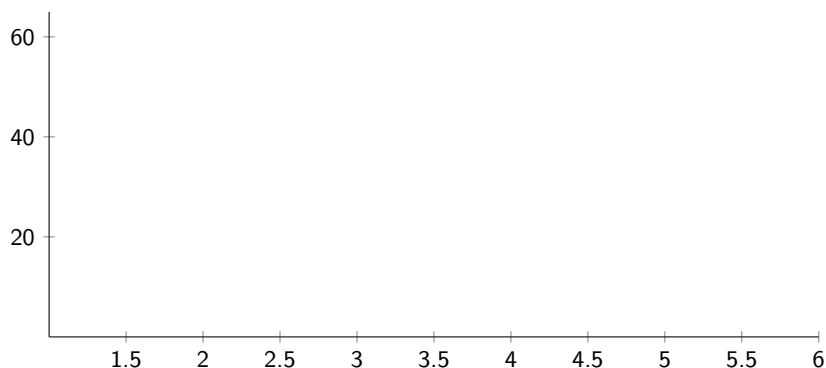
5.2 $f(x) = 1, 10, 100, 1000, 10000$.

So we can see that as we increase the argument x by one, then our function $f(x) = 10^x$ increases by a factor of 10. We call the number 10 the **base** of the exponent.

This rapid increase in the value of f is called **exponential growth**.

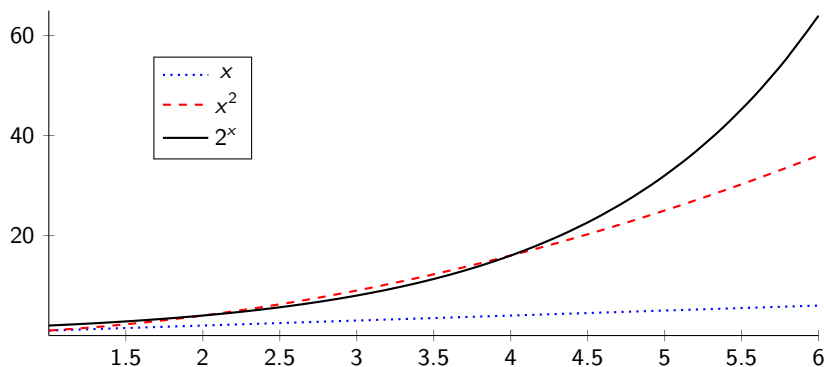
On the axes below, sketch the three functions:

$$f(x) = x, \quad f(x) = x^2, \quad f(x) = 2^x$$



Think about what the value of each function would be as x gets bigger.

5.3



You can see that the function 2^x rises much faster than the others. If we consider the point $x = 10$, then $x^2 = 100$, but $2^{10} = 1024$. Every time we increment x by one, the result multiplies by a factor of the base. This is what we mean by *exponential growth*.

A similar function is:

$$h(t) = 10^{-t}$$

What is the value of this function, for $t = 0, 1, 2, 3, 4$?

5.4 $h(t) = 1, 0.1, 0.01, 0.001, 0.0001$.

Now, as we increase the argument t by one, the value of the function $h(t) = 10^{-t}$ decreases by a factor of 10.

What do you think we call this rapid decrease in the value of h ?

5.5 We call it **exponential decay**.

Exponential growth and decay are very common things in science, because exponential functions are often the solutions to the differential equations that ultimately govern the behaviour of everything around us. We will understand that more fully next semester.

The classic chemical example of an exponential scale is the pH scale, where a change of 1 unit of pH corresponds to a factor of 10 change in the concentration of H_3O^+ ions in solution. You'll also see exponentials in equations like the **Arrhenius equation**:

$$k = A \exp\left(-\frac{E_a}{k_B T}\right)$$

or the **Boltzmann law**:

$$\frac{n_1}{n_0} = \exp\left(-\frac{E_a}{k_B T}\right)$$

Note that the notation $\exp(x)$ is an alternative form of e^x . This is another exponential, do you know what the number e is? Look it up if you don't.

5.6 $e = 2.718...$

e is a fundamental mathematical constant, like π , and is a very common base for exponentials. The reason for this is to do with what the gradient of the function e^x is, but we will talk about this more next semester.

Can you simplify the expression below, using the rules of exponentials we learnt previously?

$$\sqrt{\frac{e^{2x+2}}{e^2}}$$

5.7

$$\sqrt{\frac{e^{2x+2}}{e^2}} = \sqrt{\frac{e^{2x}e^2}{e^2}} = \sqrt{e^{2x}} = e^x$$

Easy.

So we know how to deal with exponentials. However, there is one thing we don't know. How do we solve the equation below for x ?

$$100 = 10^x$$

5.8 The solution is $x = 2$, which we can do by inspection.

What if we had a harder example, like:

$$20 = 10^x$$

There's not an obvious value of x that satisfies that equation. We need a way to 'undo' the exponential and bring down the x . That is what the **logarithm** function, $\log$, does:

$$\log_{10}(10^x) = x$$

The notation $\log_{10}$ means '*logarithm to the base 10*'. Usually, if you just see $\log$ written on it's own, it means the same thing as $\log_{10}$.

Can you work out the answer to the below?

$$\log_{10} 100 = ?$$

5.9

$$\log_{10} 100 = 2$$

If we compare with our equation from the previous frame:

$$\log_{10} 10^x = x$$

We can see that the $\log_{10}$ function basically answers the question: '*what power do I have to raise the base to, to get the number that I am taking the logarithm of?*'. $10^2 = 100$, so $\log_{10} 100 = 2$. The logarithm function is the inverse of the exponential function.

How about:

$$\log_{10} 1000 = ?$$

5.10

$$\log_{10} 1000 = 3$$

Hopefully you are getting the idea.

What about:

$$\log_{10} 0.1 = ?$$

5.11

$$\log_{10} 0.1 = -1$$

Because $10^{-1} = 0.1$.

So you can see how these logarithms work. They're commonly used when we have to deal with quantities that vary over many **orders of magnitude**. An 'order of magnitude' means a given power of 10, and links back to what we learnt about standard form in chapter 1. Two numbers like 2 and 8 are in the *same order of*

magnitude (within a power of 10 of each other), but two numbers like 2 and 200 are different by *two orders of magnitude* (two powers of 10).

The example of pH from before is excellent for demonstrating this. The definition of pH is:

$$\text{pH} = -\log_{10}[\text{H}_3\text{O}^+]$$

A solution has a concentration of hydronium ions $[\text{H}_3\text{O}^+] = 1 \times 10^{-7} \text{ mol dm}^{-3}$. What is the pH of the solution?

5.12

$$\text{pH} = -\log_{10}(1 \times 10^{-7}) = -(-7) = 7$$

What is the pH of a solution that has a concentration of hydronium ions $[\text{H}_3\text{O}^+]$ one million times larger than this (six orders of magnitude)?

5.13

$$\text{pH} = -\log_{10}(1 \times 10^{-7} \times 1 \times 10^6) = -\log_{10}(1 \times 10^{-1}) = -(-1) = 1$$

What would be the pH of a solution that has ten times fewer hydronium ions than one with pH 1?

5.14 It would have a pH of 2.

The point here is that changes in hydronium ion concentration are so huge, that it is difficult to talk about them on a linear scale. Doubling the number of hydronium ions in solution doesn't change the acidity much, but changing them by a factor of 100 does. Rather than keep talking about how we have 'one hundred times more ions', we prefer to say 'the pH changes by 2'. This is called using a **logarithmic scale**, or a **log scale** for short, and is very common. There are many other quantities that we usually put on a log scale. We'll meet some of them later.

To calculate logarithms, you either need to use a calculator or look them up in a huge 'table of logarithms' (if you went to school before 1960). Your calculator has a 'log' button that you can use for this, which will also allow you to calculate logarithms to different bases. Calculate the logarithms below:

$$\log_{10} 60, \log_2 8, \log_5 125, \log_{10} 1, \log_3 1, \log_{10} 2, \log_2 10, \log_{10}(-6)$$

Give your answers to two decimal places where appropriate.

5.15

$$1.78, 3, 3, 0, 0, 0.30, 3.32, \text{error!}$$

From these results we can see that $\log_x 1 = 0$ for any x , because $x^0 = 1$ for any x , and we can also see that **logarithms of negative numbers are undefined**. You can't take the logarithm of a negative number (or, if you do you get a result that is not a real number).

I find the expression below useful to remember the relationship between exponentials and logarithms:

$$a^x = b \longleftrightarrow \log_a b = x$$

In words, the equation above says: 'if a to the power of x is equal to b , then log to the base a of b is equal to x '. You may find another way to remember it that works better for you.

We can use the logarithm function to help us solve equations involving exponentials, as we can just take logarithms of both sides of an equation. Solve the equation below for y by this method:

$$10^y = 4$$

5.16

$$\log 10^y = \log 4 \longrightarrow y = \log 4 = 0.60$$

Can you solve the equation below, using a logarithm to the base e ? The button for this on your calculator will say ' $\ln$ '.

$$e^x = 10$$

5.17

$$\log_e e^x = \log_e 10 \longrightarrow x = 2.30$$

Functions involving e^x are so common that we give the base- e logarithm ($\log_e$) its own name and symbol: the **natural logarithm**, $\ln$. This will be on your calculator too.

Solve the equation below using natural logarithms:

$$e^{2x} = 8$$

5.18

$$\ln e^{2x} = \ln 8 \longrightarrow 2x = 2.10 \longrightarrow x = 1.05$$

Easy.

Natural logarithms, like the related exponential functions, come up a lot in chemistry. For example, in the relationship between the equilibrium constant K and Gibbs energy ΔG :

$$\Delta G = -RT \ln K$$

Can you use what you know about logarithms to rearrange this equation, and make K the subject? Remember that in the same way we use logarithms to 'undo' exponentials, we use exponentials to 'undo' logarithms.

5.19

$$K = \exp\left(-\frac{\Delta G}{RT}\right)$$

Remember $\exp(x)$ is an alternative notation for e^x . You'll often see this in books because it's neater than compressing fractions into tiny spaces for some formulae.

Don't worry if that was tricky. Let's walk through it. Firstly, we know that to get rid of the $\ln$ in the $\ln K$ we need to 'undo' it by taking the exponential:

$$\exp(\ln K) = e^{\ln K} = K$$

So, we rearrange the original equation so we're left with just $\ln K$ on one side and everything else on the other:

$$\ln K = \frac{-\Delta G}{RT}$$

Then we take exponentials of both sides, and end up at the result above.

You might notice that we end up with a function inside the exponent that has the form 'energy over RT ', a bit like in the Arrhenius or Boltzmann equations. This is not a coincidence. We will get into it later on.

Now, let's learn some algebra for manipulating logarithms that will be useful. To begin with, solve the equation below for t :

$$\log t = 2$$

Remember that when there is no base given, a bare log means base 10 logarithm.

5.20

$$t = 100$$

So, what would $\log 100 + \log 100$ be equal to?

5.21

$$\log 100 + \log 100 = 2 + 2 = 4$$

What about:

$$\log 10000 = ?$$

5.22

$$\log 10000 = 4$$

Just like $\log 100 + \log 100 = 4$. This shows $\log 10000 = \log 100 + \log 100$, which leads to the first rule of logarithms that we need to know:

$$\log_x AB = \log_x A + \log_x B$$

Where x can be any base (natural, base 10, whatever). But the base does have to be the same.

Simplify the below into one logarithm

$$\log 100 + \log 10 + \log 2$$

5.23

$$\log 100 + \log 10 + \log 2 = \log(100 \times 10 \times 2) = \log 2000$$

So that rule ($\log_x AB = \log_x A + \log_x B$) works with multiplication and addition. A similar rule exists for division and subtraction:

$$\log_x \frac{A}{B} = \log_x A - \log_x B$$

These rules can combine together. Simplify the below into one logarithm:

$$(\ln 2 - \ln 3) + \ln 4$$

5.24

$$\ln \frac{8}{3}$$

Another useful rule is this:

$$\log_x a^b = b \log_x a$$

So if you have the logarithm of something raised to a power, you can bring the power down to 'in front' of the logarithm.

Simplify the below:

$$\ln e^2 - \ln e + \ln e^{-1}$$

5.25

$$\ln e^2 - \ln e + \ln e^{-1} = 2 \ln e - \ln e - \ln e = 0$$

Can you use what you know about logarithms and exponentials to work out what the below is equal to? It is always the same, for any value of x .

$$\log_x x = ?$$

5.26

$$\log_x x = 1$$

Because $x^1 = x$.

Sometimes, (though to be fair, not that often) it will be useful to change the base of a logarithm. We can do this using the relationship:

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Use this formula to express $\ln x$ ($\log_e x$) as a multiple of a base-10 logarithm (i.e. as a multiple of $\log_{10} x$).

5.27

$$\log_e x = \frac{\log_{10} x}{\log_e 10} = \frac{\log_{10} x}{2.303}$$

We'll see this again in the problems. Let's now summarise the rules of logarithms:

- $\log AB = \log A + \log B$
- $\log \frac{A}{B} = \log A - \log B$
- $\log A^B = B \log A$
- $\log_A x = \frac{\log_B x}{\log_B A}$

Note that these rules follow directly from the rules for dealing with powers (exponentials) that we saw in chapter 2. Note that there is not a rule that will allow you to easily simplify $\log A \times \log B$.

Can you use the rules above to simplify the expression below? Don't use your calculator.

$$\frac{\ln 2 + \ln 2^4}{\log_{10} 10000}$$

5.28

$$\frac{\ln 2 + \ln 2^4}{\log_{10} 10000} = \frac{5 \ln 2}{5} = \ln 2$$

Remember that $10000 = 10^5$.

Finally, there is another useful rule that can help us when using logarithms and exponentials in practical calculations. Remember we mentioned the Gibbs equation, Arrhenius equation, and Boltzmann law, which all included a term of the form:

$$\exp\left(\frac{-E}{k_B T}\right)$$

If E is an energy in Joules, what are the units of the thing inside the exponential?

5.29

$$\frac{-E}{k_B T} \rightarrow \frac{\text{J}}{\text{J K}^{-1} \times \text{K}} = \text{dimensionless}$$

They are unitless. **Anything that you put inside a logarithm or an exponential has to be unitless.**

It does not make physical or dimensional sense to take the log or exp of a quantity with units (the reason is most easily seen if you take the series expansion of the functions – see the challenge problems from this chapter.).

Astute readers among you may now say that the definition of pH involved taking the logarithm of a concentration, which is not dimensionless. So either I'm talking rubbish, or the pH scale is rubbish. In fact, the *strict* definition of pH uses a quantity called the *activity*, which you can think of like a dimensionless concentration. Under normal circumstances this is similar enough to a normal concentration that we don't worry about it, but we can talk about that another time.

This also means that any logarithm or exponential is dimensionless. With this in mind, what are the units of k in the Arrhenius equation below?

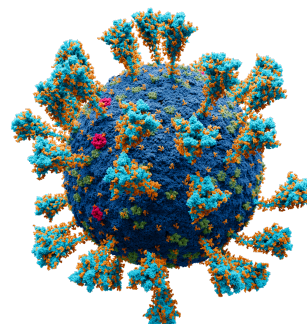
$$k = A \exp\left(-\frac{E_a}{RT}\right)$$

5.30 A bit of a trick question, but hopefully you realised that '*they are the same as the units of A*' is the answer. The actual units will depend on the kinetics of the reaction in question. A story for later.

Now it's time for the problems, to get used to dealing with exponentials and logs. Then we will talk about **trigonometric functions**.

Problem Sheet 5 - Exponentials and Logarithms

Another non-human picture today. To the right is the SARS-CoV-2 virus – the thing that caused the COVID-19 pandemic. Modelling of disease spread (epidemiology) is a classic use of applied mathematics, and it can be shown that the spread of these pandemics is, at certain times, exponential (we'll discuss this more next semester).



A pet peeve of mine is journalists misusing the phrase 'exponential growth'. It specifically refers to the case where the rate of growth is proportional to the current amount of whatever is growing. It doesn't just mean 'really fast growth'. I remember when the Suez canal was blocked by a grounded ship, and the news reports were that it was getting 'exponentially worse', when it demonstrably wasn't, it was just getting 'a bit worse'. The COVID-19 pandemic, however, is the one time where 'exponential growth' was the correct description. Shame none of the politicians believed the scientists.

1. Use the rules of logarithm algebra to rewrite the following expressions without any exponents (i.e. so every log term is in the form $a \log_b x$).

i $\log_{10} a^3$

iii $\log_{10} \left(\frac{a}{c^3} \right)$

v $\sqrt{\ln(e^9)}$

ii $\ln(ab^2)$

iv $\log_{10} \left(\frac{a^2}{1000} \right)$

vi $\ln \sqrt{e^9}$

2. Use the rules of logarithm algebra to rewrite the following expressions as a single logarithm.

i $\ln t + \ln t^2$

iii $5 \log_a x - \log_a x^3 + \log_a a^2$

v $\ln p + \ln V - \ln n - \ln T$

ii $4 \log_{10} t - \log_{10} t^3$

iv $\ln Z_1 - \ln Z_2$

vi $\ln 2 - \log_{10} 2$

3. Solve the following equations to find the value of x :

i $29 = 3^x + 2$

iii $5 = \sqrt[3]{3125}$

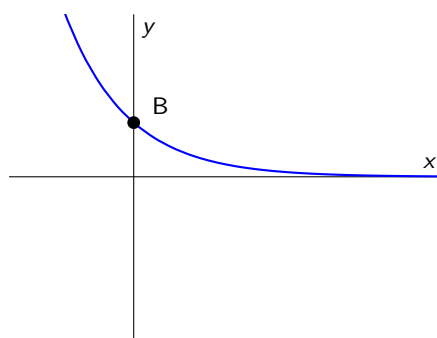
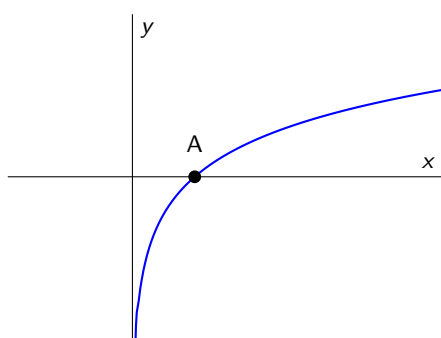
v $\log_{10}(x+7)^2 = 4$

ii $\sqrt{2^x} = 8$

iv $\ln(2e^{-x}) = 0$

vi $\frac{e^x}{e^{-x}} = 121$

4. The graphs below show a plot of $y = \ln(x)$ (left), and $y = e^{-x}$ (right). Find the coordinates of the points A and B.



Does the graph of $y = \ln(x)$ ever cross the y -axis, or the graph of $y = e^{-x}$ ever cross the x -axis? Explain your answer.

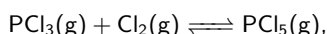
5. i Express $\log_3 x$ as a logarithm to the base 9.

- ii Express $\log_8(t^3)$ as a logarithm to the base 2.
 - iii Find x if $\log_x 100 = -2$.
 - iv Prove that $\ln x = 2.303 \log_{10} x$
6. The equilibrium constant (K_p) for a gas-phase chemical reaction is given by:

$$\Delta G = -RT \ln K_p$$

Where ΔG is the change in Gibbs energy for the reaction, T is temperature, and R is the gas constant.

- i For the reaction of phosphorus trichloride below, $\Delta G = -37.3 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.



Calculate the value of K_p .

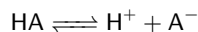
- ii Does this equilibrium favour the reactants or the products?
 - iii As the temperature is raised, how do you expect the position of equilibrium to change?
7. pH is defined as:

$$\text{pH} = -\log_{10} [\text{H}_3\text{O}^+]_{(\text{aq})}$$

- i Calculate the pH of a solution with $[\text{H}_3\text{O}^+]_{(\text{aq})} = 7.5 \times 10^{-2} \text{ mol dm}^{-3}$.
- ii Calculate $[\text{H}_3\text{O}^+]_{(\text{aq})}$ for a solution of $\text{pH} = 9.2$.
- iii 100 cm^3 of a solution with $\text{pH} = 3.2$ is diluted with 900 cm^3 of water. What is the pH of the resulting solution?
- iv 100 cm^3 of a solution with $\text{pH} = 8.5$ is diluted with 900 cm^3 of water. What is the pH of the resulting solution? (*Hint: think carefully, and use your chemical knowledge.*)

[Data: The dissociation constant of H_2O is $K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}$ at room temperature.].

8. An acid (HA) in equilibrium with its conjugate base (A^-) has the following chemical equation:



The acid dissociation constant, K_a is given by:

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

- i Take logarithms of the equation above to derive the **Henderson-Hasselbach Equation** below for the pH of a solution containing an acid at equilibrium with its conjugate base (i.e. an acidic buffer solution).

$$\text{pH} = \text{p}K_a + \log_{10} \frac{[\text{A}^-]}{[\text{HA}]}$$

- ii Calculate the pH of a solution containing an acid of concentration 0.3 mol dm^{-3} with its conjugate base at concentration 0.08 mol dm^{-3} . The acid has $K_a = 6.17 \times 10^{-8}$.
- iii For the same acid, calculate the pH of the solution when the concentration of the acid and conjugate base are equal.
- iv (Bonus) Use the Henderson-Hasselbach equation to explain why buffer solutions resist changes in pH. Can you think of anything that would limit the ability of a buffer solution to do this?

Challenge Problems

1. The exponential function e^x can be expanded as an infinite series:

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}$$

Write down a general formula for this series, and show that computing $\exp x$ where x is a quantity with units of length results in an equation that is dimensionally inconsistent.

2. Use the series expansion for e^x above to demonstrate why:

i $e^0 = 1$

ii $(e^x)^2 = e^{2x}$

iii A useful approximation to e^x , when x is small, is $1 + x$.

3. Another interesting series expansion is that of $\ln(1 + x)$:

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

i Write down a general formula for this series expansion, using sum notation.

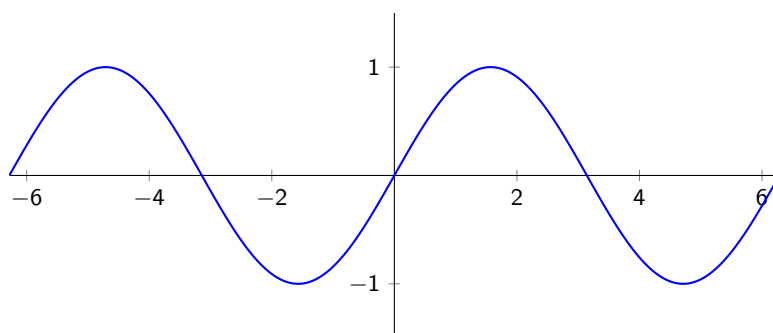
ii If x is small, what is a reasonable polynomial approximation to $\ln(1 + x)$?

iii What values of x is the series expansion above accurate for? One is simple to find, the other is more subtle.

Chapter 6

Periodic Functions and Trigonometry

Now we are going to talk about **periodic functions**. These are functions where the output oscillates back and forth between some range of values. An example is plotted below.



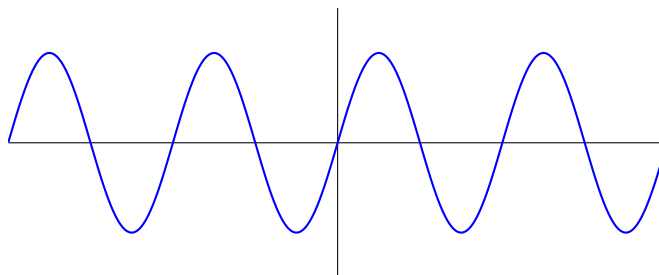
This is the **sine** function, $\sin(x)$.

In school, you probably learnt about the **trigonometric** functions $\sin(x)$, $\cos(x)$, and $\tan(x)$ in the context of computing geometry of triangles. That is a useful application of them, which we will come to, but actually the most important use they have in science is in modelling of things that oscillate – like light waves, vibrating chemical bonds, electric currents, and things like that.

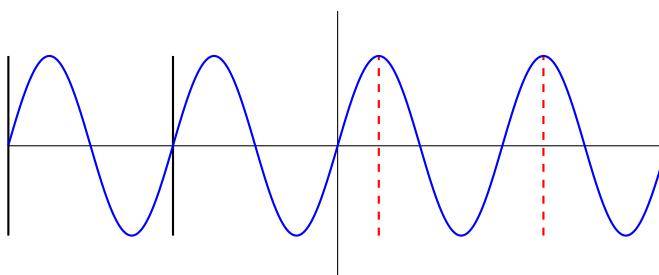
When you think about these trigonometric functions, I want you to be thinking of them as **periodic functions** that describe something that oscillates, like a wave. Waves are everywhere, as we'll see.

6.1 Periodic in this context means that the function cycles through the same set of outputs over and over again. The **period** of the function defines the length of this cycle. During a period, the function starts somewhere, goes through all the possible values it can take, and then comes back to where it started.

On the plot of the sine function below, can you mark two vertical lines that show one period of the function? There are many correct answers.



6.2 Two possible answers are shown in black (solid) and red (dashed) below. The first line starts in one place, the curve then cycles through all values of the function, and the second line is where the function gets back to where it started. The pair of black lines start and end as the function passes zero, but the red lines start and end at the peak of the function. Both sets of lines each define **one period** of the function.



If our wave had a physical interpretation, like a light or a sound wave, then ‘one period’, or ‘one cycle’ could be defined as something like the *wavelength* or *time period* of our wave.

The mathematical form of this sine function is:

$$f(x) = \sin(x)$$

Use your calculator, or memory, or the graph at the start of the chapter, to write down the value of $\sin(x)$ when $x = 0$.

6.3

$$f(0) = \sin(0) = 0$$

For other arguments, we need to take some care. In school, you’re probably used to the arguments of the trigonometric functions being angles in *degrees*. This works well for calculating the geometry of triangles, but doesn’t work very well for describing waves. Instead, we are almost always going to use the angles expressed in **radians** (with unit symbol rad, although actually radians are dimensionless). You can convert between radians and degrees easily, knowing that:

$$360^\circ = 2\pi \text{ rad}$$

Convert the following angles from degrees into radians. Leaving your answers as multiples of π is easiest.

$$45^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$$

6.4

$$\frac{\pi}{4} \text{ rad}, \frac{\pi}{2} \text{ rad}, \pi \text{ rad}, \frac{3\pi}{2} \text{ rad}, 2\pi \text{ rad}$$

We can do this by noting that if $360^\circ = 2\pi \text{ rad}$, then $180^\circ = \pi \text{ rad}$, and so on.

In general, we can define some conversion factors for converting between radians and degrees:

- To convert from radians to degrees, multiply radians by $\frac{180}{\pi}$.
- To convert from degrees to radians, multiply degrees by $\frac{\pi}{180}$.

Convert the following angles from radians into degrees:

1 rad, 3.14 rad, 10 mrad, 6 rad

6.5

57.3° , 180° , 0.57° , 343.7°

When you're using trigonometric functions on a calculator, you will need to make sure your calculator is set up to take the argument of the function as an angle in radians, not degrees. That the argument of a trigonometric function is given in radians is the default behaviour in all scientific programming libraries, so is worth getting used to.

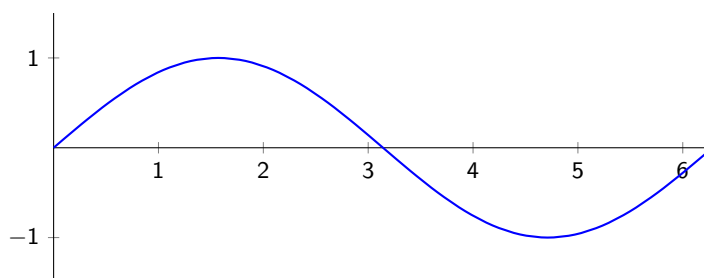
Calculate the value of $f(x) = \sin(x)$ at the following values of x (in radians):

0 , $\frac{\pi}{2}$, π , $\frac{3\pi}{2}$, 2π

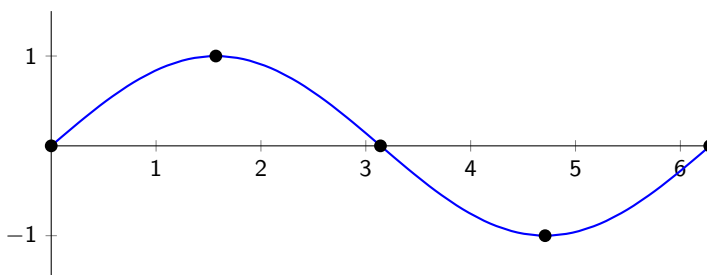
6.6

0 , 1 , 0 , -1 , 0

Annotate these points on the graph of $\sin(x)$ below.



6.7

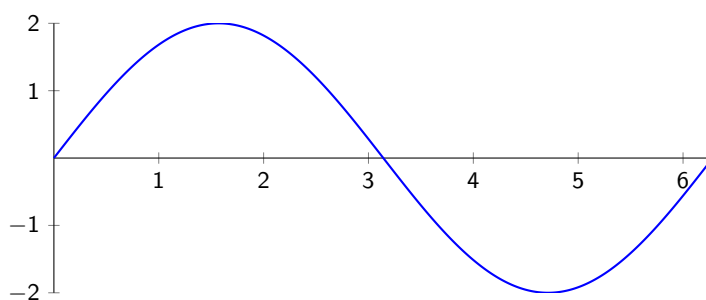


You need to know by heart how to sketch this function, and where these common points are on it.

Notice that the limits of the graph on the y axis are -1 and 1 . How would the plot above be different, if we plotted the function below instead?

$$f(x) = 2\sin(x)$$

6.8 The plot would be stretched along the y direction, like the below:



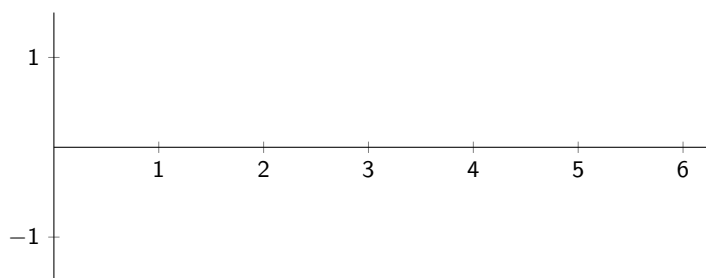
Notice that the upper and lower limits are now 2 and -2. Any number in front of our function will stretch it along the y direction like this. If our function was describing a physical wave, we could define the **amplitude** of the wave, A :

$$f(x) = A \sin(x)$$

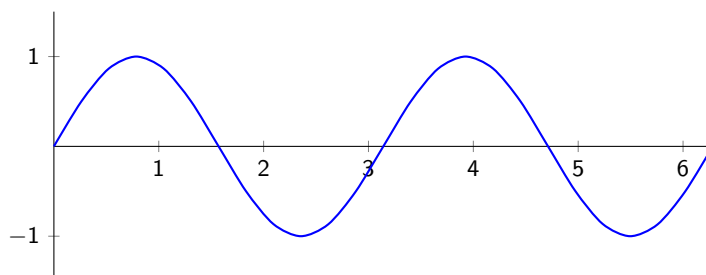
On the axes below, can you sketch the function:

$$f(x) = \sin(2x)$$

Think carefully. Plug in some values of x if you are unsure. The x axis limits run from 0 to 2π ($2\pi \approx 6.28$).

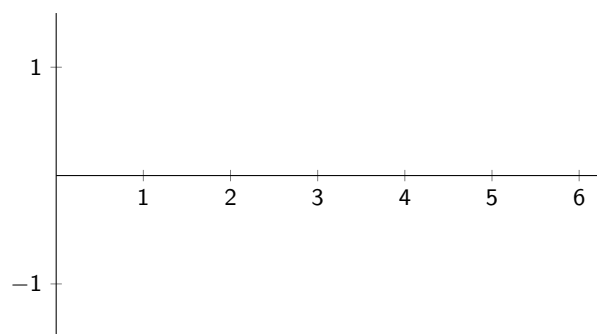


6.9

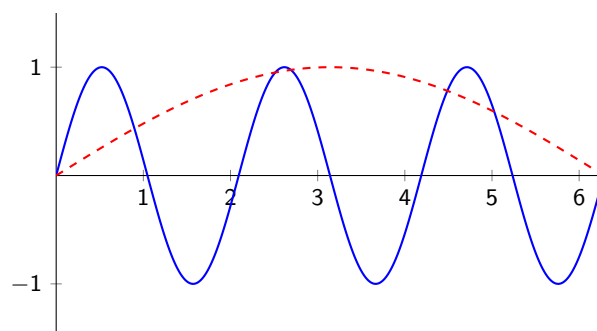


You can see that now we have two cycles of our wave visible in the same axis range. Increasing the number that multiplies the argument (the x in $\sin(x)$) will mean we get more cycles of our wave in a given axis range.

On the axes below, can you sketch the two functions: (i) $\sin(3x)$, (ii) $\sin(x/2)$?



6.10



Hopefully you can see that $\sin(3x)$ (blue, solid line) repeats itself three times, and that $\sin(x/2)$ (red, dashed line) only gets through half of a cycle, in the same range.

Physically, we can interpret any factor that multiplies the argument of our trigonometric function as a **frequency**. So we could define a wave an amplitude A and frequency ω as:

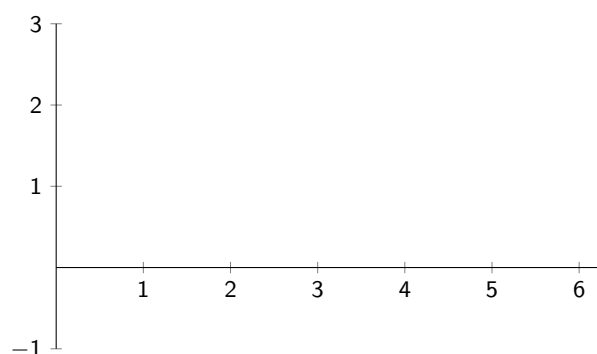
$$f(x) = A\sin(\omega x)$$

The exact units of ω will be the inverse of the units of x , which will depend on what our wave physically represents. The **wavelength** of our wave would be proportional to $1/\omega$.

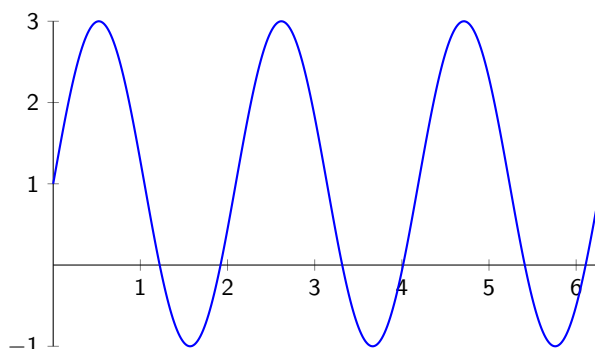
Sketch the function $g(t)$ below on the given set of axes, and label the axes. Think about what the constant $+1$ will do.

$$g(t) = 2\sin(3t) + 1$$

If you're stuck, plug in a few values of t and see if you can (literally or figuratively) connect the dots together.



6.11



Going through each part of the function ($g(t) = 2 \sin(3t) + 1$) in turn:

- The 2 at the front will change the amplitude of the wave, so that the distance from the initial zero point and the first peak is 2 units, or that the distance between a peak and a trough is 4 units.
- The 3 inside the argument will mean we fit 3 oscillations of the wave inside the range 0 to 2π .
- The +1 at the end acts a constant that shifts the whole wave up, so the initial point is at (0, 1), not (0,0).

This is difficult to see when you first encounter it, so if that was unclear, try it again work through each of the points above step by step.

What would the value of the function below be, if $x = 0$?

$$f(x) = \sin\left(x + \frac{\pi}{2}\right)$$

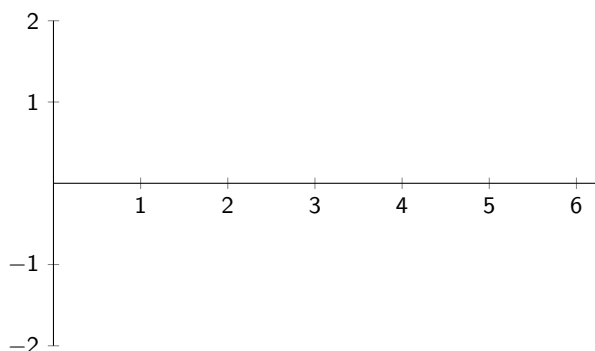
6.12

$$f(0) = \sin\left(0 + \frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1$$

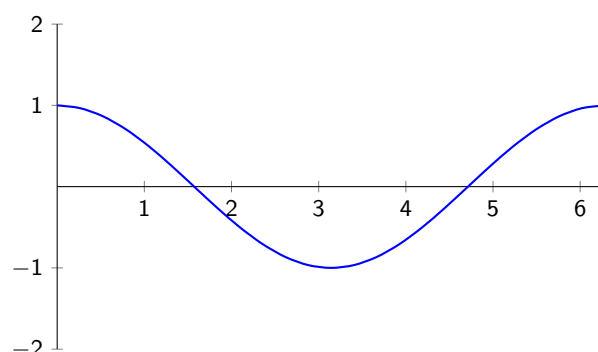
Given that you know the first point is at (0, 1), can you sketch a graph of:

$$f(x) = \sin\left(x + \frac{\pi}{2}\right)$$

On the axes below?



6.13



You can hopefully see that the difference between graphs of $\sin(x)$ and $\sin(x + \frac{\pi}{2})$ is that the additional $+\frac{\pi}{2}$ inside the function argument means that the graph *starts* $\frac{\pi}{2}$ radians later than the graph of simple $\sin(x)$. This is notoriously quite difficult to see the first time you come across it though - so try and sketch the graph of $\sin(x)$ over the one above and think about the value of it at different values of x .

For physical interpretation of the wave, the additional constant factor inside the function argument is called a **phase shift**. So, if I have (for example) a light wave $E(x)$ with amplitude A , frequency ω , and a phase shift ϕ , I could write:

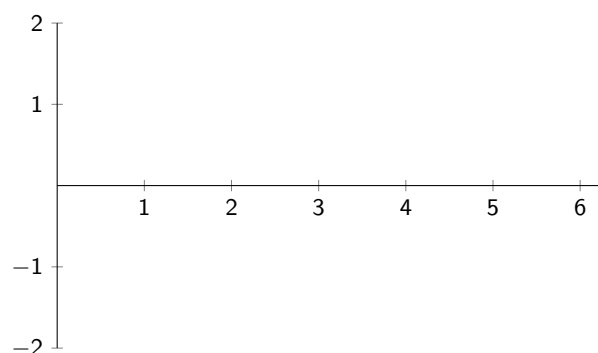
$$E(x) = A \sin(\omega x + \phi)$$

We'll talk more about physical interpretations of this later in the degree.

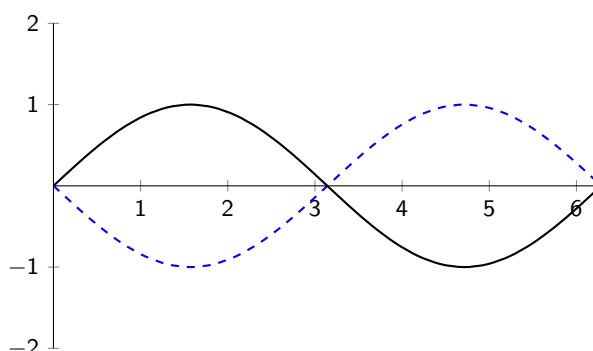
Can you sketch the functions below on the given axes?

$$f(x) = \sin(x)$$

$$g(x) = \sin(x + \pi)$$



6.14



The solid black line is $\sin(x)$, and the dashed blue is $\sin(x + \pi)$.

When two waves are like this, where peaks of one line up with the troughs of another, we call them *out of phase*. We've probably already spoken about this in CH1200, depending on where the lecture timetable falls this year...

Time to meet a new trigonometric function. The wave we saw a few frames ago:

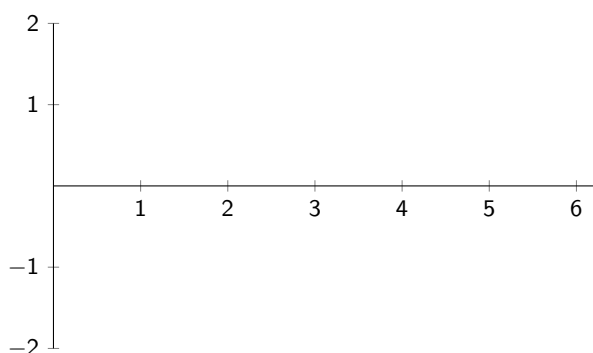
$$f(x) = \sin\left(x + \frac{\pi}{2}\right)$$

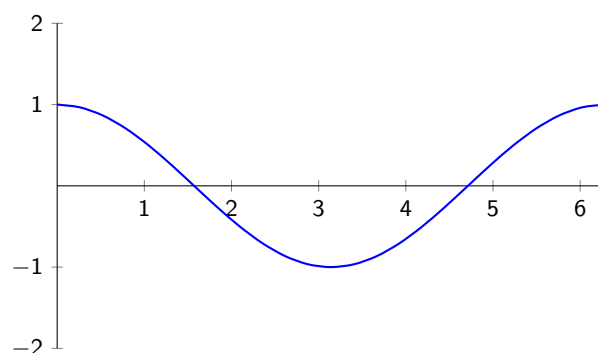
Has another name:

$$f(x) = \sin\left(x + \frac{\pi}{2}\right) = \cos(x)$$

The **cosine function**, $\cos(x)$. The cosine function has the same shape as the sine function, but is phase shifted by $\pi/2$ so that it starts at $(0,1)$.

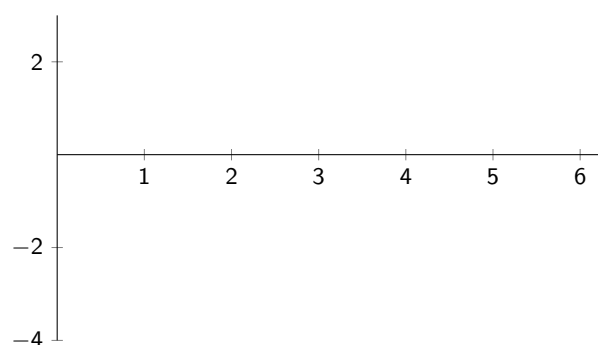
Sketch the function $f(x) = \cos(x)$ on the axes below.



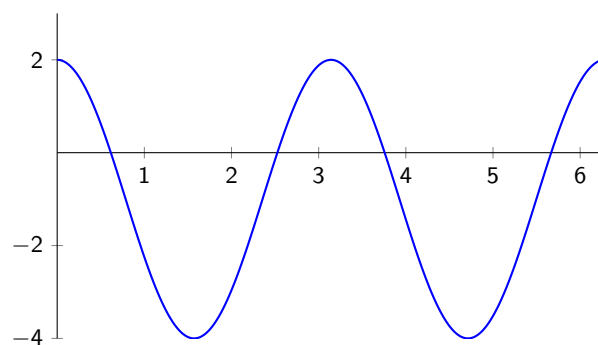


All the things we've already learnt about sketching sine functions apply to cosine functions too, so hopefully you will find it easy (or at least, not impossible) to sketch the function $h(t)$ below on the given axes.

$$h(t) = 3 \cos(2t) - 1$$



6.16

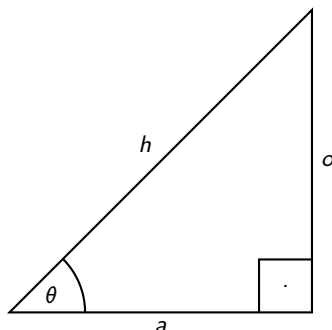


I reckon that's enough curve sketching for now (until the problems). We will see these kinds of functions over and over again. Let's now learn how to use these them in algebra.

There's a third function alongside the sine and cosine function, the **tangent** function, $\tan(x)$, which is the ratio of the sine and cosine functions:

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

These functions together are defined as the ratios of the length of sides of right-angled triangles like the one below, where the angle marked θ is related via the sine, cosine, and tangent functions to the ratios of the three sides, the hypotenuse, opposite, and adjacent (h, o, a on the plot).



There's a mnemonic for remembering how these are related which you learnt in school. What is it?

6.17 'SOHCAHTOA' – often written out as below, so it's clearer that (for example) to find out the length of the opposite, you take $\sin(\theta)$ multiplied by the hypotenuse.

$\begin{array}{ccc} \text{O} & \text{A} & \text{O} \\ \text{S H} & \text{C H} & \text{T A} \end{array}$

Using SOHCAHTOA, how would I find out the cosine of the angle, $\cos(\theta)$?

6.18

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$

This is hopefully all familiar from school.

How could you work out the length of the hypotenuse, given the two other side lengths?

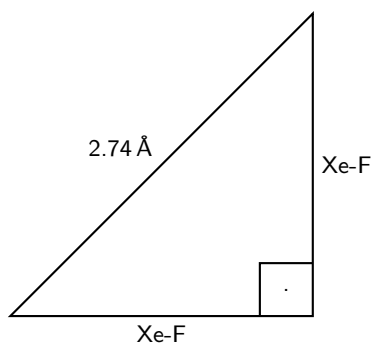
6.19 You can use **Pythagoras' Theorem**:

$$h^2 = o^2 + a^2$$

We can use these relationships to help us work out molecular geometry, which can be more useful than you might immediately think, especially in solid-state chemistry and crystallography.

Here's an example. Xenon tetrafluoride (XeF_4) is a square planar molecule, and all F–Xe–F bond angles are 90° . The straight-line distance between two fluorine atoms on the same side of the molecule is $2.74 \text{ \AA}$. What is the Xe–F bond length? *Hint: draw a sketch of the problem.*

6.20 We end up with a picture like the below:

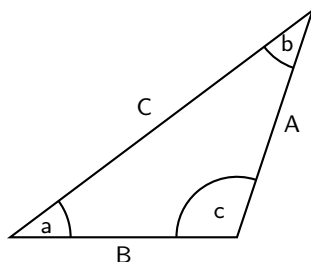


Pythagoras' theorem, or SOHCAHTOA using half the drawn triangle, lead to the answer that the Xe-F bond length is 1.94 Å. Note that when we are talking about geometry, it's more usual to refer to angles in degrees than in radians.

What are the two non-right angles in the diagram above?

6.21 They're both 45°. We know this, because the internal angles of a triangle must sum to 180°, the right angle is 90°, and the other two angles must be the same.

The rules I have just spoken about only apply for right-angled triangles. There are other rules that apply for any triangle too, though, like the one below.



One of these is the **sine rule**:

$$\frac{\sin a}{A} = \frac{\sin b}{B} = \frac{\sin c}{C}$$

Where the angles (a, b, c) and side lengths (A, B, C) are defined as above.

Another is the **cosine rule**:

$$C^2 = A^2 + B^2 - 2AB \cos(c)$$

Sketch a water molecule in the space below.

6.22 The O-H bond length in water is 95 pm, and the H-O-H bond angle is 109.5°.

Calculate the straight-line distance between the two hydrogen atoms.

6.23 You can do this in a number of ways, but one way is using the cosine rule. Calling the distance we want x :

$$x = \sqrt{(95 \text{ pm})^2 + (95 \text{ pm})^2 - 2 \times (95 \text{ pm})^2 \cos(109.5^\circ)} = 155 \text{ pm}$$

We will see some more examples of this kind of geometry calculation in the problems.

We said that our trigonometric functions, $\sin$, $\cos$, and $\tan$, are *functions*. We know from earlier on that **all functions have inverse functions**. The inverse functions, $\arcsin(x)$, $\arccos(x)$, and $\arctan(x)$ are defined so that:

$$\arcsin(\sin(x)) = x$$

$$\arccos(\cos(x)) = x$$

$$\arctan(\tan(x)) = x$$

In school you may have called these inverse functions simply $\sin^{-1}(x)$, $\cos^{-1}(x)$, and $\tan^{-1}(x)$. We tend not to like this in most serious science, because it's important to note that $\arcsin(x) = \sin^{-1}(x)$ is **not** the same as $\frac{1}{\sin(x)}$, and $\sin^{-1}(x)$ looks a bit too much like that for my liking.

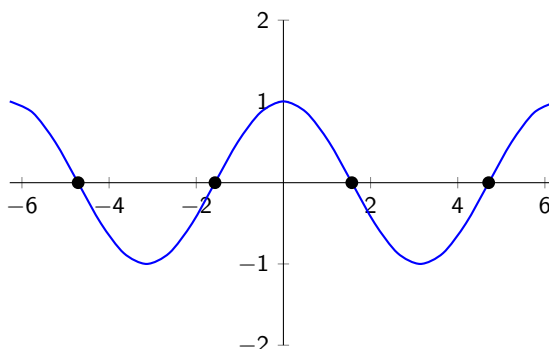
If $\cos(x) = 0$, what is x (in radians)?

6.24

$$x = 1.57 \text{ rad} = \frac{\pi}{2} \text{ rad}$$

Is this the *only* solution to the equation $\cos(x) = 0$? Think about the graph of $\cos(x)$ you drew earlier.

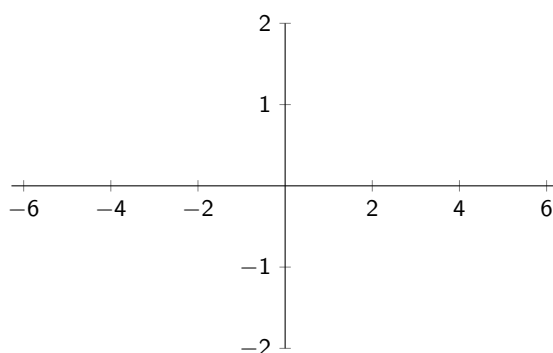
6.25 It's not. There are an infinite number of solutions, because the cosine function is periodic.



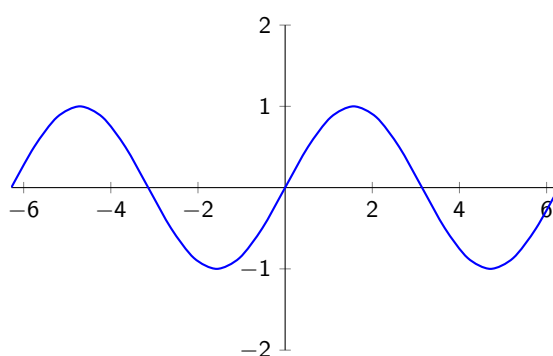
As the graph above shows, the function is zero (marked points) at $x = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$, and actually at any value of x where:

$$x = (2n + 1)\frac{\pi}{2}, \text{ where } n \text{ is an integer}$$

On the axes below, can you sketch a graph of $\sin(x)$?



6.26



What are the points x where $\sin(x) = 0$? Can you figure out a general formula for these points? Try writing out a few of them and see if you notice a pattern.

6.27 $\sin(x)$ is zero at $x = -2\pi, -\pi, 0, \pi, 2\pi, \dots$. The general formula will then be:

$$x = n\pi, \text{ where } n \text{ is an integer}$$

You'll use this fact later, when working out the energy levels of a quantised electron, which you can use to predict the colour of certain molecules.

Back to the inverse trigonometric functions, we saw that we had arcsin, arccos, and arctan. As we've just seen, these functions will always produce an infinite number of solutions, so in any practical use case either:

- The problem you're solving will put some limit on the valid solutions, or
- You just quote the smallest positive number that is the solution.

For example, solve the equation below for x :

$$\sin(2x) = 0.5$$

6.28

$$\arcsin(\sin(2x)) = \arcsin(0.5) \rightarrow 2x = 0.52 \text{ rad}$$

In that case, we had no boundary to put on the solution, so just quoted the smallest positive angle (0.52 rad) that solves it. Unless the question specifically says to give you answer in one form, I don't care if you give it in radians or degrees.

Here's another example. Solve the equation:

$$\cos(x) = 1$$

In the range $\pi < x \leq 3\pi$.

6.29

$$x = \arccos(1) = 2\pi$$

In this case, the default answer on your calculator is probably $x = 0$, but this is outside of the bounds we were given for x . The only solution that is within the bounds $\pi < x \leq 3\pi$ is $x = 2\pi$.

You can either work this out by sketching the graph (what I'd do), or by noting that the graph of $\cos(x)$ repeats every 2π radians, so if there is a solution at a value x , then there is also a solution at $x + 2\pi$, because the function is periodic.

The trigonometric functions can also all be related to each other, using the **trigonometric identities**. There are a lot of these (have a look at the Wikipedia article '*List of Trigonometric Identities*'), and you definitely don't need to learn them all. The ones I think it is worth knowing off by heart are:

$$\tan(x) = \frac{\sin(x)}{\cos(x)} \quad (\text{definition of tangent})$$

$$1 = \sin^2(x) + \cos^2(x) \quad (\text{Pythagorean identity})$$

Note here that $\sin^2(x)$ means $(\sin(x))^2$, **not** $\sin(x^2)$ (i.e. square the whole thing, not just the argument).

Solve the equation below for θ (you may find one or both of the identities above helpful):

$$2\sin^2\theta + \cos^2\theta = 1$$

6.30

$$\theta = 0, \text{ or more generally } \theta = n\pi \text{ where } n \text{ is an integer}$$

If you didn't follow that, let's break it down. First, we use our Pythagorean identity to rewrite the cosine term:

$$2\sin^2\theta + (1 - \sin^2\theta) = 1$$

Then simplify:

$$\sin^2\theta = 0$$

Then taking square roots leads us to $\sin\theta = 0$, which is an equation we have seen before (except with x rather than θ as the argument).

There are some other identities that it's useful to know exist, but are not really essential to memorise (though, if you want to, I won't stop you). These are: the **power reduction** identities, which can help simplify expressions involving powers of the trigonometric functions:

$$2\sin^2(x) = 1 - \cos(2x)$$

$$2\cos^2(x) = 1 + \cos(2x)$$

and the **angle sum** identities, which help if you have the sine or cosine of a combination of things:

$$\sin(x + y) = \sin(x)\cos(y) + \cos(x)\sin(y)$$

$$\cos(x + y) = \cos(x)\cos(y) - \sin(x)\sin(y)$$

You can derive the **angle difference** identities from these, if you think about it.

What do the angle sum identities reduce to, in the case that $x = y$?

6.31

$$\begin{aligned}\sin(2x) &= 2 \cos(x) \sin(x) \\ \cos(2x) &= \cos^2(x) - \sin^2(x)\end{aligned}$$

We'll use some of these identities later on, especially in quantum mechanics. The reason I say it is useful to know they exist, rather than memorising, is that as long as you know that *there is an equation that will help me reduce the power of a trigonometric function*, you can always look up the actual equation. This is what I do, anyway.

Finally, we can just use the trigonometric functions in equations. One classic case is in the wavefunctions of atomic orbitals. The angular part of the wavefunction of a 2p orbital, $\Psi(\theta)$, is given by:

$$\Psi(\theta) = \frac{1}{2} \sqrt{\frac{3}{\pi}} \cos(\theta)$$

An **angular node** is a point where the wavefunction passes through zero (i.e. $\Psi(\theta) = 0$).

What is the value of θ at the angular node in a 2p orbital? Does the answer make sense?

6.32

$$\theta = \frac{\pi}{2} \text{ rad}$$

Or 90° . We get here by setting the equation above to zero and solving – try it again if you didn't get the answer above. The answer should make sense, because there's a *nodal plane* in a 2p orbital at 90° to the main axis, which is what we just showed.

A final, harder, question. Can you solve the equation below, for x ?

$$2 \sin^2(x) - \sin(x) = 1$$

If you are stuck, try and make a substitution and turn it into a more familiar-looking equation.

6.33

$$\text{Either } x = \frac{\pi}{2} \text{ rad, or } x = -\frac{\pi}{6} \text{ rad}$$

We get there by making the substitution $y = \sin(x)$, which turns the original equation into:

$$2y^2 - y = 1 \rightarrow 2y^2 - y - 1 = 0$$

This quadratic equation can then be factorised or solved with the formula:

$$(2y + 1)(y - 1) \rightarrow y = 1 \text{ or } y = -\frac{1}{2}$$

Undoing the substitution leads to:

$$\sin(x) = 1, \text{ or } \sin(x) = -\frac{1}{2}$$

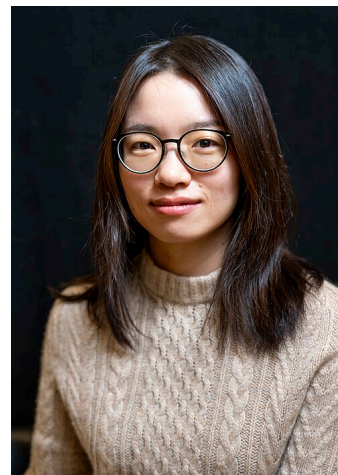
Which lead to the results given, taking the angles x nearest to zero.

That's enough trigonometry. Do the problems, and next time we will move onto thinking about **graphs**, and how to interpret them.

Problem Sheet 6 - Periodic Functions and Trigonometry

Periodic functions occur everywhere in science, because we are often dealing with things that oscillate, like waves. Even some things that don't seem like they are directly related to dynamic waves, like static images or spectra, can be expressed as combinations of periodic functions via a technique known as **Fourier analysis**. Modern IR spectroscopy, to give just one example, all relies on Fourier analysis.

In 2026, Hong Wang (pictured) won the Fields Medal – the equivalent of the Nobel Prize in mathematics – for her work on Fourier analysis, among other things (look up the 'Keakey Needle Problem'). She's about my age, which puts my modest achievements into perspective. I bet she's never written a lecture handout on 'Chemistry for Mathematicians' though.



1. Convert the following angles from degrees into radians.

i 90°

iii 30°

v 540°

ii 45°

iv 250°

vi 1°

2. Convert the following angles from radians into degrees.

i $\frac{\pi}{2}$ rad

iii 1 rad

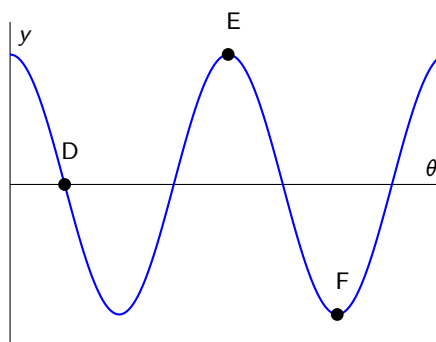
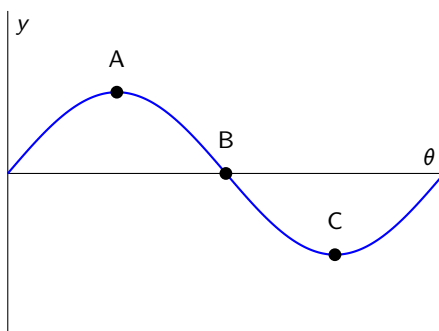
v $\frac{\pi}{8}$ rad

ii 1.5π rad

iv 20 mrad

vi $\sqrt{2}\pi$ rad

3. The graphs below show plots of $y = \sin(\theta)$ (left) and $y = 2\cos(2\theta)$ (right). Both plots are plotted with θ ranging from 0 to 2π . Find the coordinates of the points A, B, C, D, E, and F.



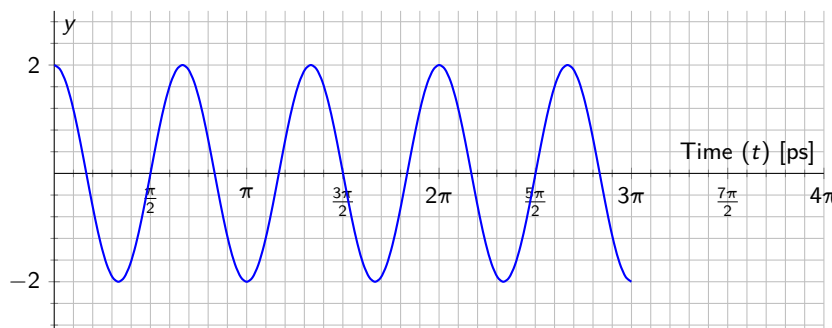
4. Sketch plots of the following functions:

i $y = 2\sin(x)$ in the range $-2\pi \leq x \leq 2\pi$.

ii $y = \sin(x + \pi)$ in the range $0 \leq x \leq 3\pi$.

iii $y = -\cos(x)$ in the range $-\pi \leq x \leq \pi$.

5. Some budget-conscious experimentalists have decided to build a time-domain IR spectrometer from scratch, rather than pay money to Bruker. The output signal from of their first measurements is below. Determine the equation of the wave, and hence state the frequency of the IR vibration they are measuring.



Hint: the wave could have an equation of the form $y = A \cos(bt + c)$.

6. **Without using a calculator**, use SOHCAHTOA and sketches of right angle triangles to find the values of θ between 0° and 90° that satisfy the following equations.

i $\sin \theta = 0.5$

iii $\tan \theta = \frac{1}{\sqrt{3}}$

v $\sin \theta = \cos \theta$

ii $\cos \theta = \frac{1}{\sqrt{2}}$

iv $\cos \theta = 0.5$

vi $\tan \theta = 1$

7. Use SOHCAHTOA to derive the following relationships.

i $\tan \theta = \frac{\sin \theta}{\cos \theta}$

ii $\sin^2 \theta + \cos^2 \theta = 1$

8. Use trigonometric relationships to simplify the following:

i $\sqrt{1 - \cos^2 \theta}$

ii $\frac{\sin^2 \theta}{1 - \sin^2 \theta}$

9. The cosine rule states that for a triangle with sides A, B, C and angles opposite each side a, b, c :

$$C^2 = A^2 + B^2 - 2AB \cos(c)$$

Verify that for a right angle triangle, this rule behaves as you would expect.

10. The molecule HOCl has a bond angle of 102.5° , an O-H bond length of 96.4 pm, and an O-Cl bond length of 168.9 pm.

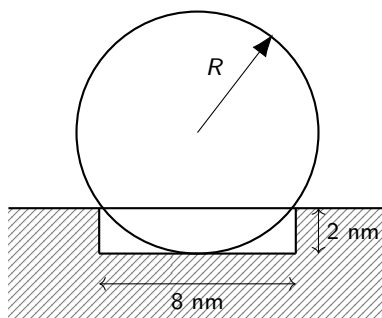
Draw a sketch of this molecule, and work out the straight-line distance between the H and Cl atoms.

11. Benzene (C_6H_6) is a planar molecule in the shape of a regular hexagon. The C-C bond lengths in benzene are $1.40 \text{ \AA}$.

i What are the internal angles of a regular hexagon, and hence of benzene?

ii Calculate the distance between any two carbon atoms that are directly opposite each other (i.e. the maximum 'length' of the overall benzene molecule).

12. The image below shows a carbon nanotube of radius R resting in a rectangular channel of width 8 nm and depth 2 nm. The nanotube rests on the bottom of the channel and touches the top two corners of the channel.



Calculate the radius of the nanotube, R .

Challenge Problems

1. A $4f_z^3$ orbital has three angular nodes. The angular part of the wavefunction, $F(\theta)$ for this orbital is:

$$F(\theta) = \frac{1}{4} \sqrt{\frac{7}{\pi}} (5 \cos^3 \theta - 3 \cos \theta)$$

Find the positions, θ , of the three angular nodes.

2. Prove (i) the cosine rule, and (ii) the sine rule.
3. A solar eclipse happens when the moon passes in front of the sun, covering the frontal area of the sun and blocking the light. One happened in August 2026, when I was writing these problems.

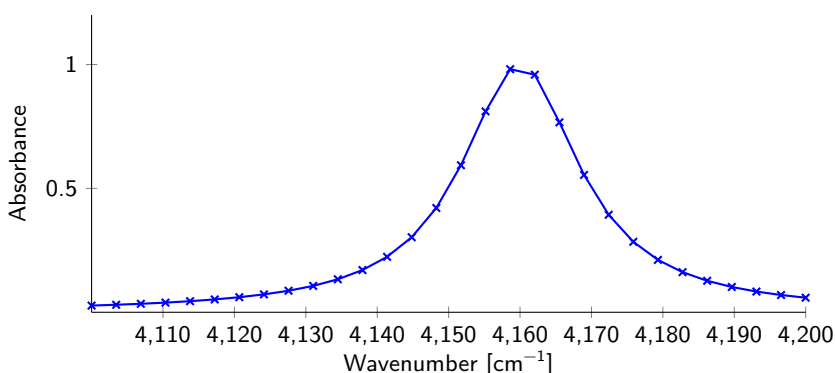
A simple model of the geometry of this eclipse is to imagine two circles of radius r , with their centres separated by a distance D .

As one circle passes in front of the other, calculate an expression for the shaded area, A , as a function of the distance, D .

Chapter 7

Graphs and Vectors

Having done a bit of graph sketching in the last few chapters, it's high time that we actually talk about graphs in science. The majority of the time when you collect data, you present that data **graphically** in the form of a graph. The data itself is numerical underneath, but we choose to represent it graphically to make trends easier to see, and to demonstrate our arguments more clearly. Here is some data, a potato-resolution vibrational spectrum of H_2 , plotted as a graph.



And here is the same data, presented as a list of numbers:

$2.7 \cdot 10^{-2}$, $3.03 \cdot 10^{-2}$, $3.42 \cdot 10^{-2}$, $3.9 \cdot 10^{-2}$, $4.47 \cdot 10^{-2}$, $5.19 \cdot 10^{-2}$, $6.08 \cdot 10^{-2}$, $7.21 \cdot 10^{-2}$, $8.69 \cdot 10^{-2}$,
0.11, 0.13, 0.17, 0.22, 0.3, 0.42, 0.59, 0.81, 0.98, 0.96, 0.77, 0.55, 0.39, 0.28, 0.21, 0.16, 0.13, 0.1,
 $8.37 \cdot 10^{-2}$, $6.97 \cdot 10^{-2}$, $5.88 \cdot 10^{-2}$

Hopefully you agree that the graph is clearer. All data that you're used to thinking of as graphs (IR spectra, mass spectra, chromatograms, etc...) is really just a list of numbers *presented as a graph*. I remember finding this fact oddly mind-blowing at an embarrassingly late stage of my scientific education. Maybe you do too.

Today we're going to learn how to present and interpret graphs, and use them to extract information. We will end by talking about **vectors**, which are related, and will set the scene nicely for the final chapter of this semester's handout.

7.1 When we plot a graph (in two dimensions), we plot something on the horizontal (x) axis and something on the vertical (y) axis. By convention:

- We plot the **independent variable** on the horizontal axis.

- The independent variable is *the one that we control*.
- We plot the **dependent variable** on the vertical axis.
 - The dependent variable is the one that *depends* on the independent variable. We don't control it directly.

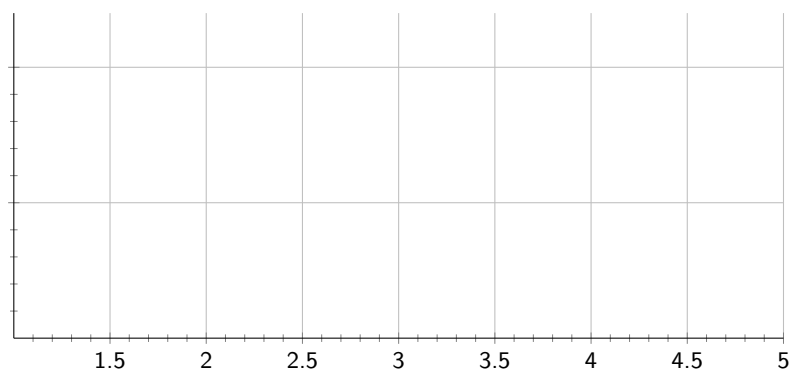
In the spectrum at the start of this chapter, the independent variable was wavenumber (we control this, by controlling the light source in our spectrometer), and the dependent variable was absorbance (this is determined by the specific molecule, in response to the wavenumber).

The data below was recorded in the teaching lab, to work out the concentration of an unknown solution. Which of the variables is the independent one, and which is the dependent one?

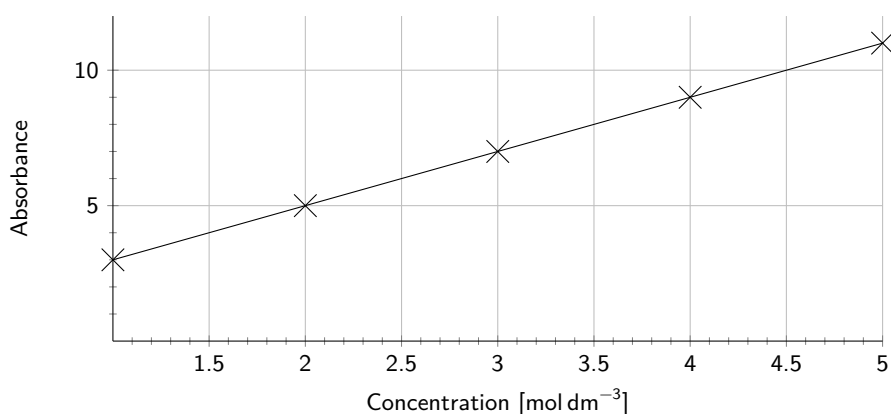
Concentration [mol dm^{-3}]	1	2	3	4	5
Absorbance	3.0	5.0	7.0	9.0	11.0

7.2 Concentration is the independent variable (the one we control). Absorbance is the dependent variable (it depends on concentration).

Can you plot the data from the frame above on the axes below?



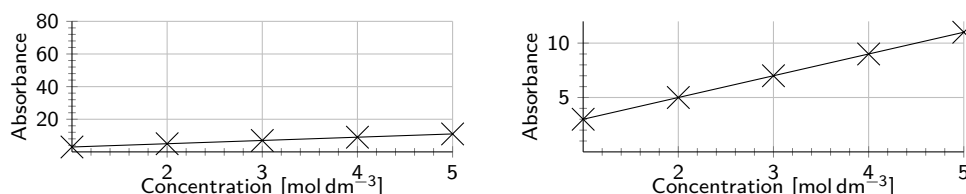
7.3



Hopefully you drew something like the graph above. Note:

- We label the axes to explain what is on each one (here, concentration and absorbance).
- If the quantity plotted on an axis has units, put the units in square brackets after the name (i.e. 'Concentration [mol dm^{-3}]').
- Set the axis *ticks* (the increment used on each axis) such that the plotted data occupies the majority of the area of the plot.

To illustrate the final point, the two graphs below show exactly the same data. Which one is clearer?



7.4 The one on the right. Hopefully that is obvious!

The point of representing data graphically is to make it easier to see, and to help you make arguments. The point of *any* data in science is to use to draw some conclusion about the world around us. If I was trying to argue the case that '*absorbance is directly proportional to concentration*', then a graph showing that absorbance is obviously increasing as concentration increases is a nice piece of evidence. That is why we draw graphs.

We can go further than this though. Do you know what kind of relationship, or trend, is plotted on the graphs of concentration and absorbance above?

7.5 A **linear** relationship is shown.

Linear relationships are easy to understand. They make straight lines on graphs, and straight lines are easy for humans to draw.

Can you write down a general equation for a linear relationship, or a straight line, between two variables y (dependent) and x (independent)?

7.6

$$y = mx + c$$

What do the m and the c represent?

7.7 m is the **gradient** of the line, and c is the **intercept**.

The gradient is the steepness (or slope) of the line, and the intercept is where it crosses the y axis.

Can you show that the intercept (c) is the value of y when the plot crosses the x axis?

7.8 $x = 0$ when the graph crosses the y axis, so:

$$y = mx + c \longrightarrow y = m(0) + c \longrightarrow y = c$$

Easy.

Do you know how calculate the gradient of a straight line? Write it down if you do.

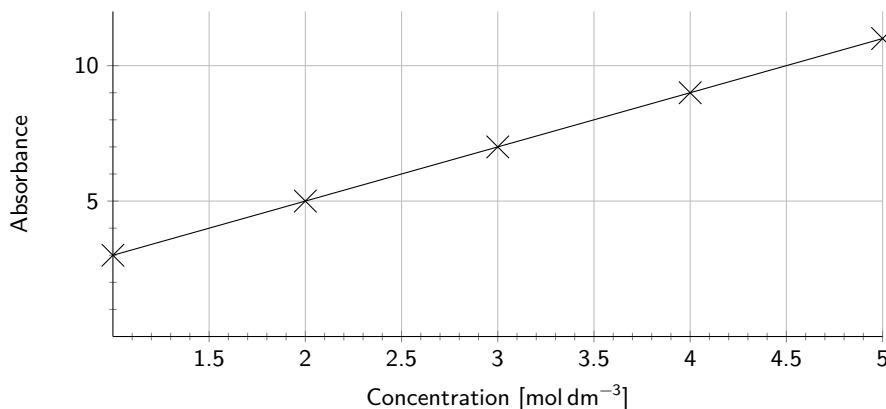
7.9 The gradient of any function (not just a straight line) is given by the *change in the dependent variable* (here we call it Δy), divided by the *change in the independent variable* (here we call it Δx):

$$\text{Gradient} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Where take two points on the graph, find the coordinates of each, (x_1, y_1) and (x_2, y_2) , and use these points to compute our gradient.

Gradients tell us about how things *change*. Scientists are usually very interested in how things change as we do things to them, so gradients are very useful things to be able to calculate. More on this next semester.

For the purposes of this chapter, we're thinking about gradients and graphs. Can you calculate the gradient, m , of the graph below?



7.10 We take two points on the graph, say (2,5) and (4,9), and plug them into the formula from above:

$$m = \frac{9 - 5}{4 - 2} = \frac{4}{2} = 2$$

For a straight line, it doesn't matter which two points you choose, the gradient is the same.

Given that the units of the quantities on the x and y axes are known, can you work out the units of the gradient? Do a dimensional analysis. Remember that absorbance is unitless.

7.11

$$\text{Units of Gradient} = \frac{\text{Units of } y}{\text{Units of } x} \rightarrow \frac{1}{\text{mol dm}^{-3}} = \text{dm}^3 \text{ mol}^{-1}$$

So for the plot given, the units of the gradient are $\text{dm}^3 \text{ mol}^{-1}$, the inverse of the units of concentration. This is because absorbance is unitless.

These units imply that the gradient has some physical meaning, and it does. Hopefully, you know by now what the relationship between absorbance and concentration is, so what is it?

7.12 The **Beer-Lambert Law**:

$$A = \epsilon c L$$

Where A is absorbance, ϵ is the absorption coefficient, c is concentration, and L is the path length the light travels through solution.

Let's say that the data we plotted earlier was recorded by measuring the absorbance in a cuvette with path length 1 cm. Can you calculate a value for the absorption coefficient, and its units? Compare the Beer-Lambert law above to the general equation of a straight line, if you're stuck.

7.13

$$\epsilon = 2 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$$

If we compare the Beer-Lambert law to the straight line equation, we can see:

$$A = \epsilon L c + 0$$

$$y = mx + c$$

Where we note that there is zero intercept in the Beer-Lambert law. We can then identify ϵL as the gradient of the Beer-Lambert plot. We know that $L = 1 \text{ cm}$ from the previous frame, so:

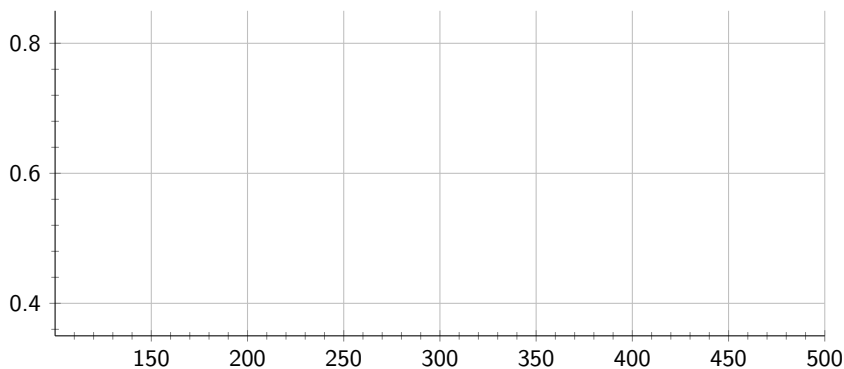
$$\epsilon = \frac{\text{Gradient}}{L} = \frac{2 \text{ dm}^3 \text{ mol}^{-1}}{1 \text{ cm}} = 2 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$$

The units for the absorption coefficient might look a bit obtuse, but they're perfectly correct. We could tidy them up to just $\text{m}^2 \text{ mol}^{-1}$, but who has time for that? Not us.

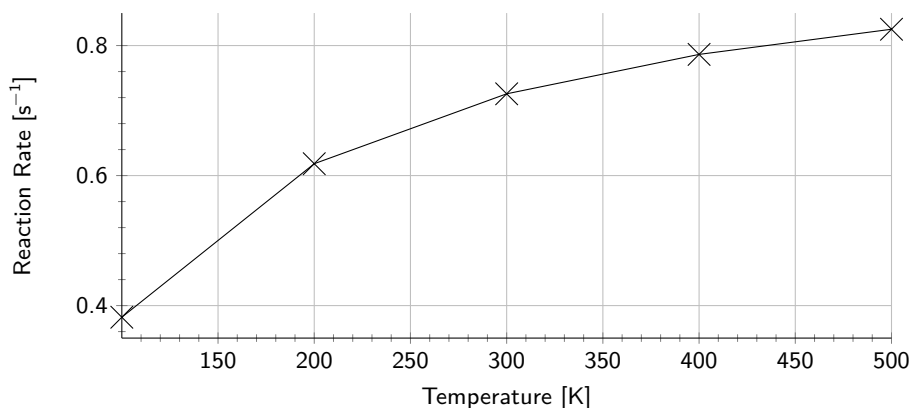
What we did here was **model** our measured data using the Beer-Lambert Law. The Beer-Lambert law gives us a way to calculate a chemical property (the absorption coefficient) from measured data of the concentration and absorbance. We call the absorption coefficient a **model parameter**, and by drawing a fit line through the data we can extract this parameter. We assume in this case that our data obeys the Beer-Lambert law, which may not always be true, but we can see that in this case we got the straight line from our data that we'd expect.

The process of drawing the fit line is essentially just varying the model parameter until the model and the data match each other as best they can. You can do this with any kind of plot, and thanks to computers, the plots don't even have to be linear. We will learn later how to model our data to any kind of function, and so extract all kinds of useful parameters. But time for another example, can you plot the data below on the axes given?

Temperature [K]	100	200	300	400	500
Reaction Rate [s^{-1}]	0.38208	0.61824	0.72577	0.78638	0.82509



7.14



I hope you remembered the axis labels.

This time, the data obviously isn't linear. You couldn't draw a straight line that reasonably fits through all the data points. This kind of **nonlinear** plot is very common, as the relationships between most things in the world are not simple linear relationships.

We can still learn things from these plots, though. We just have to fit a different line to the data, rather than a straight line. This is easy to do with a computer and we'll see it in practice separately in the course.

For now, we can try and think of what a good model for this data might be. The plot given is a plot of *reaction rate* against *temperature*. Can you think of an equation that relates these quantities?

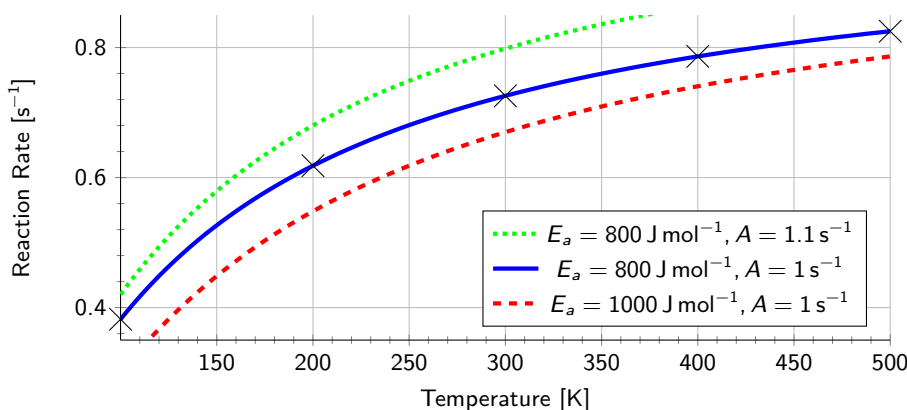
7.15 The Arrhenius Equation:

$$k = A \exp\left(-\frac{E_a}{RT}\right)$$

Where k is the reaction rate, E_a is the activation energy of the reaction, R is the gas constant, T is temperature, and A is the imaginatively named 'Arrhenius pre-exponential factor'. The equation above would be model for this data, if we assume the kinetics obey the Arrhenius equation (they don't always!).

In school, you may have learnt that a way to get around the nonlinearity of this equation is to take logs and *linearise* it. Unfortunately, nobody has done that in real science since before I was born, so we aren't going to do that now. What we do instead is just *fit the nonlinear function directly to the data using a computer*.

Explaining how this works in detail really requires that we understand some calculus (next semester), but basically the computer just changes the values of our model parameters (here E_a and A) until the model fits the data best. The plot below shows three versions of the equation above, plotted with different parameters E_a and A (different lines) together with the original data (black crosses). Which set of parameters fit the data best?



7.16 The blue solid line.

From the legend, the parameters that give us the blue solid line are $E_a = 800 \text{ J mol}^{-1}$ and $A = 1 \text{ s}^{-1}$. So, from this data, we know that the activation energy of our reaction is 800 J mol^{-1} . This method of 'change the parameters until the model curve matches the data points' is really fundamental in scientific data analysis. We will see it again when we learn some basic programming later in the course.

Another useful thing, that we've kind of already touched on several times, is **sketching graphs of functions**. For example, we might know that the concentration of product, C , from a chemical reaction over time t is given by:

$$C = C_0 \exp(-kt)$$

Where k and C_0 are constants. We may not really know what this equation is telling us, and a useful way to try and learn this is to sketch what the graph of the equation looks like. *Sketching* is not the same as *plotting* – we don't need the graph to have perfect numerical accuracy, we just need the shape and key points to be roughly right. To do this, there are a few simple steps:

1. Identify the dependent and independent variable (here t and C).
2. Work out what will happen to the graph of the independent variable at big (i.e. $t \rightarrow \infty$) and small (i.e. $t \rightarrow 0$) values of the dependent variable.
3. Work out the value at a few key points and plot them (e.g. at the intercepts of both axes, or (for next semester!) if there are maxima or minima we can identify).

Can you write down the value of C for the equation above in the case that (i) $t = 0$ and (ii) as $t \rightarrow \infty$?

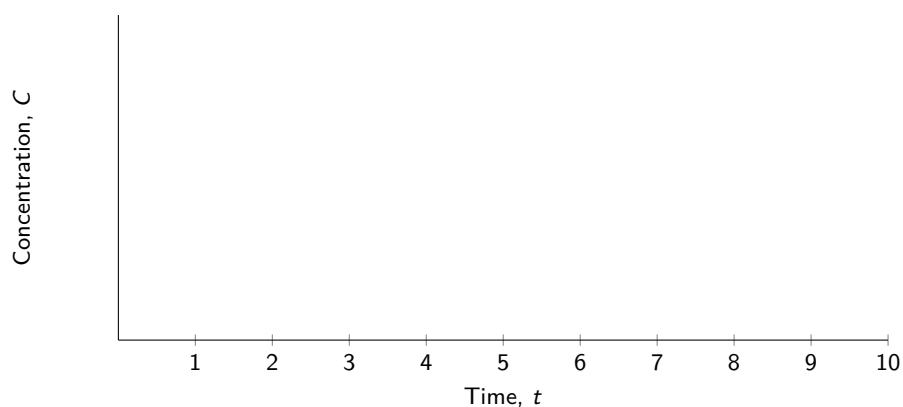
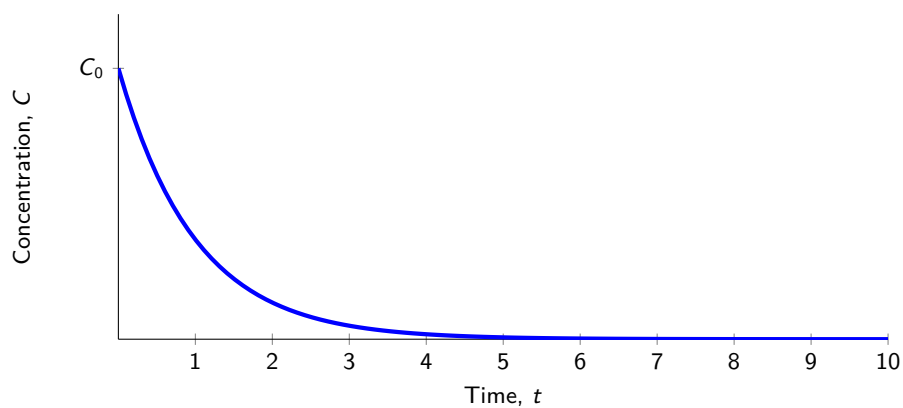
7.17 At $t = 0$, $C = C_0$ (because $e^0 = 1$). As $t \rightarrow \infty$, $C \rightarrow 0$, because $e^{-\infty} \rightarrow 0$. Think about this last point, if you're not sure why that's the case. (*Hint: plug in a few values of t with increasingly big numbers and see what happens.*)

We're about ready to try and sketch this function. We haven't said anything about the value of k , though. The value of k will obviously change the function, but actually it won't really change the *fundamental shape* of the function. For now, let's just say that $k = 1$, and we can think about what effect changing k has later.

If $k = 1$, can you work out values of the function at (i) $t = 1$, (ii) $t = 2$, and (iii) $t = 10$? Leave your answers in terms of C_0 .

7.18 We find that (i) $C = 0.37 C_0$, (ii) $C = 0.13 C_0$, and (iii) $C = 0.00004 C_0 \approx 0$.

With these, and the knowledge of the behaviour at $t = 0$ and $t \rightarrow \infty$, can you sketch our function ($C = C_0 e^{-kt}$) on the axes below? Think about what the value of C is at the y intercept, especially.

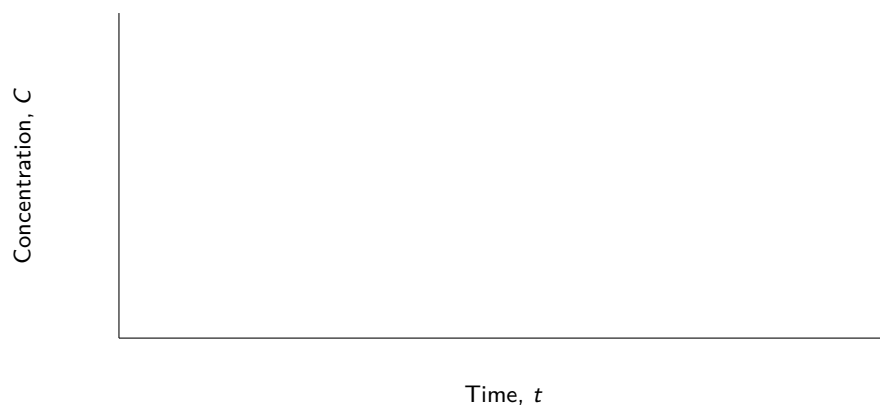
**7.19**

Anything like the above is great. The key points are:

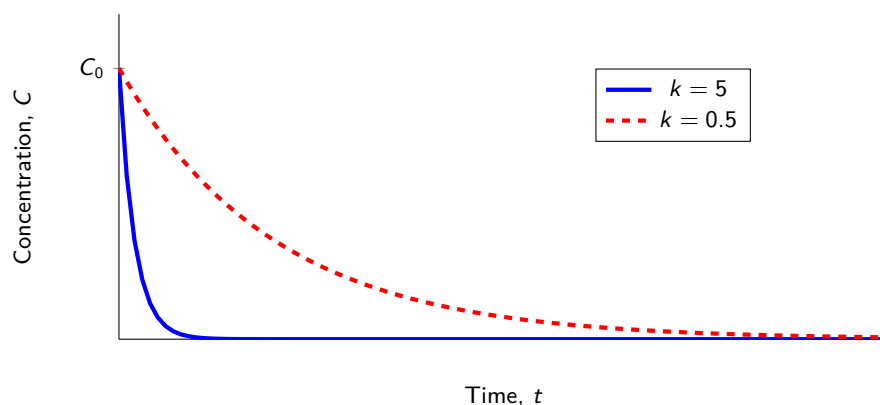
- The value of C is C_0 at $t = 0$.
- There's an exponential decay as t increases, getting to zero at about $t = 5$.
- $C \rightarrow 0$ as t gets bigger.

You don't really need to label the ticks on the x axis, I just did this to help you.

Now, on the axes below, can you sketch the same curve ($C = C_0 e^{-kt}$) in the cases that (i) $k = 5$, and (ii) $k = 0.5$? Remember we are not focussed on absolute numerical accuracy. We just need to get the relative shapes of the curves correct.



7.20



Anything like the above, where both curves start and end at the same point, but the curve with $k = 5$ decays faster than the curve with $k = 0.5$ is totally fine. If you didn't get that, calculate a few points on each curve that have the same value of t but with the different values of k . You'll see that the decay in the case where $k = 0.5$ is slower.

That's enough about making and reading graphs, until the problems. Let's use the graphs for something else to set the scene for the final chapter of this semester's maths workbook.

Think back to earlier on, when we were solving **simultaneous equations**. We learned how to solve them via substitution and elimination, and I said there was a third method we'd come to later. This method is to solve them **graphically**.

Below are two equations. Solve them simultaneously, using one of the previous methods.

$$x - 2y = 1$$

$$2x + y = 7$$

7.21 You should find that $x = 3$ and $y = 1$.

Now, notice that both of the equations can be rewritten in a form that makes it obvious they are straight lines. What are these straight lines?

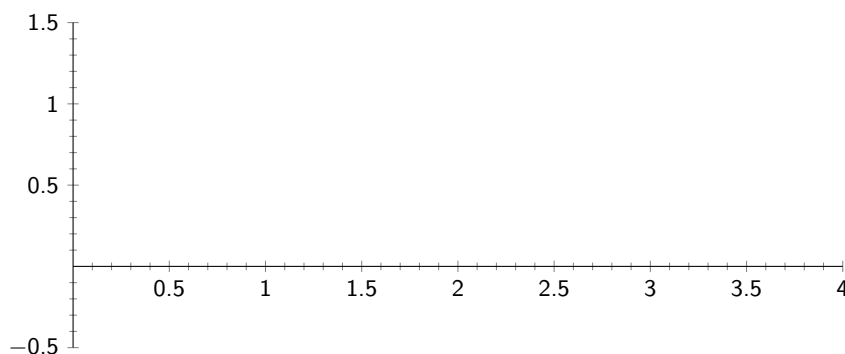
7.22

$$y = \frac{1}{2}x - \frac{1}{2}$$

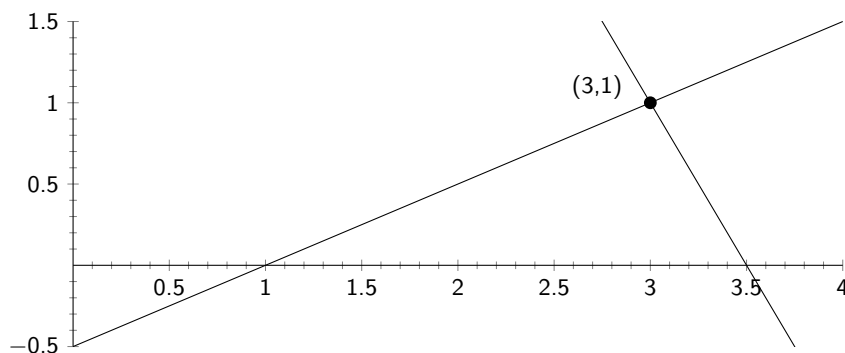
$$y = -2x + 7$$

Because we can write them as these straight line, or *linear* equations, we call the set of simultaneous equations above a **system of linear equations**. Solving systems of linear equations is a field of mathematics called **linear algebra**, which is a huge field, and underpins more science and engineering than it's possible to comprehend. All of the AI models you're probably used to using are just big linear algebra programs underneath.

Plot these straight lines on the axes below. Annotate the coordinates of the point where they cross.



7.23



You can see that the point of intersection is $(3, 1)$ – i.e. $x = 3$, $y = 1$, just like the solution we derived using algebra. This is another way to solve simultaneous equations, and it is very powerful. It works even for nonlinear equations, and for more than one equation (though in high dimensions this gets difficult to visualise!).

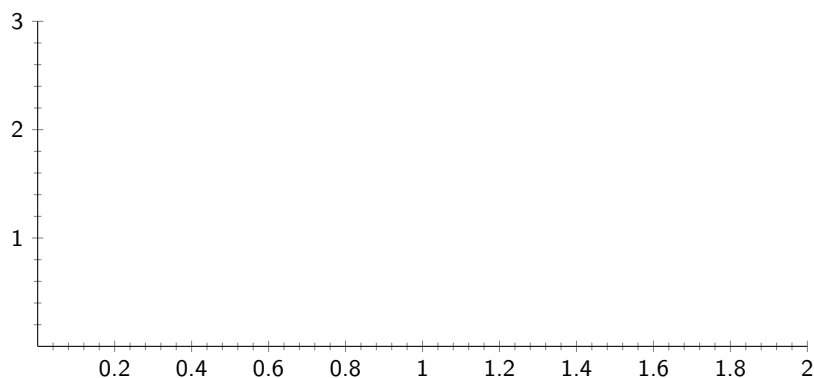
What would have to be the case for this set of equations to have *no solutions*? Think about it.

7.24 The lines would have to be *parallel*. If the lines are parallel, they never cross, and there is no solution. That's an interesting bit of mathematics to come back to later.

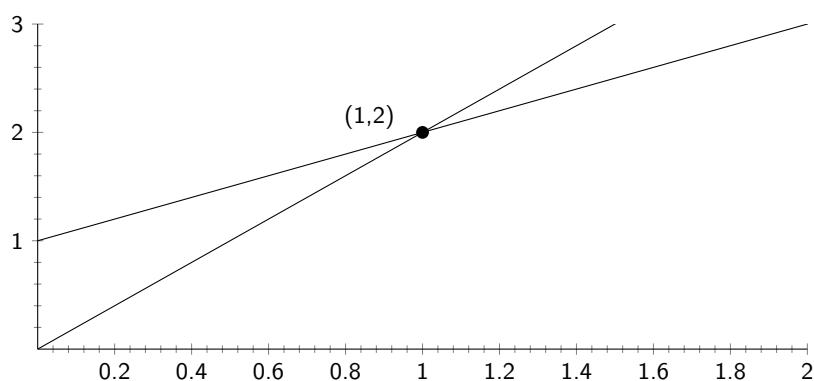
Can you solve the two equations below via a graphical method? I've given you axes to use, but you can check your answer with another method if you're stuck.

$$y - x = 1$$

$$y - 2x = 0$$



7.25



So the solution is that $x = 1$, $y = 2$. Neat.

The great Gilbert Strang, who I have shamelessly nicked this example from, calls this the **row picture** of linear equations. This is because when we draw out the equations, we are putting them in rows. Taking the first example again:

$$x - 2y = 1$$

$$2x + y = 7$$

You can see we have an equation in each row. The row picture of these equations views the solution as the point that the lines from the equations in each row meet at.

If you've ever played *Tetris* or used Excel, you'll know that *rows* tend to come with *columns*. So, if there's a *row picture*, it implies that there is also a *column picture*, and there is. What we do, unsurprisingly, is look at the *columns* in our original set of equations rather than the rows. The first of these columns, from the x terms, is:

$$\begin{pmatrix} x \\ 2x \end{pmatrix}$$

Can you see how that works? Write down all three columns.

7.26

$$\begin{pmatrix} x \\ 2x \end{pmatrix} \text{ and } \begin{pmatrix} -2y \\ y \end{pmatrix} \text{ and } \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

So we have a column for x , a column for y , and a column for the numbers at the end. We view these columns as **column vectors**.

A **vector**, generally, is anything that has both a *size* (magnitude), and a *direction*. Hopefully you met these in GCSE (but it might depend on the exam board you had...). Common vector quantities are things like *velocity*, *force*, and *acceleration* – anything that acts in a specific direction. Things that don't act in a specific direction, like *mass*, *charge*, or *volume* are called **scalars**.

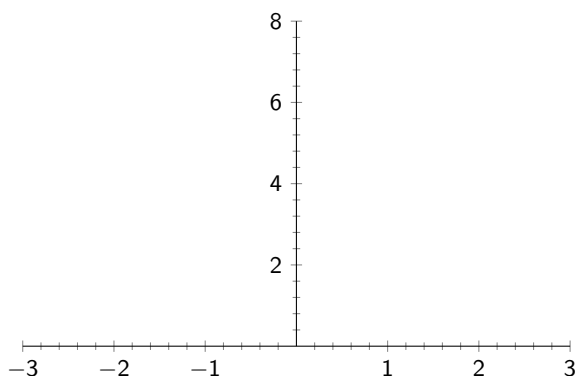
A common way to represent vectors is as arrows. The arrow points in the direction of the vector, and the length of the arrow tells you the magnitude of the vector. Given a vector like the ones above, we can visualise them as arrows easily. Let's take the vector below as an example.

$$\begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

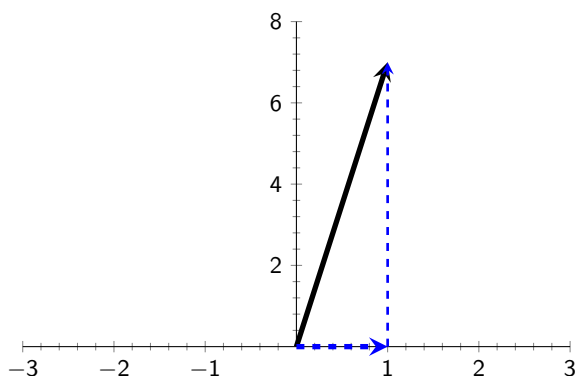
A way to turn this vector into an arrow is to realise that you can think of this vector as saying:

- Move along the x axis by 1 unit (this is the top number).
- Move along the y axis by 7 units (the bottom number).

If we do this, and then draw an arrow from where we started to where we end up, we've represented our vector as an arrow. Draw it on the axes below, starting our arrow at the origin.



7.27



To reiterate, to make this, we:

- Stepped one unit along the horizontal (x) axis.

- Stepped seven units along the vertical (y) axis.

So we are thinking of our vector:

$$\begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

As the top number meaning the number of steps along the horizontal, x , direction, and the bottom number meaning the number of steps along the vertical, y , direction. The overall effect of these steps is that we go from the point $(0,0)$ to the point $(1,7)$, as the arrow shows. Each of these steps is called a **component** of the vector. These components are shown as dashed blue lines on the plot. In two dimensions, a vector has two components. If our vector was in 3D space, it would have three, and so on.

The other vectors from our set of simultaneous equations were:

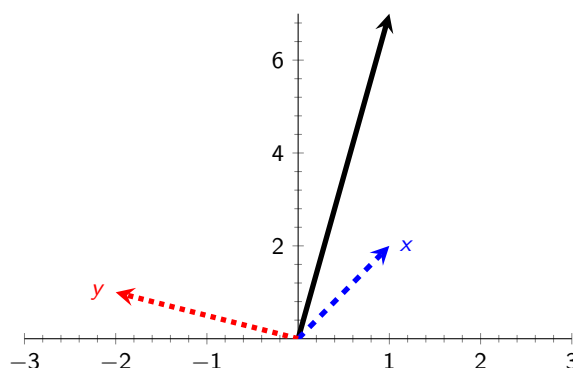
$$\begin{pmatrix} x \\ 2x \end{pmatrix} \text{ and } \begin{pmatrix} -2y \\ y \end{pmatrix}$$

We can tidy these up by factorising out the x and the y from each. This is allowed, as long as we factorise the same thing out of each component.

$$x \begin{pmatrix} 1 \\ 2 \end{pmatrix} \text{ and } y \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

Treating the factorised x and y as just labels of the resulting vectors for now, can you also draw these two vectors as arrows on the plot above?

7.28



Where the vectors for x and y are shown in blue (dashed) and red (dotted) respectively.

We can see that in the *column picture*, we have turned our set of equations into a set of vectors. This may seem like a bit of an extravagant thing to do, but vectors are useful, so let's see how to use them to solve these equations. We will look at the method now, and come back to cement our understanding of vectors thoroughly next time.

Firstly, note that we can add two vectors just by adding up their components:

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

Can you add the two vectors below?

$$\begin{pmatrix} 4 \\ -2 \end{pmatrix} + \begin{pmatrix} 5 \\ 7 \end{pmatrix} = ?$$

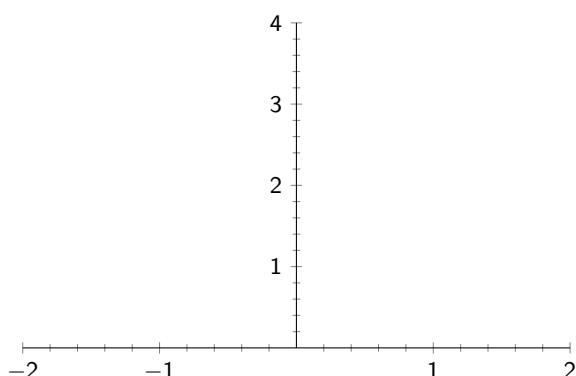
7.29

$$\begin{pmatrix} 4 \\ -2 \end{pmatrix} + \begin{pmatrix} 5 \\ 7 \end{pmatrix} = \begin{pmatrix} 9 \\ 5 \end{pmatrix}$$

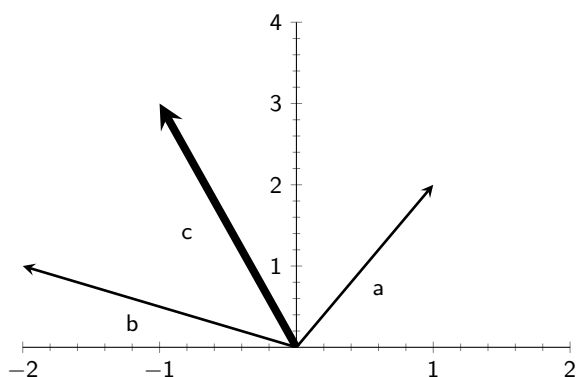
Geometrically, adding two vectors is like taking the arrow of each vector and laying them end to end. Let's see that using the vectors from earlier:

$$\underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_a + \underbrace{\begin{pmatrix} -2 \\ 1 \end{pmatrix}}_b = \underbrace{\begin{pmatrix} -1 \\ 3 \end{pmatrix}}_c$$

Draw the two vectors being added, and the resulting vector, on the axes below. Label them a , b , and c as in the equation above.



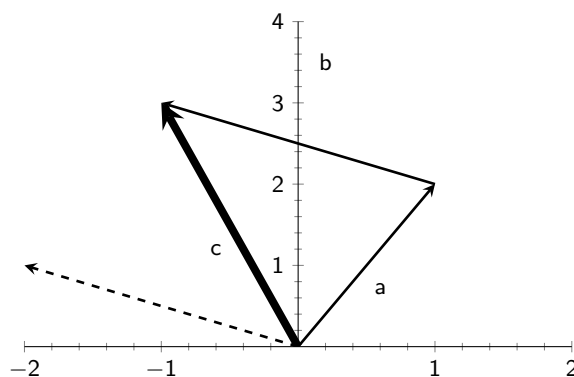
7.30



I've made the resulting vector c bold, for illustration. Can you see that we get to the same point if we take:

- A horizontal step of 1 unit and a vertical step of 2 units, then a horizontal step of -2 units, and a vertical step of 1 unit (starting at the origin, and following vector a then vector b).
- A horizontal step of -1 unit, and then a vertical step of 3 units (starting at the origin, and following vector c).

These two things are the same, the first point is the addition of the two vectors $a + b$, and the final point is the final vector c . It looks like this, if we move the vector $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$ so that it starts at the tip of the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$



I've left the original position of the vector we moved (b) as a dashed line. Imagine just moving it so the base of it starts at the tip of the other arrow. Can you see how that works? Meditate on it if not, or try sketching lots of arrows.

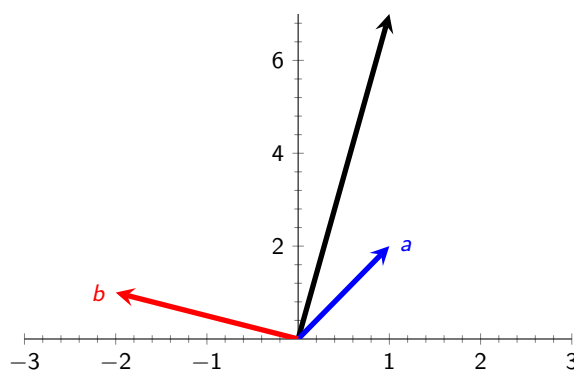
Let's get back to simultaneous equations. In the *column picture*, we had our equations turned into the vectors:

$$\begin{pmatrix} x \\ 2x \end{pmatrix} \text{ and } \begin{pmatrix} -2y \\ y \end{pmatrix} \text{ and } \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

We can write these in the form of our original equations:

$$x \underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_a + y \underbrace{\begin{pmatrix} -2 \\ 1 \end{pmatrix}}_b = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

Sketched, with some artful colour coding, these looked like the below.



The question I have for you now, is:

Is there a way you can combine the vectors a and b , in blue and red, so that you generate the resulting vector (in black)? Put another way, are there any specific values of x and y that make the equation below true?

$$x \begin{pmatrix} 1 \\ 2 \end{pmatrix} + y \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

7.31 If you think about it, you find:

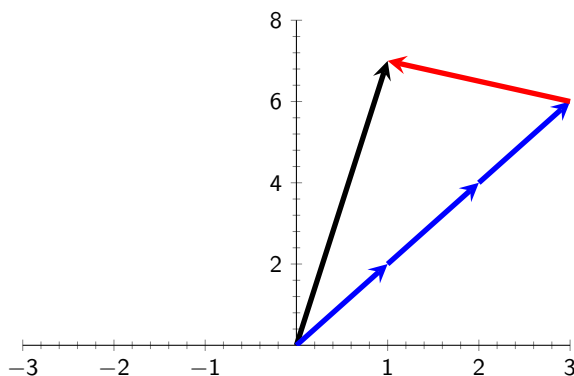
$$3 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 1 \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

Because:

$$\begin{pmatrix} 3 \\ 6 \end{pmatrix} + \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

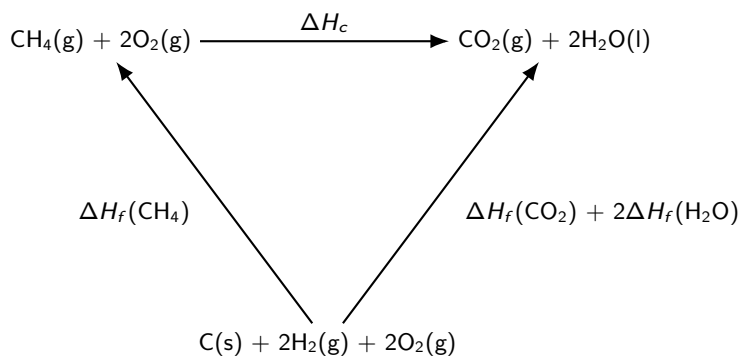
So we've just stumbled across another way to solve simultaneous equations. I said we were going to see a third, but we've got a fourth! Let no one say that the £10k per year isn't value for money.

We can visualise this solution by adding the a vector to itself 3 times, and then adding the b vector to the result once.



Can you see how that works? This method might seem a bit cumbersome compared to the others, but it ends up being useful when we get to higher dimensions. You'll be glad you've seen it, by the time we get to 3rd year quantum chemistry.

The main point here, though, is to get used to dealing with basic vectors. You're already good at this, even if you don't realise it. Do you remember how to work out ΔH_c in the Hess cycle below?



7.32 You follow the arrows from the reactants ($\text{CH}_4 + 2\text{O}_2$) to the products ($\text{CO}_2 + 2\text{H}_2\text{O}$). We take the direction into account, so we get:

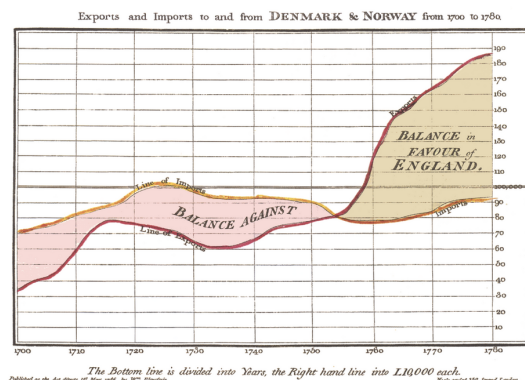
$$\Delta H_c = -\Delta H_f \text{CH}_4 + \Delta H_f \text{CO}_2 + 2\Delta H_f \text{H}_2\text{O}$$

It is just like adding up vectors, but each vector is one of our enthalpies. And before I get pedantic comments, I know enthalpy is not a vector quantity. The following the arrows around the cycle, is the bit that is like vectors. Once I understood this, Hess cycles became a lot easier than I had been led to believe in Sixth Form.

That's enough for now. Don't worry if you were fine with the graph stuff but got a bit lost with the vectors. We will continue with them next time, as we finish the semester with a first, basic, look at **matrices**. I can't think of a better Christmas present.

Problem Sheet 7 - Graphs

While writing this I realised I didn't know who invented the idea of a 'graph' showing data. It turns out no one seems to really know, but the earliest recorded use is surprisingly recent, from Scottish economist **William Playfair** in the 1780s. Below is one of his graphs, showing the balance of trade between England, Norway, and Denmark. Apparently around 1760, we started exporting loads of wool to them, hence the dramatic increase in exports



This problem sheet entirely on graphs, we will come to the stuff about vectors we saw at the end of chapter 7 in problem sheet 8, once we've got more familiar with it.

1. **Sketch** graphs of the following functions $q(t)$ (i.e. q on the y-axis and t on the x-axis). Annotate key points on each graph in terms of the constants a and b , where appropriate.

i $q = b\sqrt{t} + a$

iv $q = \frac{at+b}{2t}$

ii $\frac{1}{q} = at - \frac{1}{b}$

v $aq t^2 = b$

iii $q = a - \ln bt$

vi $q = a \exp\left(-\frac{b}{t}\right)$

2. In the previous question, t is time, measured in seconds, and q is a concentration, measured in mol dm^{-3} . For each function in parts (i) to (vi) of Q1, state the units of the constants a and b .
3. A student protester dropped an egg off the top of the Attenborough Tower. The height of the egg $H(t)$ (measured in metres) varies with time t (measured in seconds) according to the equation below:

$$H(t) = H_0 - \frac{1}{2}gt^2$$

This equation comes from Newton's equations of motion.

- i Identify the units and physical meaning of the quantity H_0 .
 - ii Identify the units and physical meaning of the quantity g .
 - iii Sketch a graph of $H(t)$, annotating key points, and comment on the shape of the graph.
 - iv (Bonus) Would this model of the falling egg be accurate even if the Attenborough tower was many orders of magnitude larger than it is? Explain your answer.
4. **Radiocarbon dating** is a technique that uses the radioactive decay of ^{14}C to calculate the age of things made of organic matter. Living plants absorb ^{14}C from the air, and when they die they stop, at which point the amount of ^{14}C starts to decay. Measuring the amount of ^{14}C thus allows the age of things to be measured.

The radioactivity, A , decays over time t according to:

$$A = A_0 \exp(-kt) \quad (7.1)$$

Where A_0 is the radioactivity of a modern piece of organic matter and k is the radioactive decay constant of ^{14}C .

- a) The radioactive **half life**, $t_{\frac{1}{2}}$ is defined as the time taken for the radioactivity of a sample to decay from A_0 to $A_0/2$. Use Equation 7.1 to show that the half life of ^{14}C is given by:

$$t_{\frac{1}{2}} = \frac{\ln 2}{k}$$

- b) Sketch a graph of Equation 7.1 in the case that (a) k is small, and (b) k is large.
 c) Take natural logarithms of both sides of Equation 7.1 and sketch a graph of $\ln A$ against t . Comment on the difference between this graph and the graphs from the previous part question.
5. The Stefan-Boltzmann law gives the relationship between the *radiant exitance* E of an object to its temperature T of the body:

$$E = sT^4$$

Where $s = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ is the Stefan-Boltzmann constant. This equation is commonly used to estimate the temperature of stars.

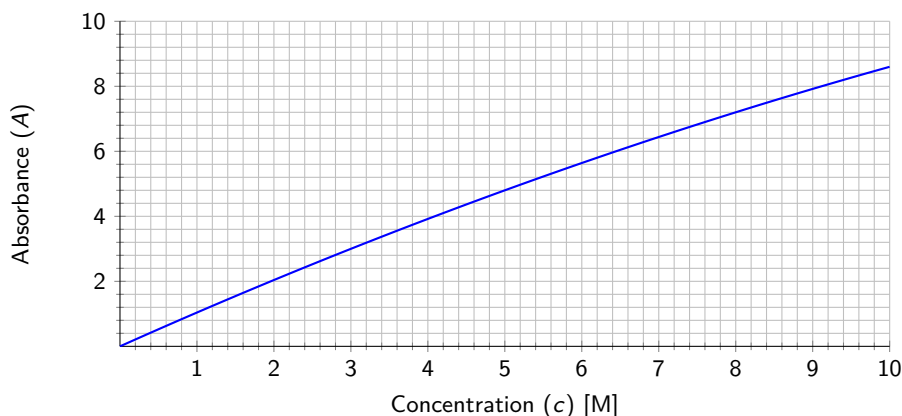
- i Derive the units of the radiant exitance that make the equation above dimensionally consistent.
 ii Sketch a graph of (a) E against T and (b) $\log E$ against $\log T$, and comment on the difference.
 iii By considering the amount of energy hitting the Earth from the Sun, the Stefan-Boltzmann law can be used to derive an expression that relates the temperature of the Earth T_e to the temperature of the Sun T_s :

$$\left(\frac{T_e}{T_s}\right)^2 = \frac{R_s}{2D}$$

Where $T_s = 5777 \text{ K}$ is the temperature of the Sun, $R_s = 695\,500 \text{ km}$ is the radius of the Sun, and $D = 149\,600\,000 \text{ km}$ is the distance between the Earth and the Sun.

Calculate the temperature of the Earth.

- iv The measured temperature of the surface of the earth is 288 K . Comment on this result in light of your answer to part (iii), and suggest a reason for the difference in temperatures.
6. The plot below shows the absorbance A , of a solution of dye measured as a function of the concentration c of the dye. The absorption coefficient of the solution $\epsilon = 1 \text{ m}^{-1} \text{ M}^{-1}$, and the path length of the cuvette used is 1 mm .



The following questions make use of the Beer-Lambert Law.

- i Sketch the variation of A with c that you would expect from the Beer-Lambert law.
 ii Does the plot behave as you expect from the Beer-Lambert law, over the whole concentration range c ? Identify the concentration range where the solution behaves ideally.
 iii Why does the real plot of absorbance deviate from the Beer-Lambert prediction? Suggest a plausible physical reason.

Challenge Problems

Sketch graphs of the following functions $F(x)$, which are commonly used in science. Any variable other than F or x is a constant.

1. The *Sinc Function*, which is used widely in signal processing and data analysis:

$$F(x) = \frac{\sin x}{x}$$

2. The *Gaussian Function*, which is used extensively in all kinds of data analysis and modelling:

$$F(x) = A \exp\left(-\frac{(x-b)^2}{c^2}\right)$$

3. The *Lorentzian Function*, which is used extensively in spectroscopy for modelling lineshapes:

$$F(x) = \frac{Ac^2}{(x-b)^2 + c^2}$$

4. The *Poisson Distribution*, which is used often in statistics and photon counting:

$$F(x) = \frac{b^x e^{-b}}{x!}$$

Note that the Poisson distribution is only defined for positive integer values of x .

Chapter 8

Vectors and Matrices

Finally, we are going to take a first look at **linear algebra**. For some reason, this is **not** in the GCSE or A level mathematics, so it will be new to you, unless you've done A level further maths. However, this doesn't mean it is difficult, as I hope you'll agree soon.

STEM subjects revolve around collecting data, analysing data, and presenting data. The data we collect *always* boils down to lists and tables of numbers. Fundamentally, that is what **vectors** and **matrices** are: lists and tables of numbers. We can use the same vectors and matrices to solve systems of equations and model complicated chemical reactions, compute molecular structures, calculate dosage limits and half-lives of drugs in the bloodstream, and much more.

As a concrete example, over the next two years in the course, you'll learn about **molecular orbital theory**, which involves calculating the energies (E) of molecules by solving systems of simultaneous equations with the form:

$$\begin{aligned}(H_{11} - E)c_1 + H_{12}c_2 &= 0 \\ H_{21}c_1 + (H_{22} - E)c_2 &= 0\end{aligned}$$

The details aren't important for now, but basically the two variables c_1 and c_2 tell you how much each atom in the molecule contributes to bonding. For big molecules, with more atoms, we end up with many more rows of equations here. Linear algebra is eventually going to give us a powerful way to solve these equations, which is how all computational chemistry in the real world is done.

Gilbert Strang (who's books I highly recommend) describes linear algebra as *pure gold*. In chemistry, it really underpins the theory of the whole subject. We can't finish the topic today, but we will start it and finish it over the next couple of years. Hopefully this chapter will give you enough of a flavour of the fundamentals that you'll be able to carry on later, at least.

Let's begin by recapping what we talked about last time with vectors.

8.1 A **vector** is a quantity that has both a *magnitude* and *direction*. **Vector quantities** in science are things like *velocity* and *force*, where the direction that they are applied in matters. **Scalar quantities** are things like *mass* and *charge*, where there is no obvious 'direction' that the quantity acts in.

Classify the below as scalar or vector quantities.

Acceleration, energy, temperature, momentum.

8.2 Acceleration and momentum are *vectors*, energy and temperature are *scalars*.

We saw last time that we can write a two-dimensional vector, $\mathbf{v}$ as a column of two numbers, like:

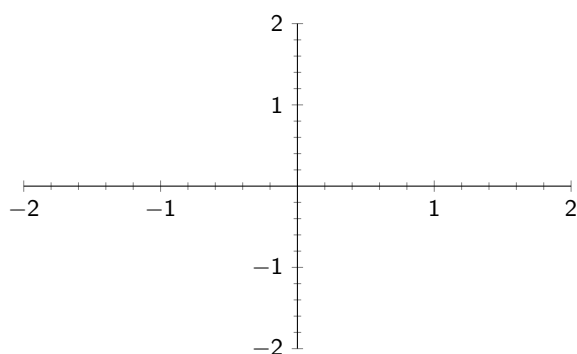
$$\mathbf{v} = \begin{pmatrix} a \\ b \end{pmatrix}$$

It's common to use bold lettering (i.e. $\mathbf{v}$ not v) to represent vectors in typed mathematics, or to draw an arrow over the variable ($\vec{v}$). We will use bold letters here.

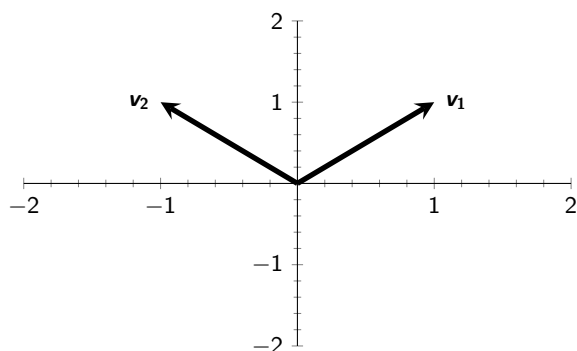
We also saw previously that we can represent this vector in Cartesian coordinates (the usual x, y coordinates) by starting from the origin and move along the x axis by a units, and along the y by b units. We call the quantities a and b the **components** of the vector.

On the axes below, sketch the vectors:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } \mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$



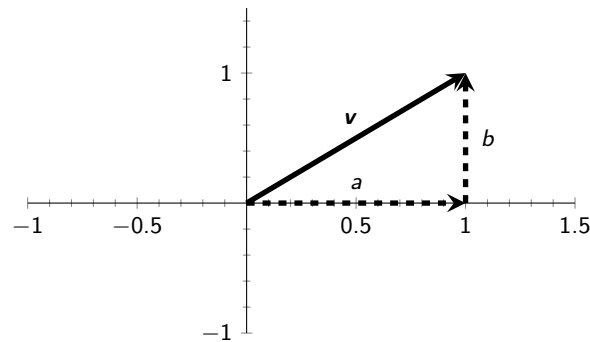
8.3



Note that it's just a convention that we start at the origin. The vector itself doesn't have a start point, it just has a direction and magnitude. The magnitude of the vector is the length of the arrow. The notation for the magnitude of a vector $\mathbf{v}$ is $|\mathbf{v}|$, just like we've seen for the absolute value of a number in chapter 1.

Below is a zoomed in version of the plot above showing just $\mathbf{v}$, with the components a and b highlighted as dashed lines. Can you think of a way to calculate the magnitude $|\mathbf{v}|$?

Remember that the magnitude is just the length of the vector, and the two components are at right angles to each other. Think about triangles.



8.4

$$|\mathbf{v}| = \sqrt{a^2 + b^2}$$

Just by using Pythagoras' theorem.

This definition also works in more than two dimensions, just by adding additional components. If we had a vector in 3D space, with 3 components a , b , and c , what would the magnitude of this vector be?

8.5

$$\sqrt{a^2 + b^2 + c^2}$$

This is sometimes called taking the **Euclidean norm** of the vector, and you can use this in conversation with your friends in maths and physics to impress them, as it implies you know about other kinds of vector norm but have judiciously chosen this one. Fake it until you make it, has always been my philosophy.

We can generalise this idea of the length of a vector to higher dimensions by adding more terms (but we can't visualise them as easily). Note that you still take the square root, even with three terms. We will only deal with two-dimensional vectors in this course.

We've already seen in the last chapter that we can add vectors to each other by adding their components:

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$$

We can also multiply a vector by a scalar:

$$A \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} Ac \\ Ad \end{pmatrix}$$

Let's practice. Given the two vectors $\mathbf{v}$ and $\mathbf{w}$ below:

$$\mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

Calculate the following:

i $\mathbf{v} + \mathbf{w}$

iii $2\mathbf{v}$

ii $\mathbf{v} - \mathbf{w}$

iv $3\mathbf{v} - 2\mathbf{w}$

8.6

i $\mathbf{v} + \mathbf{w} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

ii $\mathbf{v} - \mathbf{w} = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$

$$\text{iii } 2\mathbf{v} = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$$

$$\text{iv } 3\mathbf{v} - 2\mathbf{w} = \begin{pmatrix} -7 \\ -1 \end{pmatrix}$$

Work through them methodically again if you didn't get those. Taking combinations of multiples of vectors like this is called taking **linear combinations** of vectors. You'll hear that term a lot, especially in quantum chemistry.

We've seen how to add vectors to each other and multiply vectors by scalars. We haven't seen how to multiply vectors by vectors. There are multiple ways of multiplying vectors together, but the most important one for us is called the **scalar product** or **dot product**, denoted by a dot:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} c \\ d \end{pmatrix} = ac + bd$$

The output of this product is not a vector, it's a number (a scalar). We will get into the general geometric interpretation of the dot product in the problems.

What is the dot product of the vectors $\mathbf{v}$ and $\mathbf{w}$ from above?

8.7

$$\mathbf{v} \cdot \mathbf{w} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 2 \end{pmatrix} = (-1 \times 2) + (1 \times 2) = 0$$

The dot product being zero has an important physical meaning – **it means that the two vectors being multiplied are orthogonal (at right angles) to each other.**

Let's now get back to the idea of **systems of linear equations**. Here is a 2×2 system of linear equations: two equations, and two unknown variables x and y . All the other quantities are constants.

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

We can write this in a different form using vectors and **matrices**. A **matrix** is like a vector, but one that has two or more rows and two or more columns:

$$\begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

Can you see the correspondence between the matrix form (above) and the 'two equation' form above that? Try to write the specific 2×2 system of linear equations below in matrix form.

$$x - 2y = 1$$

$$2x + y = 7$$

8.8

$$\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

If you didn't get that, don't worry – we will see a systematic way to find it shortly.

The matrix in this case is:

$$\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$$

This is a 2×2 matrix, as it has two rows and two columns. Each row or column contains numbers, and these numbers are called **elements** of the matrix. There are four elements of a 2×2 matrix. In general, matrices can be of any size or shape, but we will only deal with square matrices in this course.

The elements of this matrix correspond to the **coefficients** in front of the x and y in our original set of equations, so we call it the **coefficient matrix** of our system of equations.

What is the coefficient matrix of the following set of equations?

$$\begin{aligned} 2x - y &= -3 \\ 4x + y &= -3 \end{aligned}$$

8.9

$$\begin{pmatrix} 2 & -1 \\ 4 & 1 \end{pmatrix}$$

We can add also matrices of the same size and shape by just adding the corresponding elements together, like we did with vectors.

Add the two matrices below together.

$$\begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} = ?$$

8.10

$$\begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The matrix we get at the end here, which has ones on the *diagonal* elements, and zeros everywhere else, is called the **identity matrix**, $\mathbf{I}$.

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The identity matrix is the matrix equivalent of the number 1. We'll see why later on.

Returning to our linear equations, we saw that one of our original sets of equations expressed in matrix form was:

$$\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

On the left hand side, we have a matrix multiplied by a vector. The way to think of this is as each *row* of the matrix being multiplied (dot product) by the vector. The answer is then a new vector, where each component is a dot product with one row of the matrix.

$$\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (1 \times x) + (-2 \times y) \\ (2 \times x) + (1 \times y) \end{pmatrix}$$

Which gets us back to our original set of equations (check and see).

Multiply the following matrix M by the vector $\mathbf{v}$:

$$M = \begin{pmatrix} 2 & -2 \\ 3 & -1 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

8.11

$$\begin{pmatrix} 2 & -2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} (2 \times 2) + (-2 \times 3) \\ (3 \times 2) + (-1 \times 3) \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

This process generalises to multiplying matrices by matrices, which we can do provided that the number of rows in the first matrix is equal to the number of columns in the second:

$$\begin{pmatrix} \text{Row 1} \\ \text{Row 2} \end{pmatrix} \begin{pmatrix} \text{Col 1} & \text{Col 2} \end{pmatrix} = \begin{pmatrix} \text{Row 1} \cdot \text{Col 1} & \text{Row 1} \cdot \text{Col 2} \\ \text{Row 2} \cdot \text{Col 1} & \text{Row 2} \cdot \text{Col 2} \end{pmatrix}$$

You can see that when multiplying two 2×2 matrices, the four elements in the result are the dot products of the rows of matrix 1 with the columns of matrix 2. Let's do an example – calculate the below:

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 0 & -2 \end{pmatrix} = ?$$

8.12

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ -1 & -1 \end{pmatrix}$$

Work through it slowly if you didn't get that.

Let's now try and multiply the matrices in the opposite order. Work out the below:

$$\begin{pmatrix} -1 & 1 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = ?$$

8.13

$$\begin{pmatrix} -1 & 1 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ -2 & -4 \end{pmatrix}$$

Notice that the resulting matrix is different from in the previous step. For matrices, **the order in which we multiply them matters**.

This is not what normally happens with numbers: $3 \times 2 = 2 \times 3$. For two matrices, A and B , then in general $AB \neq BA$. In the special case that $AB = BA$, we say that the matrices **commute with each other**.

So, we have our set of equations written in matrix form (I promise we are getting to solving them):

$$\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

We could write this as a product of a matrix M and a vector $\mathbf{v}$, that is equal to another vector $\mathbf{b}$:

$$M\mathbf{v} = \mathbf{b}$$

Write down the full form of M , $\mathbf{v}$ and $\mathbf{b}$, so we know what we're doing.

8.14

$$M = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

Which go into the equation:

$$M\mathbf{v} = \mathbf{b}$$

Crucially, the equation above would be the equation whether we had a 2×2 matrix, or a 3×3 , or a 100×100 matrix. The method we will see works in the same way.

Ultimately, what we want to do to solve the equations is to find x and y . In matrix form, this means finding the vector $\mathbf{v}$, because $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$.

If you had the equation below, and M , $\mathbf{v}$, and $\mathbf{b}$ were normal algebraic quantities, how would rearrange it to find $\mathbf{v}$?

$$M\mathbf{v} = \mathbf{b}$$

8.15 You'd divide both sides by M , leaving $\mathbf{v}$ on its own.

The issue with this here, is that unlike numbers, **in general there is no operation for 'division' by a matrix**. But all is not lost. What is the answer to the below?

$$\frac{3}{3} = ?$$

8.16

$$\frac{3}{3} = 1$$

This is the same thing as:

$$3 \times \frac{1}{3} = 1$$

Here, $\frac{1}{3}$ is the **inverse** of 3. For a variable X , the inverse of X is X^{-1} and is defined such that:

$$XX^{-1} = 1$$

All numbers have inverses, and the inverse of every number is the reciprocal of the number.

For matrices, the same idea applies, except that rather than $XX^{-1} = 1$ we have $MM^{-1} = I$, where I is the **identity matrix** we saw previously (remember I said this was like the matrix equivalent of the number 1? This is why).

Multiply the matrices below:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix} = ?$$

8.17

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Given what you have read in the previous step, what does this imply about the two matrices below?

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \text{ and } \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$$

8.18 They are **inverses** of each other.

However, **not all matrices have inverses**. That means that for a matrix M there is not always a matrix M^{-1} such that:

$$MM^{-1} = I$$

If there *is* a matrix M^{-1} , then we say that M is **invertible**. If there isn't, we say that M is **singular**.

If M is invertible, then our system of linear equations is easily solved by multiplying both sides of the equation by M^{-1} :

$$\begin{aligned} M^{-1}M\mathbf{v} &= M^{-1}\mathbf{b} \\ \rightarrow I\mathbf{v} &= M^{-1}\mathbf{b} \\ \rightarrow \mathbf{v} &= M^{-1}\mathbf{b} \end{aligned}$$

We'll see how to calculate inverse matrices shortly. For now, remember our original set of equations:

$$\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

The inverse matrix we need is:

$$\begin{pmatrix} 1/5 & 2/5 \\ -2/5 & 1/5 \end{pmatrix}$$

Multiply the original equations by this inverse matrix, and find the values of x and y .

8.19 Multiplying both sides by the inverse :

$$\underbrace{\begin{pmatrix} 1/5 & 2/5 \\ -2/5 & 1/5 \end{pmatrix}}_{M^{-1}} \underbrace{\begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}}_M \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{\mathbf{v}} = \underbrace{\begin{pmatrix} 1/5 & 2/5 \\ -2/5 & 1/5 \end{pmatrix}}_{M^{-1}} \underbrace{\begin{pmatrix} 1 \\ 7 \end{pmatrix}}_{\mathbf{b}}$$

We end up with:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{5} & \frac{2}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix} \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

Look at the LHS of the above equation, what do you get if you multiply it out? i.e. if you multiply:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ? \\ ? \end{pmatrix}$$

8.20

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (1 \times x + 0 \times y) \\ (0 \times x + 1 \times y) \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

i.e. any quantity multiplied by the identity matrix remains the same. It is just like 'multiplying by one' with numbers.

Coming back to our original equation:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{5} & \frac{2}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix} \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

We've seen how to multiply out the LHS. Can you multiply out the RHS?

8.21

$$\begin{pmatrix} \frac{1}{5} & \frac{2}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{pmatrix} \begin{pmatrix} 1 \\ 7 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

So, using this, and our answer from multiplying out the LHS previously, can you work out the vector $\mathbf{v}$, and the values of x and y ?

8.22

$$\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

i.e. $x = 3$ and $y = 1$. The answer we've got in many different ways over the last two chapters. This is now the fifth way!

The nice thing about this method, is that **if we can find the inverse matrix**, then this method will work for a system of equations of any size.

In general, we said that inverse matrices don't always exist, so it would be nice to know if there is a way to tell if they exist or not before we go to the trouble of trying to calculate them. We can do this by calculating something called the **determinant** of a matrix.

The **determinant** $|M|$, or $\det(M)$, of a 2×2 matrix M is a number that we calculate from the matrix as follows:

$$|M| = \det(M) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

The notation is either $|M|$ or $\det(M)$. I will stick with $\det(M)$ so we don't get confused between determinants and magnitudes. We are only going to worry about 2×2 matrices and 2×2 determinants in this course.

Remember that the determinant is just a number. Calculate the determinant of the matrix below:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

8.23

$$\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1 \times 1 - 0 \times 0 = 1$$

Easy. The determinant has some geometrical meaning, but we can come to that another time. The important thing is to remember that it's just a number.

We are almost there. For a 2×2 matrix M :

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

The **inverse matrix**, M^{-1} is defined as:

$$M^{-1} = \frac{1}{\det(M)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

So you see that in the 2×2 case, we get the inverse matrix by swapping around the positions and signs of some of the elements of the original matrix (the matrix we get here is called the *adjugate matrix*, for connoisseurs), and then dividing the whole thing by the determinant of the original matrix.

Find the inverse matrix of the matrix:

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

8.24

$$\frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Just by applying the formula from the previous step.

How could we check that this inverse is correct? Think about what MM^{-1} must be equal to.

8.25 We can check that the inverse we've derived, multiplied by the original matrix, gives us back the identity matrix.

Do that now, and check it works.

8.26

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Nice.

Ok, so let's go back to thinking about inverse matrices. Remember the formula for an inverse 2×2 matrix:

$$M^{-1} = \frac{1}{\det(M)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

What would happen to the inverse matrix, in the case that $\det(M) = 0$?

8.27 If $\det(M) = 0$, then we'd end up dividing by zero as we divided by $\det(M)$. We can't do this, and so if $\det(M) = 0$, then M^{-1} **does not exist**, and M is **singular**.

This will be a useful thing once we come to molecular orbital theory later in the course, as it will allow us find solutions to some equations called the **secular equations** that are scientifically interesting. I'll give you my notes for CH3203 if you are really interested.

Does the matrix below have an inverse?

$$\begin{pmatrix} x & 4x \\ 5 & 20 \end{pmatrix}$$

8.28 **No**, because the determinant is $20x - 20x = 0$.

Let's put it all together to round this up. Solve the system of linear equations below using matrices:

$$\begin{aligned}y - x &= 2 \\ y - 2x &= 3\end{aligned}$$

Remember the steps:

1. Write in matrix form.
2. Find the inverse of the coefficient matrix.
3. Multiply both sides of the equation by the inverse.

I'm not guiding through each step now, as we're almost at the end of the handout. Try it, and don't worry if you get stuck. Look at the answer below if you are really stuck.

8.29 Walking through it step by step, in matrix form we have:

$$\begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Our coefficient matrix, M has a determinant:

$$\det(M) = -2 - (-1) = -1$$

So the matrix does have an inverse ($\det(M)$ is not zero). The inverse matrix, M^{-1} is then:

$$\frac{1}{-1} \begin{pmatrix} -2 & 1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix}$$

Multiply the original equation by the inverse on both sides:

$$\begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Tidying up, leads to:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Check that this solution solves the original equation, and then proceed.

8.30 I hope you checked it.

This method of finding inverse matrices is one way of solving systems of linear equations. The nice thing about it is that it is scalable to systems of any size, and computers are fast at computing inverse matrices (there are actually even faster methods, like *elimination*, but we can look at these another time). We will see how to do these calculations in practical cases later in the degree.

That's it for new material this semester. **Please do the problems and the revision problems** so that you are ready for the exam in January. Next semester, we continue with maths and look at **calculus**. Calculus is brilliant and one of my favourite things. See you there.

Problem Sheet 8 - Vectors and Matrices

Unlike graphs, the history of matrices stretches back several thousand years. Humans have always been looking for ways to solve systems of equations, and in ancient history a lot of this work was motivated by trying to find the optimal way to plant your crops, given constraints on area, seed, labour, and so on. This was first written down by the Ancient Chinese, and has been used ever since.

Linear algebra using matrices is the foundation of all modern scientific computing, and especially machine learning and artificial intelligence. In chemistry, **Werner Heisenberg** developed the matrix formulation of quantum mechanics, which is largely what we use in quantum chemistry. He also worked on the German nuclear weapons program during WW2, but ended up being investigated by the Nazi party for not being sufficiently pro-Hitler in his views.



1. Identify whether the following are vector or scalar quantities.

- | | | |
|------------|----------------|----------------------|
| i Momentum | iv Temperature | vii Electric Current |
| ii Power | v Force | viii Molarity |
| iii Time | vi Velocity | ix Dipole Moment |

2. Sketch the following vectors in the $x - y$ plane (i.e. treating the top component of the vector as the x component, and the bottom as the y component, starting at the origin).

- | | | |
|---|--|---|
| i $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ | iii $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ | v $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ |
| ii $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ | iv $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ | vi $\begin{pmatrix} -2 \\ -3 \end{pmatrix}$ |

3. Three vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are defined as:

$$\mathbf{u} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

Sketch the three vectors, and then calculate the quantities below.

- | | | |
|--------------------------------|---|------------------------------------|
| i $\mathbf{u} + \mathbf{v}$ | iv $\mathbf{u} - 3\mathbf{w}$ | vii $\mathbf{u} \cdot \mathbf{v}$ |
| ii $\mathbf{u} + \mathbf{w}$ | v $\mathbf{v} + 2\mathbf{u} + \mathbf{w}$ | viii $\mathbf{u} \cdot \mathbf{u}$ |
| iii $2\mathbf{v} - \mathbf{u}$ | vi $\mathbf{v} - 2\mathbf{u} - 4\mathbf{w}$ | ix $\mathbf{u} \cdot \mathbf{w}$ |

Do you notice anything about the dot products in the last three parts, considering the sketches of the vectors you've made?

4. Calculate the matrix products below:

- | | |
|--|--|
| i $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ | iii $\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 3 & -4 \end{pmatrix}$ |
| ii $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} -1 & -2 \\ -3 & -4 \end{pmatrix}$ | iv $\begin{pmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{pmatrix} \begin{pmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{pmatrix}$ |

Don't be put off by the last one, just be methodical and remember some trigonometry!

5. Calculate the determinants of the matrices below:

i $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

iii $\begin{pmatrix} 1 & 1 \\ x^2 & x \end{pmatrix}$

ii $\begin{pmatrix} 1 & -3 \\ -3 & 1 \end{pmatrix}$

iv $\begin{pmatrix} y & y \\ 2 & 1 \end{pmatrix}$

6. By expressing the set of equations in matrix form, and multiplying by the appropriate inverse matrix, solve the pairs of simultaneous equations below:

i $y = 2x + 3$ and $y = -4x - 3$

ii $y = 3 - x$ and $y = 2 - 2x$

iii $y = 5 - 5x$ and $y = 10x + 5$

iv $y = x$ and $y = -x - 3$

Hint: follow the steps in the handout, and write every step out slowly and methodically.

7. Do the simultaneous equations below have a solution? Find it, if so, or explain why they do not.

$$y = x + 1$$

$$y = x + 2$$

8. In molecular orbital theory, the energies, E , of the molecular orbitals of H_2 are given by a set of matrix equations known as the **secular equations**:

$$\begin{pmatrix} \alpha - E & \beta \\ \beta & \alpha - E \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Where α and β are constants that relate to the orbitals involved in the bonding.

This kind of matrix equation (which is equal to the zero vector), only has an interesting solution in the case that:

$$\det \begin{pmatrix} \alpha - E & \beta \\ \beta & \alpha - E \end{pmatrix} = 0$$

- Calculate the determinant described above, leaving your answer in terms of α , β , and E .
- Set the determinant equal to zero. What kind of polynomial in E have we obtained, and how many solutions to this polynomial should there be?
- Solve the resulting polynomial equation for E .
- Hence explain why there are both *bonding* and *anti-bonding* orbitals in H_2 .

Challenge Problems

1. In each case below, identify the 2×2 matrix M that will perform the following transformations:

i $M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

iii $M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ y \end{pmatrix}$

ii $M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$

iv $M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$

2. A common use of matrices is to describe transformations, such as rotations and reflections, of objects. On a set of axes, sketch the vector:

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Now apply each of the following transformation matrices M to this vector $\mathbf{v}$, to obtain a new *transformed* vector $\mathbf{v}'$, as:

$$\mathbf{v}' = M\mathbf{v}$$

In each case, sketch the resulting vector $\mathbf{v}'$ and describe the transformation done by M (rotation, reflection, scaling, etc...).

i $M = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

iii $M = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

v $M = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

ii $M = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$

iv $M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

vi $M = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$

3. The matrix in part (vi) of Q2 was a *rotation matrix*. For the 2×2 case, an anticlockwise rotation of a vector by an angle θ is given by the matrix R :

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

- i Calculate R in the specific cases that $\theta = 0^\circ, 45^\circ, 90^\circ$, and 180° . Do the results make sense, given the sketches in Q2?
- ii The determinant of a transformation matrix represents the amount by which the transformed object is scaled by the transformation. Calculate the determinant of the matrix R , and explain the result.
- iii Apply R to the vector $\begin{pmatrix} x \\ y \end{pmatrix}$ and explain the result in the context of trigonometry.
- iv What is the inverse of the matrix R ?

Problem Sheet 9 - Revision

These problems are for revision. A lot of them have been taken from past CH1204A exam papers.

You will find that in general, **the questions on the exam are easier than the harder questions from the earlier problem sheets**. So, if you can do all the preceding problems in sheets 1-8 (not including challenge problems), you will do well on the exam.

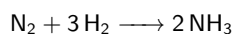
Note that you also have access to the last three years of exam papers, and we will go through some of these towards the end of the semester.

1. Solve the following equations:

i $1155^2 + 1656^2 = x^2$

ii $\frac{2}{-1+\sqrt{x}} = \frac{1+\sqrt{x}}{2}$

2. In the Haber process (is that still about the only thing they teach in GCSE chemistry?), the reaction:



Has an enthalpy change $\Delta H = -92.4 \text{ kJ mol}^{-1}$ and a change in Gibbs free energy of $\Delta G = -33.2 \text{ kJ mol}^{-1}$ at a temperature of 298 K.

- Given that $\Delta G = \Delta H - T\Delta S$, calculate the entropy change ΔS at 298 K.
 - Assuming that ΔH and ΔS are independent of temperature, calculate the temperature at which $\Delta G = 0 \text{ kJ mol}^{-1}$ for the Haber process.
 - What is the equilibrium constant, K for the reaction when $\Delta G = 0 \text{ kJ mol}^{-1}$?
3. Absorption spectroscopy is used to measure the concentration of an absorber $[x]$ dissolved in solution. The absorbance, A , of the solution is defined as in the Beer-Lambert law:

$$A = \epsilon[x]L$$

Where ϵ is the absorptivity (absorption coefficient) of the absorber and L is the distance light travels through the solution. The absorptivity depends on the wavelength, λ of light used to measure the absorbance, which is why molecules have distinct absorption spectra.

The absorbance of a mixture of two absorbers, x_1 and x_2 , is given by the sum of their individual absorbances:

$$A = A_1 + A_2 = \epsilon_1[x_1]L + \epsilon_2[x_2]L$$

The absorbance of a solution containing both titanium and vanadium ions was measured at two different wavelengths in a spectrometer with $L = 1 \text{ cm}$. The total absorbance of the solution was $A = 1.252$ measured at 400 nm and $A = 1.023$ measured at 460 nm. The table below gives the absorptivity of titanium and vanadium ions at these two wavelengths, in units of $\text{mol}^{-1} \text{ dm}^3 \text{ cm}^{-1}$.

Species	ϵ at 400 nm	ϵ at 460 nm
Titanium	6350	3200
Vanadium	1650	2200

Use these data to calculate the concentrations of titanium and vanadium in the solution.

4. The frequency, ν of a vibrating bond in a diatomic molecule is given by:

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

Where k is the force constant of the bond (imagine the bond as a spring holding the atoms together, k is the stiffness of the spring), and μ is the reduced mass of the two atoms.

Calculate the force constant for HCl, given that $\nu = 8.97 \times 10^{13} \text{ Hz}$ and $\mu = 1.614 \times 10^{-27} \text{ kg}$.

5. The fraction of catalytic adsorption sites, θ , occupied by molecules on the surface of a metal catalyst is given by Langmuir's isotherm:

$$\theta = \frac{Kp}{1 + Kp}$$

Where K is an equilibrium constant and p is the gas pressure. Find K in the case that $\theta = 0.2$ at a pressure of 133 mbar. Note that θ is dimensionless.

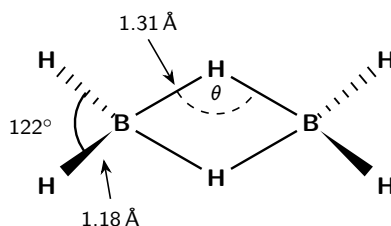
6. The equilibrium constant K for the reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ is defined as:

$$K = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

The initial concentrations of both H_2 and I_2 are 0.1 mol dm^{-3} , and there is no HI present at the start of the reaction. The value of K is 57.

Calculate the concentrations of all three species present in the reaction once the mixture has reached equilibrium. *Hint: You may find it helpful to call the concentration of reactant that reacts x , so that the concentration of reactant at equilibrium is $0.1 - x$.*

7. A distillation produces ethanol at a constant rate of $15 \text{ cm}^3 \text{ min}^{-1}$. The distillate is collected into a 250 cm^3 conical flask.
- How long does it take for the flask to fill with ethanol?
 - The time taken to fill the flask was measured with an uncertainty of $\pm 10 \text{ s}$, and the volume of the conical flask has an uncertainty of $\pm 2 \text{ cm}^3$. What is the uncertainty on the flow rate of ethanol produced by the distillation?
8. The structure of diborane, B_2H_6 , is shown below.



- Calculate the distance between the two terminal hydrogen atoms bonded to the same boron atom.
 - The distance between the two boron atoms is 1.74 Å . Calculate the angle θ between the two B-H bonds at a bridging hydrogen atom.
9. A right angle triangle has a hypotenuse with side length d^2 , and the other two sides have length $4d$ and 15 respectively. Sketch the triangle, calculate the value of d , and hence determine the lengths of all the sides.
10. Sketch a graph of the function:

$$y = 3 \sin 3x$$

For values of x between 0 and 2π . Hence determine the values of x for which $y = 3$.

11. The periodicity of trigonometric functions means that phase-shifted functions (like $\sin(x + \phi)$) can often be expressed in terms of non-phase shifted functions (like $\sin(x)$). Letting $P = \sin(x)$, express the following in terms of P :

i $\sin(x + 2\pi)$

ii $\sin(x + \pi)$

iii $\sin(x - \pi)$

iv $\cos(x - \frac{\pi}{2})$